



MATH SURVEY

Official Publication of the Stuyvesant Math and Computer Science Department since 1927

2023-2024
Winter Edition

Contents

Page No.

Contributor's List	i
Editor's Note	ii
Foreword	iii
... <i>by Mr. Stan Kats</i>	iii
1 Puzzles	1
TrueFalse Challenge <i>by Ethan Sharma</i>	1
Solutions	5
A Reflection on Mathematicians <i>by Mikayla Lin</i>	16
Solutions	19
Tower of Hanoi <i>by Patrick Was</i>	21
Solution	23
Algebraic Constructions <i>by Patrick Was</i>	27
Solutions	28
Winter Crossword <i>by Vincent Chen</i>	30
Solutions	32
2 Mathematics Articles	34
Soddy Circles, Inversion, and Descartes' Circle Theorem <i>by Ethan Sharma</i>	34
Fourier Transforms <i>by David Jiang</i>	58
P-adic Numbers <i>by Tanvir Ahmed</i>	64
Cantor's Infinities <i>by Jacob Levin</i>	85
The Layers of Cake-Cutting <i>by Shaon Anwar</i>	90
Galois and Quintic Formula <i>by Patrick Was</i>	93
Magic Squares <i>by Ivy Zhu</i>	98
Cool Things You Need to Know About Triangles <i>by Qinwen Zheng</i>	105
Rise of the Cosine <i>by Elizabeth Burnell</i>	115
3 Computer Science Articles	
GANs <i>by Han Xiao</i>	119
Regression NLP and Proximal Policy Optimization <i>by Naowal Rahman</i>	129
How Does AI Audio Generation Work? <i>by Caden Khuu</i>	137
Securing Your Cyber Security: What is Hacking? <i>by Aditya Anand</i>	140

Stuyvesant Math Survey

2023-2024 Winter Edition

The Official Mathematics and Computer Science Publication
of the Stuyvesant High School Math Department since 1927

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Editor's Note

This winter, we are excited to share with you the 2023-2024 Winter Edition of *Math Survey*! This edition is filled with thought-provoking articles on math and computer science as well as fun and challenging puzzles. Thank you so much to our writers for their hard work researching and exploring interesting topics in math and computer science. A big thanks also to our staff members and editors for their dedication to our publication.

We would also like to thank our faculty advisor, Mr. Rubinstein, for his guidance and support. We would like to extend our appreciation to the Parents' Association, for without their generous support we would not be able to print hard copies. Finally, a massive thank you to Mr. Kats for contributing a foreword and for always offering us advice.

Amanda Louie, Daniel Potievsky
February 2024

Foreword

As a student of Brooklyn Tech, Mr. Stan Kats was heavily influenced to pursue mathematics by his teacher, Larry Zimmerman. He was a member of the New York City Math Team, and after graduating high school, he continued his association with the NYC Math Team by being a coach. Mr. Kats was a Stuyvesant Math Teacher for 8 years and a coach for the Stuyvesant Math Team. He is currently a writer for the NYSML Competition and is the Head Coach for the New York City Math Team. He is currently teaching at Brooklyn Technical High School.

Math, to me, is the most beautiful subject in the world. The person who made it come alive for me was my high school teacher, Larry Zimmerman. When I saw him teach, it was like watching an artist at work. I feel fortunate to have experienced teaching like this, and I feel sorry that so many people in the modern-day schooling system go through their education without ever seeing teaching mathematics elevated to the level of art.

The chalk was his paint, the blackboard was his canvas, and every day in room 2W7, Mr. Zimmerman would paint a masterpiece. He would talk about Divisibility Rules for Primes, Pell Equations, Napoleon Triangles, Invariance, and an array of other topics. He piqued everyone's curiosity and allowed us time not only to process the information, but to try to make our own insights and conclusions.

When I started teaching, I wanted so much to be like him. I even tried breaking the chalk the way he did. I quickly learned the importance of finding your own voice, but one constant remained - giving students time to think and make their own insights.

With each passing year, I relax the pace at which I teach my curriculum. Part of this is to give students more time to process information and part of it is because I keep finding new observations to talk about. That is part of what makes math so beautiful - after teaching a lesson for the 40th time in my life, I'm still discovering something new.

That's what makes the *Math Survey* so great - writers can pick a topic and explore it. It's not about stating all the results, or copying something from a book, article, or Wikipedia page, but about sharing something you love, why you love it, and what you've noticed about it. In a day and age of rote learning and handouts, being able to sit down and work on some math using paper and pen is the best form of enrichment.

To the readers, find something that piques your interest and think about it. Put pen to paper and write out your thoughts. If what you're thinking about is a math problem and you solve it, that's great, but just because you solved a problem doesn't mean you're done with it. Try to look

for solutions in two, three, or seven different ways. And when you find a new way of looking at a problem, share it with others. It's what makes this community so worthwhile.

Stan Kats
Math Teacher
February 2024

TrueFalse Challenge

Ethan Sharma

Welcome to the TrueFalse Puzzle! The goal of this puzzle is to guess a predetermined positive integer n , or to narrow it down to as few choices as possible. You will be given 10 statements, and you have to figure out which are true and which are false. Based on your answer to each statement, you will be given a hint towards finding n . The harder the question, the more restrictive hints. Note that the correct answer doesn't necessarily correspond to the choice that results in fewer possible choices for n . A calculator probably won't help you with these problems, but will be useful for finding n .

Here is an example statement:

0. Preliminary information

True or False Statement	
If the statement is true, this is true	If the statement is false, this is true

Now that you know how this all works, take a go at the problems and good luck!

The solution is on page 5.

1.

Given three sets P, Q , and R , $P \cup Q \cup R = P \cap Q \cap R$ if and only if $P = Q = R = \emptyset$.

$$10^6 < n < 10^{12}$$

$$n < 10^9$$

2. Let $f(x), g(x)$ be real-valued functions such that $f(x) = \frac{1 - x^{100}}{1 + x^2}$ and $g(x) = \frac{1 + x^{100}}{1 - x^2}$.

The maximum of $f(x)$ and the minimum of $g(x)$ are the same value.

The last two digits of n in base 2 are 00.

The last three digits of n in base 3 are 000.

3.

There exists at least one non-degenerate circle tangent to any three distinct lines in a plane.

n , when expressed in base 63, has 4 digits.

n , when expressed in base 36, has 5 digits.

4. Let the choose function be defined as $\binom{n}{k} = \frac{n!}{k!(n-k)!}$.

$$\sum_{i=k}^n \binom{i}{k} = \binom{n+1}{k+1}.$$

n is one of the terms of an infinite arithmetic sequence with $a_1 = 1$ and $a_2 = 2024$.

n is one of the terms of an infinite geometric sequence with $a_1 = 1$ and $a_{2024} = 2$.

5.

There exists a real number k such that $\cos\left(\frac{2\pi n}{k}\right)$ never repeats a value in the range $[-1, 1]$ across all positive integers n .

Let H_k be the hint you chose for problem k , and N_k be the hint you didn't choose.

Instead of H_3 , use hint N_3 .

Instead of H_1 , use hint N_1 .

6. Let $f(x) = x^2 - 8x + 14$ and $g(a, b, p) = \gcd(a + b, \frac{a^p + b^p}{a + b})$, where p is an odd prime.
(Problem Credit: Patrick Was)

There exists no x_1, x_2 such that $f(x_1) = f(x_2) = p$ and $g(a_1, b_1, p) \neq g(a_2, b_2, p)$ for all integers a_1, b_1, a_2, b_2 that satisfy $a_1 + b_1 = x_1$ and $a_2 + b_2 = x_2$.

n is divisible by 701.

n is divisible by 107.

7. $\triangle ABC$ is an isosceles right triangle with right angle at B . Let point X_1 be the foot of the altitude from B to AC , and let X_2 be the foot of the altitude from X_1 to BC . Then, for positive integer $n > 2$, let X_n be the foot of the altitude from X_{n-1} to segment $X_{n-2}C$. Consider the incenters O_1 of $\triangle ABX_1$, O_2 of $\triangle BX_1X_2$, and O_n of all triangles $\triangle X_nX_{n+1}X_{n+2}$.

For integers $k > 0$, all incenters O_{2k-1} are collinear, all O_{2k} are collinear, and these two lines intersect at C .

Let $b_k = \prod_{i=1}^k \left(1 + \frac{1}{i}\right)^2$.

$n \equiv b_{2023} \pmod{9}$

$n \equiv b_{2024} \pmod{9}$

8. Let $f^n(x)$ be f iterated n times onto itself, so $f^3(x) = f(f(f(x)))$. Consider the function $f(x) = ||x| - \pi| - \pi|$.

$\lim_{n \rightarrow \infty} f^n(x) = \arccos(\cos x)$.

$\log_5 n > 8(\log_5 2) + 9$

$\log_2 n < 9(\log_2 10) - 1$

9. Let $f(x) = \frac{\lceil x \rceil}{\lfloor x \rfloor}$, where $\lceil x \rceil$ and $\lfloor x \rfloor$ are the ceiling and floor functions, respectively. The ceiling function returns the least integer greater than or equal to x , and the floor function returns the greatest integer less than or equal to x for any real input x . For example, $\lceil 2.2 \rceil = 3$ and $\lfloor 1.7 \rfloor = 1$.

The following integral converges:

$$\int_1^\infty \left(f(x) - \frac{1}{f(x)} \right) dx$$

$$n \equiv \sum_{k=1}^{18} 2k \binom{18}{k} \pmod{19}$$

$$n \equiv \sum_{k=1}^{20} k \binom{20}{k} \pmod{19}$$

10. Let $f(x, y) = \frac{1}{xy(x^2 + y^2)}$.

The integral

$$\int_1^\infty \int_1^\infty f(x, y) dx dy$$

converges to a value Z that satisfies $e^{nZ} = m$ for some positive integers n and m .

Let S be the set of possible values of n you have from other clues.

n is the element of S with the minimum sum of digits.

n is the element of S with the maximum sum of digits.

TrueFalse Challenge: Solution

Ethan Sharma

1 Solutions

The value of n is: 895001500

1. False. We can see that, if $P = Q = R = \emptyset$, then we know $P \cup Q \cup R = P \cap Q \cap R = \emptyset$. However, this is not true the other way around. The union of the three sets will contain all values in any of the sets, and the intersection of the three will contain elements common to all three. Thus, if $P = Q = R$, then $P \cup Q \cup R = P \cap Q \cap R = P = Q = R$, since each of the values are contained in each set, and no set contains more than the common values. For example, consider $P = Q = R = \{1, 2\}$. Then $P \cup Q \cup R = \{1, 2\}$ (all values in any set) and $P \cap Q \cap R = \{1, 2\}$ (all values common to all three sets). We can see that $P \cap Q \cap R = \emptyset$ is one case of this condition.
2. True. Let $a = x^2$, and note that we must have $a \geq 0$ since $x^2 \geq 0$. Then, we have

$$f(x) = \frac{1 - a^{50}}{1 + a} \qquad g(x) = \frac{1 + a^{50}}{1 - a}$$

We can prove that, for any positive rational number $k = \frac{a}{b}$ and nonnegative real numbers x and y , that

$$\frac{a - x}{b + y} \leq \frac{a}{b} \leq \frac{a + x}{b - y}.$$

Intuitively, when we decrease the numerator and increase the denominator, both of those actions decrease the value of a rational number, and increasing the numerator and decreasing the denominator increases the value. We can prove the first statement by making a series of reversible steps that result in a statement that is always true.

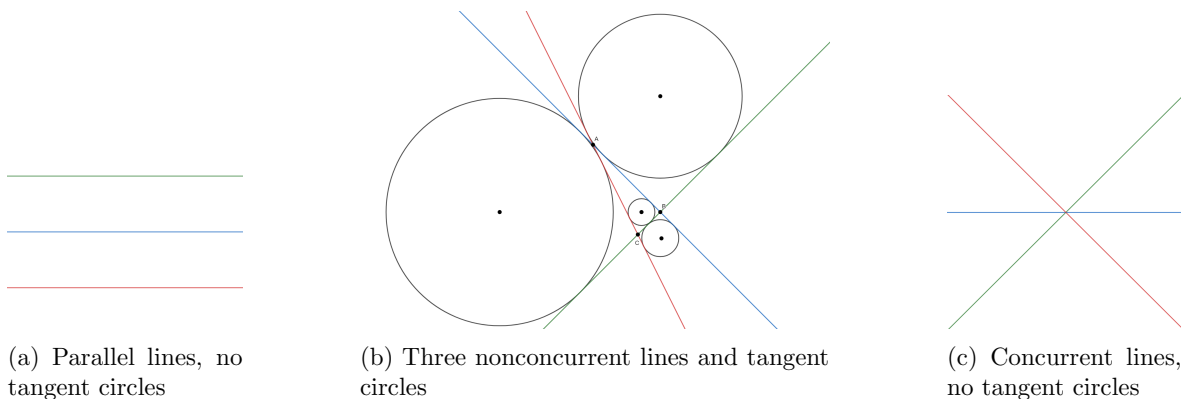
$$\begin{aligned} \frac{a - x}{b + y} &\leq \frac{a}{b} \\ b(a - x) &\leq a(b + y) \\ ab - bx &\leq ab + ay \\ -bx &\leq ay \end{aligned}$$

Which is always true since a, b are positive and x, y are nonnegative. The proof for $\frac{a}{b} < \frac{a+x}{b-y}$ follows by a similar method. If $k < 0$, the statement is not always true, but that case isn't relevant to this question. For all x, a and a^{50} are nonnegative, and we have

$$f(x) = \frac{1 - a^{50}}{1 + a} \leq \frac{1}{1} \leq \frac{1 + a^{50}}{1 - a} = g(x).$$

We have found an upper bound on $f(x)$ and a lower bound on $g(x)$, and since these are both achievable at $x = 0$, we have found the maximum of $f(x)$ and the minimum of $g(x)$ to be $\frac{1}{1} = 1$, so the statement is true.

3. False. To have a circle tangent to three distinct lines in a plane, there needs to exist at least one section of the plane that is bounded by all three lines. For example, for the lines $y = x$, $y = -x$, and $x = 1$, the section to the right of the origin is bounded by all three lines and so we can find a tangent circle (in this case there are others too!). However, this isn't true for any set of three lines. No circle can be tangent to any three concurrent lines, since there is no area bounded by all three lines. Another contradictory case is three distinct parallel lines. There are no circles tangent to all three tangent lines, because a tangent circle has to lie strictly on one side of a line, which means any circle tangent to one of the parallel lines cannot be tangent to both of the other two.



Problem 3: Tangent circles to three distinct lines (Would appreciate if this could be remade, they dont look very professional)

4. True. The statement is analogous to the Hockey Stick Identity. We can see for small values that this works, for example with $n = 5$ and $k = 3$:

$$\binom{3}{3} + \binom{4}{3} + \binom{5}{3} = 1 + 4 + 10 = 15 = \binom{6}{4}.$$

The general statement can be proven with induction. For the base case, take $n = k$:

$$\sum_{i=k}^k \binom{i}{k} = \binom{k}{k} = 1 = \binom{k+1}{k+1} = \binom{n+1}{k+1}.$$

Now, we assume that the statement is true for some $n = m$, where $m \geq 1$ is an integer, and we want to show that this implies the statement is true for $n = m + 1$. We have by assumption that

$$\sum_{i=k}^m \binom{i}{k} = \binom{m+1}{k+1},$$

and we want to show this implies

$$\sum_{i=k}^{m+1} \binom{i}{k} = \binom{m+2}{k+1}.$$

We can take the last term from this sum and get

$$\sum_{i=k}^{m+1} \binom{i}{k} = \left(\sum_{i=k}^m \binom{i}{k} \right) + \binom{m+1}{k} = \binom{m+1}{k+1} + \binom{m+1}{k},$$

with the last step due to the inductive assumption. From Pascal's Identity, $\binom{n}{k} + \binom{n}{k+1} = \binom{n+1}{k+1}$, so we can rewrite

$$\binom{m+1}{k+1} + \binom{m+1}{k} = \binom{m+2}{k+1},$$

which completes the induction.

5. True. If k is rational, $\cos\left(\frac{2\pi n}{k}\right)$ will repeat values, and if k is irrational, it won't. First, let k be a rational number. Then, we want to show that $\cos\left(\frac{2\pi n}{k}\right)$ repeats over positive integers n . This means, for two distinct positive integers q and r , we must have

$$\cos\left(\frac{2\pi q}{k}\right) = \cos\left(\frac{2\pi r}{k}\right).$$

The full condition for $\cos(x) = \cos(y)$ is if $x = \pm y + 2\pi d$ for some integer d , since $\cos(x)$ is an even function so $\cos(x) = \cos(-x)$. We have that, for some integer m ,

$$\begin{aligned}\frac{2\pi q}{k} &= \pm \frac{2\pi r}{k} + 2\pi m \\ \frac{q}{k} &= \pm \frac{r}{k} + m \\ \frac{1}{k}(q \mp r) &= m \\ q \mp r &= km\end{aligned}$$

Since k is rational, there are infinitely many integers m such that km is an integer, then we can find (q, r) such that the equation is true. For example, let $k = \frac{3}{4}$, then $q = 4$ and $r = 1$ gets the solution

$$\cos\left(\frac{32\pi}{3}\right) = \cos\left(\frac{8\pi}{3}\right)$$

Which is true since $\frac{32\pi}{3} = \frac{8\pi}{3} + 8\pi$ (meaning $m = 4$, which agrees with what we had before). We have shown that, if k is rational, $\cos\left(\frac{2\pi n}{k}\right)$ repeats value over positive integers n . However, if k is irrational, this doesn't hold. Let k be any irrational number. Then, we want to show if there exists distinct integers q and r such that

$$\cos\left(\frac{2\pi q}{k}\right) = \cos\left(\frac{2\pi r}{k}\right).$$

Using the same condition that $\cos(x) = \cos(y)$ if and only if $x = \pm y + 2\pi d$ for some integer d , we must have, for some integer m ,

$$\begin{aligned}\frac{2\pi q}{k} &= \pm \frac{2\pi r}{k} + 2\pi m \\ \frac{q}{k} &= \pm \frac{r}{k} + m \\ \frac{1}{k}(q \mp r) &= m \\ q \mp r &= km\end{aligned}$$

Since q and r are integers, $q \mp r$ must also be an integer, and since k is irrational and m is an integer then km must be irrational. It is not possible for an integer to be irrational, so there is no possible way for $\cos\left(\frac{2\pi q}{k}\right) = \cos\left(\frac{2\pi r}{k}\right)$ if k is irrational. Thus, the statement is true, since $\cos\left(\frac{2\pi n}{k}\right)$ will never repeat a value over its range when k is irrational.

6. False. First, we attempt to find a general form for $g(a, b, p)$. We can notice a factoring similar to a geometric series:

$$\begin{aligned}\frac{a^p + b^p}{a + b} &= \frac{a^{p-1} + \frac{b^p}{a}}{1 + \frac{b}{a}} \\ &= a^{p-1} \frac{1 + (\frac{b}{a})^p}{1 + \frac{b}{a}} \\ &= a^{p-1} - a^{p-2}b + a^{p-3}b^2 + \dots - ab^{p-2} + b^{p-1}.\end{aligned}$$

We can use the Euclidean Algorithm by first evaluating the expression $(\text{mod } a + b)$. Notice that $a \equiv -b \pmod{a + b}$ and that $p - 1$ is even.

$$(-b)^{p-1} - (-b)^{p-2}b + (-b)^{p-3}b^2 + \dots + b^{p-1} \equiv (-1)^{p-1}pb^{p-1} \equiv pb^{p-1} \pmod{a + b}$$

By the Euclidean Algorithm, we simply get:

$$\gcd\left(\frac{a^p + b^p}{a + b}, a + b\right) = \gcd(pb^{p-1}, a + b)$$

We also know that $\gcd(a, b) = 1$, so $\gcd(b, a + b) = 1$, letting us further simplify:

$$\gcd(pb^{p-1}, a + b) = \gcd(p, a + b)$$

We now have a way to relate the quantity $a + b$ and the odd prime p . We know that, if $p \mid (a + b)$, then $\gcd(p, a + b) = p$. Otherwise, we have $\gcd(p, a + b) = 1$.

Next, we can factor $f(x)$ with the following manipulation:

$$\begin{aligned}f(x) &= x^2 - 8x + 7 + 7 \\ f(x) &= (x - 1)(x - 7) + 7\end{aligned}$$

Since we are given the condition that $f(x_1) = f(x_2) = p$, we must consider values for which $f(x)$ attains a prime value. Notice that $f(7) = f(1) = 7$ from our factoring, which gives us equations $a_1 + b_1 = 7$ and $a_2 + b_2 = 1$. Since, for $p = 7$, we have $a_1 + b_1 = p$, then $\gcd(a_1 + b_1, p) = 7$, while $\gcd(a_2 + b_2, p) = 1 \neq 7$. Since we have found a counterexample to the statement, it must be false.

7. True. Figure 2 is the construction of the diagram by the problem statement. We want to show that all O_i lie on one of the two dotted rays. Note that all triangles $\triangle ABX_1$, $\triangle BX_1X_2$, $\triangle X_nX_{n+1}X_{n+2}$ are isosceles right triangles, so they are all similar to each other. The key observation is that we have two dilations (or homotheties) with respect to point C .

First, we want to show that the triangles $\triangle ABX_1$ and $\triangle X_iX_{i+1}X_{i+2}$ (where i is an odd positive integer) are images of each other under a homothety with center C . Notice that corresponding vertices lie on the same ray: X_i and A , as well as X_{i+2} and X_i , lie on $\overrightarrow{AC}$, and X_{i+1} and B lie on $\overrightarrow{CB}$. Next, there must be a fixed ratio of dilation between any two triangles in the sequence. Consider $\triangle ABX_1$ and $\triangle X_1X_2X_3$. Altitude X_1X_2 bisects BC , so

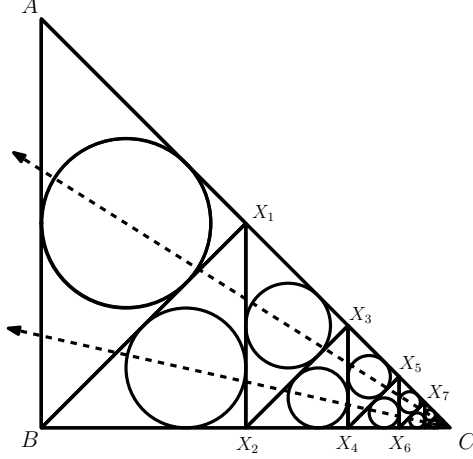


Figure 2

$\frac{CX_2}{BC} = \frac{1}{2}$, and $\triangle ABC \sim \triangle X_1X_2C$, so $\frac{X_1X_2}{AB} = \frac{1}{2}$ as well. Since both $\triangle ABX_1$ and $\triangle X_1X_2X_3$ are isosceles right triangles, they must be similar, and we have shown this similarity is with factor $\frac{1}{2}$. This can be done with any pair of triangles $\triangle X_iX_{i+1}X_{i+2}$ and $\triangle X_jX_{j+1}X_{j+2}$, where i and j are odd integers, and get a constant scale factor of $\frac{1}{2^{i-j}}$.

Similar steps will show that $\triangle BX_1X_2$ and $\triangle X_iX_{i+1}X_{i+2}$, for positive *even* integers i , are images of each other under homothety centered at C . Under dilation, corresponding points are collinear with the center of dilation. The incenter of a triangle corresponds to the incenter of the image of that triangle, so for any dilation of a triangle, the incenter is collinear with the incenter of the image and the center of dilation. Since we have two separate dilations, we have two lines that the incenters will lie on. For integer $k > 0$, all O_{2k-1} lie on triangles of the form $\triangle X_{2k-1}X_{2k}X_{2k+1}$ or $\triangle ABX_1$, which we have shown to all be dilations of each other, thus all O_{2k-1} are collinear since they correspond under dilation. Similarly, all O_{2k} correspond under dilation, and both of these dilations have center C , so the lines formed by these two sets of incenters must intersect at C .

8. True. We can represent both $\arccos(\cos x)$ and $\lim_{n \rightarrow \infty} f^n(x)$ as piecewise functions. First, $\arccos(x)$ returns values between 0 and π , taking inputs from -1 to 1 . $\arccos(\cos x)$ extends this range to $\mathbb{R}$, since the domain of $\cos(x)$ is $\mathbb{R}$ with range $[-1, 1]$. Since $\arccos(x)$ only outputs positive values, $\arccos(\cos x)$ will not return x when $x < 0$, rather, it returns $-x$. For all inputs x , we have $\cos(x) = \cos(\theta)$, for some $-\pi < \theta < \pi$, since $\cos(\theta)$ achieves the entire range of $[-1, 1]$ on that interval. So, we have

$$\arccos(\cos \theta) = \begin{cases} \theta & 0 \leq \theta \leq \pi \\ -\theta & -\pi \leq \theta \leq 0 \end{cases}$$

Now, we can extend this to all real x . Each x corresponds to some θ such that $-\pi < \theta < \pi$ and $\cos(x) = \cos(\theta)$, and this implies that $x = \theta + 2\pi k$ for some integer k . Now, we can rewrite the piecewise function, for all integers k , as

$$\arccos(\cos x) = \begin{cases} x - 2\pi k & 2\pi k \leq x \leq 2\pi(k+1) \\ -x + 2\pi k & 2\pi(k-1) \leq x \leq 2\pi k \end{cases}$$

Now for $\lim_{n \rightarrow \infty} f^n(x)$. Consider $g(x) = |x|$, and we are repeatedly applying the transformation $F(x) = ||x - \pi| - \pi|$ onto it. What does $F(x)$ do to a function? Well, first it translates the function π units down, reflects everything under the y axis back above it, then repeats this. We know that

$$g(x) = |x| = \begin{cases} x & x \geq 0 \\ -x & x < 0 \end{cases}$$

Applying $F(g(x))$ (which is $f(x)$), we now have

$$F(g(x)) = \begin{cases} -x - 2\pi & x \in (-\infty, -2\pi] \\ x + 2\pi & x \in [-2\pi, -\pi] \\ -x & x \in [-\pi, 0] \\ x & x \in [0, \pi] \\ -x + 2\pi & x \in [\pi, 2\pi] \\ x - 2\pi & x \in [2\pi, \infty) \end{cases}$$

Repeated application of $F(x)$ to $g(x)$ will make $F^n(g(x))$ alternate between $x - 2\pi k$ and $-x + 2\pi k$ (for changing k) over intervals of length π . Applying $F(x)$ infinitely allows this pattern to continue indefinitely in both directions, so there will be no interval $(-\infty, -2\pi k]$ or $[2\pi k, \infty)$ in the piecewise representation of $F^k(g(x))$ as k tends to infinity.

$$F^k(g(x)) = \begin{cases} -x - 2\pi k & x \in (-\infty, -2\pi k] \\ x + 2\pi k & x \in [-2\pi k, -2\pi(k-1)] \\ -x - 2\pi(k-1) & x \in [-2\pi(k-1), -2\pi(k-2)] \\ \vdots & \vdots \\ x - 2\pi(k-1) & x \in [2\pi(k-2), 2\pi(k-1)] \\ -x + 2\pi k & x \in [2\pi(k-1), 2\pi k] \\ x - 2\pi k & x \in [2\pi k, \infty) \end{cases}$$

We can see that, comparing $F^k(g(x))$ and $F^{k+1}(g(x))$, they are identical on the range $[-2\pi k, 2\pi k]$, and this is true for all k . As $k \rightarrow \infty$, $-2\pi k \rightarrow -\infty$ and $2\pi k \rightarrow \infty$, so as long as $\lim_{n \rightarrow \infty} F^n(g(x))$ exists, it is reasonable to say that $\lim_{n \rightarrow \infty} F^n(g(x)) = \arccos(\cos x)$ as their piecewise representations will be identical. Notice that, if we had $F(x) = |x - \pi|$, this would not be true since $F^{2k}(g(0)) = 0$ while $F^{2k+1}(g(0)) = \pi$, so the function would alternate based on the parity of how many times F is applied and therefore the limit would not exist. However, since $F(x) = ||x - \pi| - \pi|$, the function does not alternate (the parity of applications of $|x - \pi|$ is always even), and instead converges to the piecewise function

$$\lim_{n \rightarrow \infty} F^n(g(x)) = \begin{cases} x - 2\pi k & 2\pi k \leq x \leq 2\pi(k+1) \\ -x + 2\pi k & 2\pi(k-1) \leq x \leq 2\pi k \end{cases} = \arccos(\cos x).$$

We know that $\lim_{n \rightarrow \infty} f^n(x) = \lim_{n \rightarrow \infty} F^n(g(x))$, since the only difference is that $f(x)$ will have extra absolute value signs every application (for example, $f^2(x) = ||||x| - \pi| - \pi| - \pi| - \pi|$ which is the same as $F^2(g(x)) = ||||x| - \pi| - \pi| - \pi| - \pi|$). So, $\lim_{n \rightarrow \infty} f^n(x) = \lim_{n \rightarrow \infty} F^n(g(x)) = \arccos(\cos x)$, meaning the statement is true.

9. False. We are trying to figure out if the integral

$$\int_1^\infty \left(\frac{\lceil x \rceil}{\lfloor x \rfloor} - \frac{\lfloor x \rfloor}{\lceil x \rceil} \right) dx$$

converges or diverges. This setup is pretty intimidating, but instead of an integral we can consider an infinite sum. We can see that, if x is an integer, then $\lceil x \rceil = \lfloor x \rfloor = x$, so $\frac{\lceil x \rceil}{\lfloor x \rfloor} - \frac{\lfloor x \rfloor}{\lceil x \rceil} = 1 - 1 = 0$. Otherwise, the quantities of $\lceil x \rceil$ and $\lfloor x \rfloor$ do not change when x is between two integer values. So, when we integrate over a range of real numbers, $f(x) - \frac{1}{f(x)}$ will be constant for each interval between two integers (which is a range of length 1). Then, finding the area on this range would be a sum of rectangles of width 1 and height $f(x) - \frac{1}{f(x)}$. Since $\lceil x \rceil$ and $\lfloor x \rfloor$ always return integers, we can convert the integral into a discrete sum of rectangle areas.

Let n be the integer such that $n < x < n+1$ (we do not consider when x is an integer because it adds nothing to the area). Then, $\lfloor x \rfloor = n$ and $\lceil x \rceil = n+1$. We start at $n = 1$ (from the integral bounds), so the sum becomes

$$\sum_{n=1}^{\infty} \left(\frac{n+1}{n} - \frac{n}{n+1} \right) = \sum_{n=1}^{\infty} \left(1 + \frac{1}{n} - \left(1 - \frac{1}{n+1} \right) \right) = \sum_{n=1}^{\infty} \left(\frac{1}{n} + \frac{1}{n+1} \right).$$

We can rewrite this in terms of the harmonic series:

$$\sum_{n=1}^{\infty} \left(\frac{1}{n} \right) + \sum_{n=1}^{\infty} \left(\frac{1}{n+1} \right) = \sum_{n=1}^{\infty} \left(\frac{1}{n} \right) + \sum_{n=1}^{\infty} \left(\frac{1}{n} \right) - 1 = -1 + 2 \sum_{n=1}^{\infty} \left(\frac{1}{n} \right).$$

This sum must diverge since the harmonic series is divergent, and since this discrete sum is equivalent to the given integral, the integral diverges as well, making the statement false.

10. True. We are tasked to compute

$$\int_1^{\infty} \int_1^{\infty} \frac{1}{xy(x^2 + y^2)} dx dy.$$

The statement asks if this is equivalent to the value Z that satisfies $e^{nZ} = m$, which is $\frac{\ln m}{n}$ for some positive integers n and m . We will deal with the inner integral first, that is

$$\int_1^{\infty} \frac{1}{xy(x^2 + y^2)} dx.$$

When integrating multivariable functions with respect to one variable, all other variables are considered constant, since we are only changing x . So, we will deal with y as a constant, which means it does not depend on x . So, we can write

$$\int_1^{\infty} \frac{1}{xy(x^2 + y^2)} dx = \frac{1}{y} \int_1^{\infty} \frac{1}{x(x^2 + y^2)} dx.$$

Now, we will solve for the indefinite integral of the integrand. Let $u = x^2 + y^2$, then $du = 2x dx$ (remember y is constant). We can write $dx = \frac{du}{2x}$ and $x^2 = u - y^2$ to get

$$\int \frac{1}{x(x^2 + y^2)} dx = \int \frac{1}{xu} \frac{du}{2x} = \int \frac{1}{2x^2 u} du = \frac{1}{2} \int \frac{1}{(u - y^2)(u)} du.$$

Now, we can use partial fraction decomposition, solving for A and B given

$$\frac{1}{(u - y^2)(u)} = \frac{A}{u} + \frac{B}{u - y^2}.$$

This gets

$$\begin{aligned} A(u - y^2) + Bu &= 1 \\ u(A + B) - Ay^2 &= 1 \end{aligned}$$

A and B must be constants since the denominator is linear in both terms. This means we must have

$$\begin{aligned} A + B &= 0 \\ -Ay^2 &= 1 \end{aligned}$$

Solving, we get $A = -\frac{1}{y^2}$ and $B = \frac{1}{y^2}$. So, the integral becomes

$$\frac{1}{2} \int \left(-\frac{1}{uy^2} + \frac{1}{y^2(u - y^2)} \right) du = \frac{1}{2y^2} \int \left(\frac{1}{(u - y^2)} - \frac{1}{u} \right) du.$$

Using the fact that $\int \frac{1}{x} dx = \ln|x|$, we find the indefinite integral to be

$$\frac{1}{2y^2} \int \left(\frac{1}{(u - y^2)} - \frac{1}{u} \right) du = \frac{\ln|u - y^2| - \ln|u|}{2y^2} + C = \frac{\ln|\frac{u - y^2}{u}|}{2y^2} + C = \frac{\ln|1 - \frac{y^2}{u}|}{2y^2} + C = \frac{\ln|1 - \frac{y^2}{x^2 + y^2}|}{2y^2} + C.$$

Now, we can evaluate the definite integral. We have

$$\int_1^\infty \frac{1}{x(x^2 + y^2)} dx = \left[\frac{\ln|1 - \frac{y^2}{x^2 + y^2}|}{2y^2} + C \right]_{x=1}^{x=\infty} = \lim_{N \rightarrow \infty} \left(\frac{\ln|1 - \frac{y^2}{N^2 + y^2}|}{2y^2} + C \right) - \left(\frac{\ln|1 - \frac{y^2}{1 + y^2}|}{2y^2} + C \right)$$

As $N \rightarrow \infty$, $\frac{y^2}{N^2 + y^2} \rightarrow 0$, so $\ln|1 - \frac{y^2}{1 + y^2}|$ approaches $\ln(1) = 0$, and the constants C cancel out. Then,

$$-\frac{\ln|1 - \frac{y^2}{1 + y^2}|}{2y^2} = -\frac{\ln|1 - \left(1 - \frac{1}{y^2 + 1}\right)|}{2y^2} = \frac{-\ln \frac{1}{y^2 + 1}}{2y^2} = \frac{\ln|y^2 + 1|}{2y^2}.$$

Now, plugging this back in to the original integral, we get

$$\int_1^\infty \frac{1}{y} \int_1^\infty \frac{1}{x(x^2 + y^2)} dx dy = \int_1^\infty \frac{1}{y} \frac{\ln|y^2 + 1|}{2y^2} dy = \int_1^\infty \frac{\ln|y^2 + 1|}{2y^3} dy = \frac{1}{2} \int_1^\infty \frac{\ln|y^2 + 1|}{y^3} dy.$$

We will finish this off with integration by parts. Let $u = \ln|y^2 + 1|$ and $dv = \frac{dy}{y^3}$. Then, $du = \frac{2y}{y^2 + 1} dy$ and $v = -\frac{1}{2y^2}$. Then, by integration by parts,

$$\int \frac{\ln|y^2 + 1|}{y^3} dy = -\frac{\ln|y^2 + 1|}{2y^2} - \int -\frac{1}{2y^2} \frac{2y}{y^2 + 1} dy = -\frac{\ln|y^2 + 1|}{2y^2} + \int \frac{1}{y(y^2 + 1)} dy.$$

We can solve for the second integral with partial fraction decomposition once again:

$$\frac{1}{y(y^2 + 1)} = \frac{A}{y} + \frac{B}{y^2 + 1}$$

From this, we get

$$\begin{aligned} A(y^2 + 1) + By &= 1 \\ Ay^2 + By + A &= 1 \end{aligned}$$

To match the constants, we take $A = 1$, which gets $B = -y$, so we have $\frac{1}{y(y^2+1)} = \frac{1}{y} - \frac{y}{y^2+1}$, then

$$\int \frac{1}{y(y^2+1)} dy = \int \left(\frac{1}{y} - \frac{y}{y^2+1} \right) dy = \ln|y| - \frac{1}{2} \ln|y^2+1| + C = \ln \left| \frac{y}{\sqrt{y^2+1}} \right| + C.$$

Now we can solve for the definite integral, getting

$$\begin{aligned} \int_1^\infty \frac{\ln|y^2+1|}{y^3} dy &= \left[-\frac{\ln|y^2+1|}{2y^2} + \ln \left| \frac{y}{\sqrt{y^2+1}} \right| + C \right]_{y=1}^{y=\infty} \\ &= \lim_{N \rightarrow \infty} \left(-\frac{\ln|N^2+1|}{2N^2} + \ln \left| \frac{N}{\sqrt{N^2+1}} \right| \right) - \left(-\frac{\ln|2|}{2} + \ln \left| \frac{1}{\sqrt{2}} \right| \right) \\ &= \lim_{N \rightarrow \infty} \left(-\frac{\ln|N^2+1|}{2N^2} + \ln \left| \frac{N}{\sqrt{N^2+1}} \right| \right) + \frac{\ln|2|}{2} - \frac{1}{2} \ln \left| \frac{1}{2} \right| \\ &= \lim_{N \rightarrow \infty} \left(-\frac{\ln|N^2+1|}{2N^2} + \ln \left| \frac{N}{\sqrt{N^2+1}} \right| \right) + \ln(2) \end{aligned}$$

As $N \rightarrow \infty$,

$$-\frac{\ln|N^2+1|}{2N^2} \approx -\frac{\ln|N^2|}{2N^2} = -\frac{2\ln|N|}{2N^2} = -\frac{\ln|N|}{N^2}.$$

Since the growth of N^2 outpaces $\ln|N|$, this value tends to 0 (a more formal argument would use L'Hopitals). Also, as $N \rightarrow \infty$,

$$\ln \left| \frac{N}{\sqrt{N^2+1}} \right| \approx \ln \left| \frac{N}{\sqrt{N^2}} \right| = \ln \left| \frac{N}{|N|} \right| = \ln|1| = 0.$$

So, the limit evaluates to 0, and we get the value of the integral to be

$$\int_1^\infty \int_1^\infty \frac{1}{xy(x^2+y^2)} dx dy = \frac{1}{2} \int_1^\infty \frac{\ln|y^2+1|}{y^3} dy = \boxed{\frac{\ln(2)}{2}},$$

which means the statement is true for positive integers $m = n = 2$.

2 Finding n

1. $n < 10^9$.
 - Puts bounds on n .
2. The last two digits of n in base 2 are 00.
 - n is divisible by 4.
3. n , when expressed in base 36, has 5 digits.
 - Bounds n , $36^5 \leq n < 36^6$.
4. n is one of the terms of an infinite arithmetic sequence with $a_1 = 1$ and $a_2 = 2024$.
 - n is of the form $1 + 2023k$, for nonnegative integer k .
5. Instead of H_3 , use hint N_3 .
 - Disregard the base 36 bounds. Now, $63^4 \leq n < 63^5$.
6. n is divisible by 107.
 - $n = 107k$ for some positive integer k .
7. $n \equiv b_{2023} \equiv \prod_{i=1}^{2023} \left(1 + \frac{1}{i}\right)^2 \pmod{9}$.
 - We can show that $b_k = (k+1)^2$. We have that $\prod_{i=1}^k \left(1 + \frac{1}{i}\right)^2 = \left(\prod_{i=1}^k \frac{i+1}{i}\right)^2$, and the inner product is a telescoping product that converges to $k+1$. So, $b_k = (k+1)^2$, and we have $n \equiv 2024^2 \equiv (45^2 - 1)^2 \equiv 1 \pmod{9}$.
8. $\log_5 n > 8(\log_5 2) + 9$
 - Since the exponential function 5^x is strictly increasing (and continuous), we know that if $\log_5 n > 8(\log_5 2) + 9$, then $5^{\log_5 n} > 5^{8(\log_5 2) + 9}$. This simplifies, by log rules, to $n > 2^8 \cdot 5^9 = 5 \cdot 10^8$.
9. $n \equiv \sum_{k=1}^{20} k \binom{20}{k} \pmod{19}$
 - I claim that $\sum_{i=1}^k i \binom{k}{i} = k2^{k-1}$. Using a combinatorial argument, the left side is all the ways to choose any positive number of people from a group of k , then designating one person to lead the chosen. This is equivalent to choosing one person to lead in the beginning, then each of the $k-1$ others are either in the group or not, which can be done in $k2^{k-1}$ ways. This can also be shown algebraically. So, $n \equiv 20 \cdot 2^{19} \pmod{19} \equiv (1)(2 \cdot 2^{18}) \equiv 2 \pmod{19}$. We know $2^{18} \equiv 1 \pmod{19}$ from Fermat's Little Theorem.
10. n is the element of S with the minimum sum of digits.

Since $10^9 > 63^5$ and $5 \cdot 10^8 > 63^4$, we have $5 \cdot 10^8 \leq n < 63^5$. We also know that $n \equiv 0 \pmod{4}$, $n \equiv 1 \pmod{2023}$, $n \equiv 1 \pmod{9}$, $n \equiv 0 \pmod{107}$, and $n \equiv 2 \pmod{19}$. Since $\gcd(4, 2023, 9, 107, 19) = 1$, by the Chinese Remainder Theorem, these equivalences guarantee a unique $n \pmod{4 \cdot 2023 \cdot 9 \cdot 107 \cdot 19 = 148059324}$, in this case $n \equiv 6645556 \pmod{148059324}$. The possible values of n that satisfy this and the bounds are 598882852, 746942176, and 895001500. The sum of the digits of each of these are 55, 46, and 28 respectively, so $n = \boxed{895001500}$.

A Reflection on Mathematicians

Mikayla Lin

Description

This puzzle is in the style of a puzzle hunt, in which information is given without instructions. The goal of the puzzle solver is to figure out how to use the information to extract a final string.

It seems as if strange symbols have infiltrated your math history test. And with a weird grid in the back too. It reminds you of a crossword, but you suspect it's not so straightforward. Perhaps this grid calls for a change in perspective.

The solution is on page 19.

Puzzle

1. Greek mathematician known for the principle of buoyancy
 - $\Uparrow$ Archimedes
 - $\Downarrow$ Mobius
 - $\Leftarrow$ Lagrange
 - $\Rightarrow$ Noether
2. Mathematician famous for his work on topology with a one-sided strip named after him
 - $\Uparrow$ Turing
 - $\Downarrow$ Mobius
 - $\Leftarrow$ Poincare
 - $\Rightarrow$ Nash
3. Italian mathematician whose sequence is widely used in nature and art
 - $\Uparrow$ Fibonacci
 - $\Downarrow$ Leibniz
 - $\Leftarrow$ Cantor
 - $\Rightarrow$ Fermat
4. German mathematician and physicist who made significant contributions to abstract algebra
 - $\Uparrow$ Germain

- $\Downarrow$ Hawking
 - $\Leftarrow$ Noether
 - $\Rightarrow$ Einstein
5. French mathematician who laid the foundations of algebraic topology
- $\Uparrow$ Pythagoras
 - $\Downarrow$ Ramanujan
 - $\Leftarrow$ Hawking
 - $\Rightarrow$ Poincare
6. Nobel Prize-winning mathematician known for his work in game theory
- $\Uparrow$ Ramanujan
 - $\Downarrow$ Descartes
 - $\Leftarrow$ Galois
 - $\Rightarrow$ Nash
7. Father of modern computer science and codebreaker during World War II
- $\Uparrow$ Turing
 - $\Downarrow$ Pythagoras
 - $\Leftarrow$ Lagrange
 - $\Rightarrow$ Einstein
8. German philosopher and mathematician who co-invented calculus
- $\Uparrow$ Riemann
 - $\Downarrow$ Ramanujan
 - $\Leftarrow$ Leibniz
 - $\Rightarrow$ Galois
9. Mathematician who introduced set theory and the concept of infinity
- $\Uparrow$ Descartes
 - $\Downarrow$ Ramanujan
 - $\Leftarrow$ Cantor
 - $\Rightarrow$ Lagrange
10. An ancient Greek mathematician credited with a theorem about right-angled triangles
- $\Uparrow$ Galois
 - $\Downarrow$ Hawking
 - $\Leftarrow$ Pythagoras
 - $\Rightarrow$ Germain

A Reflection on Mathematicians: Solution

Mikayla Lin

Answers

1. Greek mathematician known for the principle of buoyancy
 - $\Uparrow$ Archimedes
2. Mathematician famous for his work on topology with a one-sided strip named after him
 - $\Downarrow$ Mobius
3. Italian mathematician whose sequence is widely used in nature and art
 - $\Uparrow$ Fibonacci
4. German mathematician and physicist who made significant contributions to abstract algebra
 - $\Leftarrow$ Noether
5. French mathematician who laid the foundations of algebraic topology
 - $\Rightarrow$ Poincare
6. Nobel Prize-winning mathematician known for his work in game theory
 - $\Rightarrow$ Nash
7. Father of modern computer science and codebreaker during World War II
 - $\Uparrow$ Turing
8. German philosopher and mathematician who co-invented calculus
 - $\Leftarrow$ Leibniz
9. Mathematician who introduced set theory and the concept of infinity
 - $\Leftarrow$ Cantor
10. An ancient Greek mathematician credited with a theorem about right-angled triangles
 - $\Leftarrow$ Pythagoras

Now using the numberings on the grid, fill in each of the mathematicians. However the arrow indicates the direction in which they should be written. (Ex: $\Leftarrow$ Nash means left to right: Nash, $\Rightarrow$ Noether means right to left: rehteonN). After filling in each of the questions you should get the image on the right. Now reading the highlighted letters from left to right and top to bottom you should get: **TOPOLOGY**, the final answer.

					S				M			I			
				R	E	H	T	E	O	N		C			
					D				B			C			
					E		P	O	I	N	C	A	R	E	
					M				U			N			
					I				S			O			
		N	A	S	H					G		B			
					C			Z	I	N	B	I	E	L	
					R					I		F			
	R	O	T	N	A	C				R					
										U					
			S	A	R	O	G	A	H	T	Y	P			

Tower of Hanoi

Patrick Was

Awaking from your 5 hour sleep, you emerge from your bed dreading today—and then you realize it's Monday, the day you're supposed to present for your history class. Not only are you unprepared, you haven't even started your slideshow or opened up the rubric. Because you're a math nerd who takes AP French, you immediately come up with a presentation idea: **The Tower of Hanoi**.

The premise of the puzzle is to find the minimum possible number of moves to move n **discs**, which are strictly decreasing in size from top to bottom, from one of k **pegs** to another with the following restrictions:

- Discs may only be transferred between pegs one move at a time.
- No larger disc may be placed on top of a smaller disc.
- Only the top disc of a peg may be moved to another peg.
- Define a move to be a transfer of a disc from one peg to another.

The most common variation of this puzzle involves 3 pegs and 3 discs, but it is nearly just as likely to see 3 pegs and any other positive integer n .

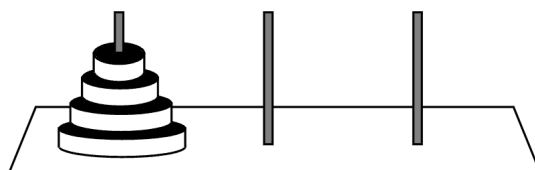


Figure 1: Unsolved tower with 3 pegs and 4 discs. (Wolfram MathWorld)

Questions:

1. What is the minimum number of moves required to transfer 3 discs to another peg given that there are 3 pegs?
2. Let $H(n, k)$ represent the minimum number of moves to transfer n discs given k pegs. What are $H(4, 3)$? $H(5, 3)$? $H(10, 3)$?
3. Express $H(n, 3)$ in terms of n .
4. Consider $H(x, x)$.

- (a) Are there any restrictions for the outputs of $H(x, x)$? Do these encompass all the restrictions of $H(n, k)$?
 - (b) What is $H(4, 4)$? What about $H(5, 5)$? $H(10, 10)$?
 - (c) Express $H(x, x)$ in terms of x .
5. Is it possible to solve the standard Tower of Hanoi in an even number of moves? The moves do not necessarily have to be optimal.
6. Define $g(n) = \min(H(n, k))$ as k varies.
 - (a) Find $g(3)$, $g(5)$, and $g(15)$.
 - (b) What is the minimum value of k for which $g(n) = H(n, k)$?
 - (c) Express $g(n)$ in terms of n .
7. Consider a variation of this puzzle with the following additions:
 - Every disc now has an orientation; its top surface is black, and its bottom surface is white.
 - The original position of every disc has its black surface facing upwards.
 - Every move requires a reorientation of the disc onto its opposite surface.
 - The base of a peg is colored and counts as a surface.
 - At any point in the game, no two surfaces with the same color may be touching.

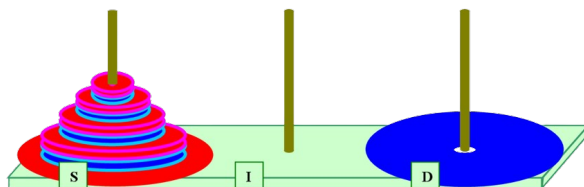


Figure 2: Magnetic tower, with a red side facing upwards, the base of the pegs are also colored here.

Define $S(n)$ to be the minimum number of moves required to move n discs with 3 pegs and consider the coloring of the pegs to be white-white-black or black-white-black.

- (a) Find $S(2)$, $S(3)$, and $S(5)$.
Consider $S(3)$.
- (b) Express $S(n)$ in terms of n .

References

Levy, Uri. "The Magnetic Tower of Hanoi." *ArXiv* abs/1003.0225, 2010.
<https://api.semanticscholar.org/CorpusID:35090750>.

Weisstein, Eric W. "Tower of Hanoi." From MathWorld—A Wolfram Web Resource, Feb 2024.
<https://mathworld.wolfram.com/TowerofHanoi.html>

The solution is on page 23.

Tower of Hanoi: Solution

Patrick Was

1. 7

Move the smallest disc onto either the first or second peg, and place the middle one onto the remaining peg. For example's sake, place the smallest on the third peg and the middle one on the second peg. This is 2 moves so far.

Then, place the small disc on the medium disc. This allows the large disc to be moved to the third peg. This is another 2 moves.

Place the small disc onto the first peg. The medium disc can then be placed on the large disc, and the small disc can finally be placed on top. These are the last 3 moves.

$$2 + 2 + 3 = \boxed{7}$$

2. $H(4, 3) = 15$; $H(5, 3) = 31$; $H(10, 3) = 1023$

In order to not make the solution too lengthy (as would be the case for $H(10, 3)$), consider the middle case, $H(5, 3)$. Instead of starting the solution all over again, we can consider what would happen if $H(4, 3)$ was already completed and we simply added another disc. If had a tower of 4 discs on the second peg, and we placed our new disc on the first peg, we can find how many moves it would take to solve this.

For brevity, the sequence will be written as xT , where x is the number of the disc that is being moved, with 1 being the shortest and 5 being the longest, and where T is the peg to which that x is being moved, denoted by L, M, R , or left, middle, right.

5R, 1L, 2R, 1R, 3L, 1M, 2L, 1L 4R, 1R, 2M, 1M, 3R, 1L, 2R, 1R.

$$H(4, 3) + 16 = 31$$

If we wanted to solve $H(4, 3)$, we could take the solution to $H(3, 3)$ and add the additional 8 moves.

An even more concise way is to consider that the only moves you have to take are solving $H(n - 1, k)$, moving the new disc, and then resolving $H(n - 1, k)$ on the new peg.

3. $2^n - 1$

A recursive argument can be made. Looking at the previous question, we see that $H(n+1, 3) = 2H(n, 3) + 1$. We can prove this recursive argument and get a general formula using both

logic and induction.

From observation, it appears that $H(n, 3) = 2^n - 1$. We have already proved this for the base case of $n = 3$, and for $n = 1$, we get $2^1 - 1 = 1$, which must clearly be true.

Let $H(n, 3) = 2^n - 1$ be true for some $i > 0$. Let $S_n = H(n, 3)$. Then, with $i + 1$ discs, we know that the first set of moves would be to move all the i discs off of the $(i + 1)^{\text{th}}$ disc. Then, move the $(i + 1)^{\text{th}}$ disc onto another peg, and finally put all i remaining discs from the stack back on top of that disc. This results in a total of $2S_i + 1$ moves.

$$S_{i+1} = 2S_i + 1 = 2(2^i - 1) + 1 = 2^{i+1} - 1$$

It can also be shown that for any disc k , where k is the disc's original position from bottom to top (ex: the largest disc = n), it takes $2^k - 1$ moves to transfer it to completion.

$$\sum_{k=1}^n 2^{k-1} = 2^n - 1$$

Both methods demonstrate that for some positive integer n , $H(n, 3) = \boxed{2^n - 1}$

4. (a) $H(1, 1), H(2, 2)$

It is impossible to move one disc onto another peg if there is only one peg. Similarly, if there are two discs and two pegs, the smaller disc can be moved onto the second peg, but nothing else can be done.

Similarly, it should make sense that for $H(n, 2)$, if $n \geq 2$, there are no solutions.

- (b) $H(4, 4) = 9; H(5, 5) = 11; H(10, 10) = 21$

Consider the middle question, $H(5, 5)$. We know that it will take 4 moves to move the top 4 discs onto their own pegs. However, we still need to move disc 5 onto its own peg. We can move one of the smaller discs on top of the larger discs, making room for disc 5 to go on another peg. We then need to take the small disc and place it onto the peg that disc 5 just left. This is an additional 3 moves. Redoing the puzzle then takes another 4 moves.

$$4 + 4 + 3 = 11$$

- (c) $2x + 1$

Using the logic from the previous question,

$$H(x, x) = (x - 1) + 3 + (x - 1)$$

$$H(x, x) = \boxed{2x + 1}$$

5. It is not possible to solve the Tower of Hanoi in an even number of moves, even in a nonoptimal setting.

We see that in an optimal setting, the number of moves it takes to complete the game is $2^n - 1$, which will always be an odd number for any positive integer n . Taking a nonoptimal number of moves to complete the Tower of Hanoi simply means making a mistake in placing a disc where it does not belong; this will always be n moves of mistakes and another n moves of fixing this mistake (it does not matter when this mistake is fixed). We have also strictly defined a move to be a transfer of a disc, so picking a disc up and placing it back down does not count as a move. Summing the optimal number of moves and the number of moves correlating to mistakes results in an odd integer.

6. (a) $g(3) = 5$, $g(5) = 9$, $g(15) = 29$

The logic is similar to question 4. Consider $g(5)$. We can move the top 4 discs onto their own pegs, then move 5 onto its own peg, and resolve the other 4 discs.

$$4 + 1 + 4 = 9$$

- (b) $n + 1$

We can see that the problem that we had in 4 is eliminated if we were to add an additional peg.

- (c) $2n - 1$

We can consider the optimal way to play the game for any k such that $g(n) = H(n, k)$, but since this will also hold true for all integers greater than or equal to k , we can think about this problem with an infinite number of pegs. In a situation like this, it would be best to place every piece out on its own peg so that no backtracking has to be taken, similar to (b).

$n - 1$ moves will need to be taken to get each of the $n - 1$ discs onto its own peg. The bottom disc can then be moved onto any other available peg, which takes precisely 1 move. Finally, since each disc is on its own peg, moving every other disc onto the bottom disc only takes another $n - 1$ moves.

$$(n - 1) + 1 + (n - 1) = \boxed{2n - 1}$$

7. (a) $S(2) = 4$, $S(3) = 13$, $S(5) = 121$

Consider the simplest case, $S(2)$. Let the orientation of the pegs be black-white-white (this will take the same number of moves as black-white-black).

Place the top disc onto another peg. In this solution, it will be placed in the middle peg. Then, move the larger disc into the last position. Finally, the orientation of the middle disc must be swapped, as it is currently white on top and flipping it would break the rules if it were to be placed on top of the larger disc. So, to re-orient it, flip it onto the black peg, and then transfer it onto the larger one. This results in 4 moves.

(b) $\frac{3^n - 1}{2}$

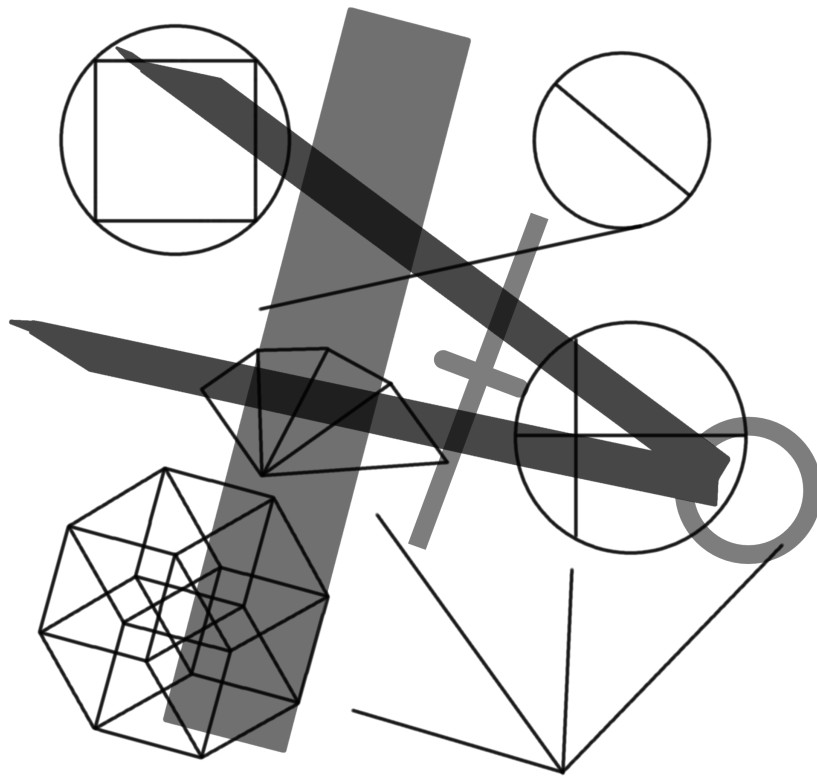
For every disc, the number of moves it takes to get it to its optimal position is 3^{k-1} .

$$\sum_{k=1}^n 3^{k-1} = \boxed{\frac{3^n - 1}{2}}$$

For additional information, such as a general form for $H(n, k)$, check out the Frame-Stewart Algorithm. It has not been proven optimal, though it is generally considered to be.

Algebraic Constructions

Patrick Was



Assume that all of the following figures are constructed with only a straightedge and compass:

1. Is it possible to construct a square with an area equal to a given circle?
2. Is it possible to construct a cube double the volume of a unit cube?
3. Is it possible to construct any n -dimensional cube twice the volume of its unit representation?
4. Is it possible to trisect any given angle using constructions?

The solution is on page 28.

Algebraic Constructions: Solution

Patrick Was

Assume that all of the following figures are constructed with only a straightedge and compass:

1. Is it possible to construct a square with an area equal to a given circle? **No.**

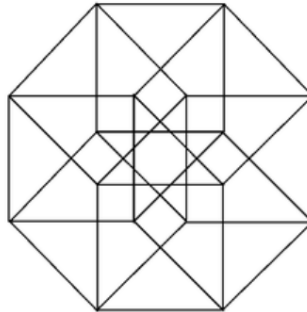
We know that π is not an algebraic number. In other words, it cannot be expressed as the root of any polynomial with integer coefficients. Given that constructible points must satisfy the roots of some polynomial (further detailed in the next solution), we know that we cannot construct a square that is some sort of multiple of π .

2. Is it possible to construct a cube double the volume of a unit cube? **No.**

It is not possible to double the volume of a unit cube. By doubling the volume of the unit cube, a new cube with volume 2 would need to be constructed. Therefore, a side with length $\sqrt[3]{2}$ would need to be constructed.

Taking a Galois approach with this, we could consider how points are formed in the first place. If we wanted to form the point $(\sqrt[3]{2}, 0)$ (or even the other pair, $(0, \sqrt[3]{2})$), then we have to consider how points in general are formed. Using a straight edge and compass, it is possible to construct circles going through two predefined points or do the same with lines, which is more useful in our case. A newly defined point can also arise from the intersection of two such lines or circles. We know from this that the points must satisfy some (at most) quadratic such that binary operations on the previous points can lead to the new point being defined. The degree of the field extension that corresponds to each point must therefore be at most 2 but at least 1. We can therefore conclude that the degree of the field extension over $\mathbb{Q}$ for any of our constructed points must be a power of 2.

However, the minimum splitting field for the polynomial $x^3 - 2$, which is the polynomial used to construct the point at $\sqrt[3]{2}$, must be $\mathbb{Q}(\sqrt[3]{2})$, which is clearly of degree 3, and this does not satisfy the premise that the constructed point must be a power of 2.



A 2D rendering of a 4D cube

3. Is it possible to construct any n -dimensional cube twice the volume of its unit representation? **Not unless n is a power of 2.**

Seeing the result we got before, we know that we can construct any point that has a degree field extension of a power of 2. This means that squares, tesseracts, and any 2^n -dimensional cube can be constructed such that they are double the area/volume of the unit cube.

Alternate reasoning is that we know that if a is constructible, so is $\sqrt{a}$. We can see that this is true because of the side lengths of a right triangle. If $\sqrt{a}$ is constructible, then so is $\sqrt[4]{a}$ and so on.

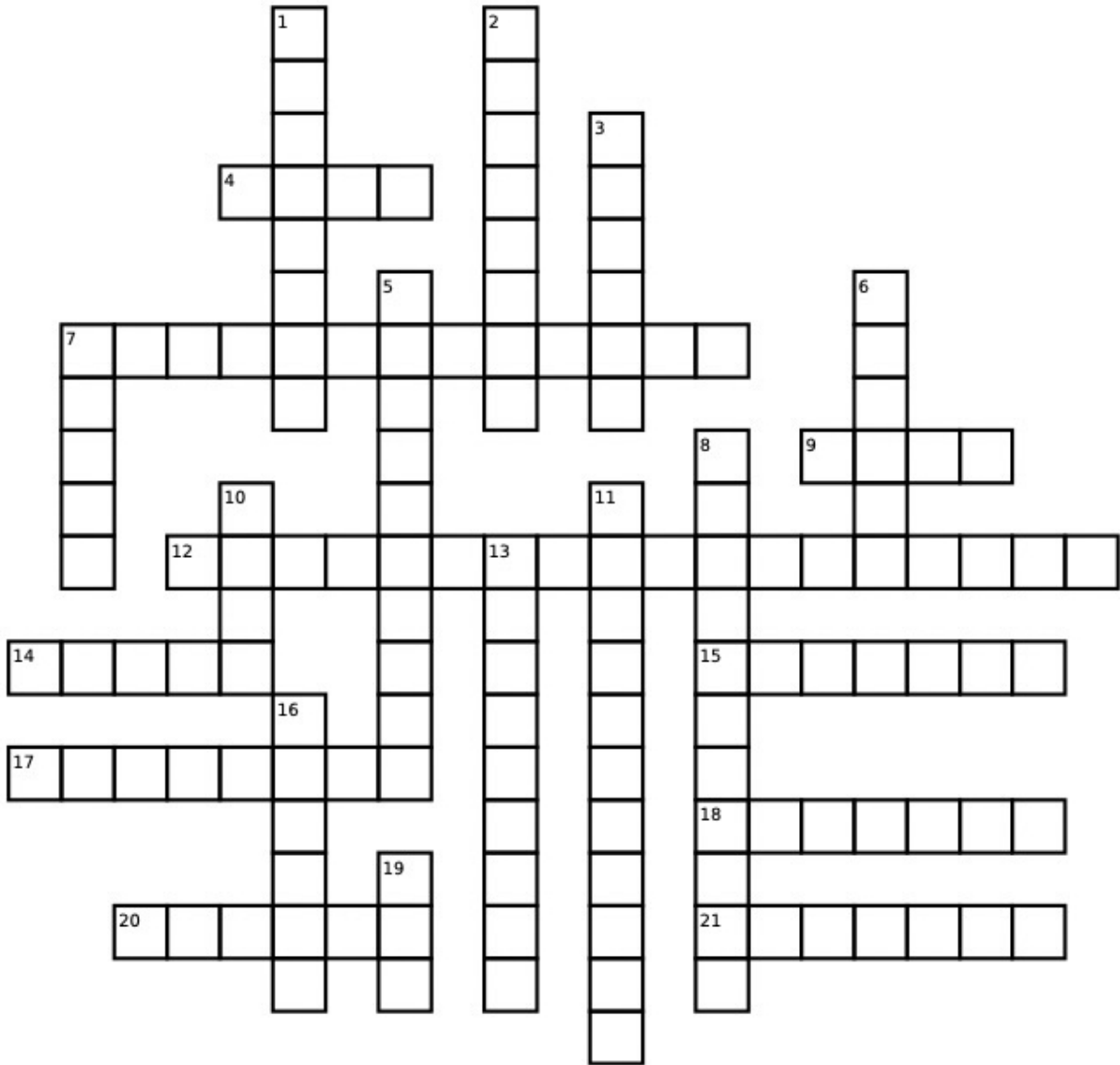
4. Is it possible to trisect any given angle using constructions? **No.**

Though there are certain angles that are obviously constructible, such as an angle of 90° , it is not possible to trisect any arbitrary one. Using a compass and straight edge, it is possible to apply any binary operations on a given angle. However, for similar reasoning as constructing lines, the field extension of a constructible angle must be of power 2.

The smallest irreducible polynomial that can be created using $\cos 20^\circ$, or an angle of 20 degrees, is $8x^3 - 6x - 1$, which is clearly of degree 3 in the rationals, implying that a 60° angle cannot be trisected. This immediately proves that any arbitrary angle cannot necessarily be trisected.

Winter Crossword

Vincent Chen

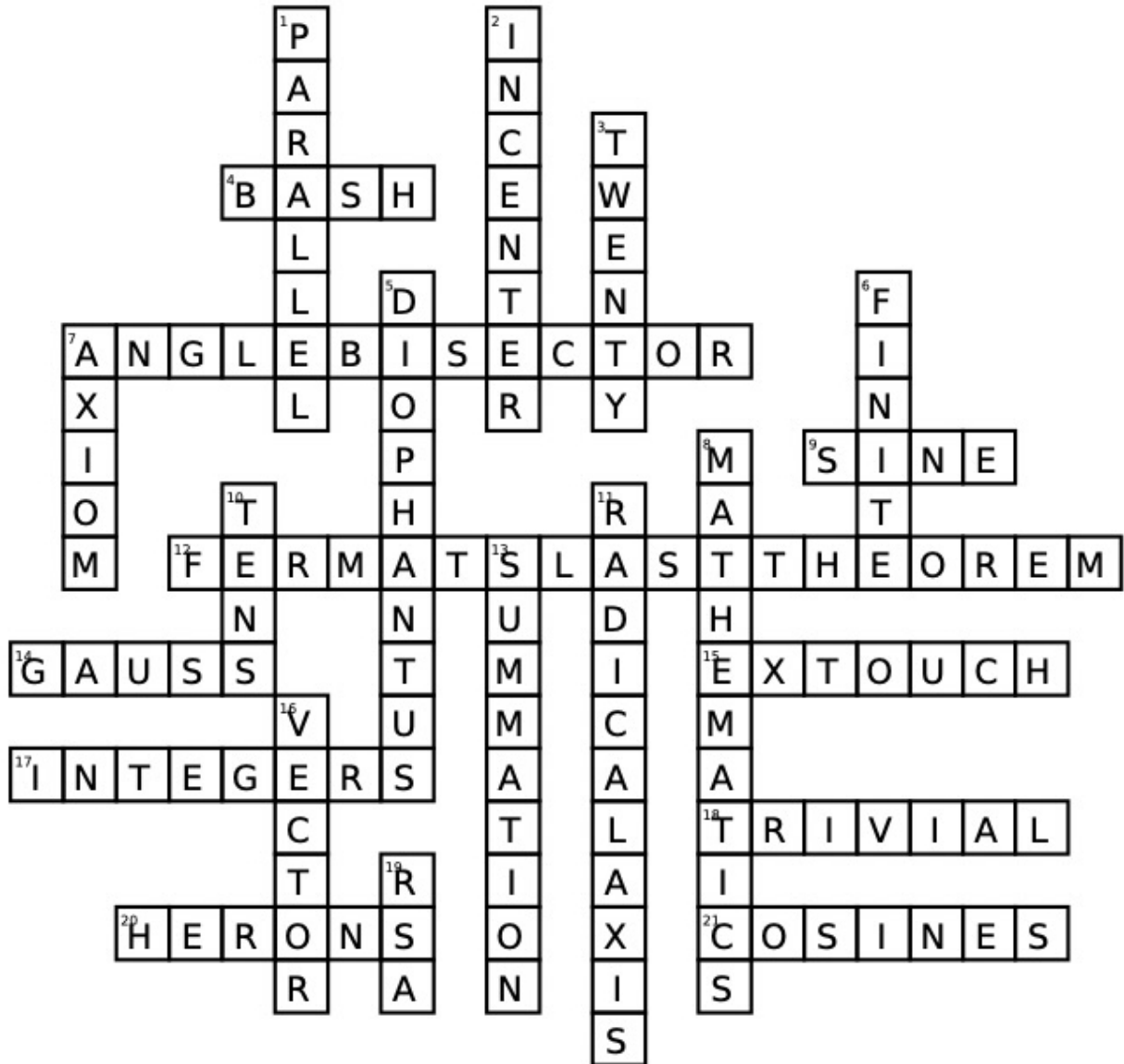


Down 1. concerns Euclid's fifth postulate 2. point equidistant from all three sides of a triangle 3. the height of this crossword 5. creator of a group of equations with integer solutions 6. countable 7. taken for granted to be true, in geometry 8. the focus of this publication 10. what the decimal system counts by 11. a line perpendicular to the line of centers of 2 circles 13. denoted with an uppercase sigma 16. "these have magnitude and direction" 19. an encryption technique made by Ron, Adi, Leonard	Across 4. a brute-force technique used in competition math 7. a line consisting of 2 points equidistant to 2 given lines 9. a basic wave function used to find height 12. famous theorem in a footnote of Arithmetica 14. first to prove the Fundamental Theorem of Algebra 15. name of a triangle formed by the excircles 17. the floor function does not modify the value of these 18. name of an inequality that states $x^2 \geq 0$ 20. formula to find a triangle's area given its lengths 21. uses SAS to find the length of a triangle's side
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The solution is on page 32.

Winter Crossword – Solution

Vincent Chen



Down 1. PARALLEL 2. INCENTER 3. TWENTY 5. DIOPHANTUS 6. FINITE 7. AXIOM 8. MATHEMATICS 10. TENS 11. RADICALAXIS 13. SUMMATION 16. VECTOR 19. RSA	Across 4. BASH 7. ANGLEBISECTOR 9. SINE 12. FERMATSLASTTHEOREM 14. GAUSS 15. EXTOUCH 17. INTEGERS 18. TRIVIAL 20. HERONS 21. (LAW OF) COSINES
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Soddy Circles, Inversion, and Descartes' Circle Theorem

Ethan Sharma

The Kiss Precise (1936)

Frederick Soddy

*For pairs of lips to kiss maybe
Involves no trigonometry.
This not so when four circles kiss
Each one the other three.
To bring this off the four must be
As three in one or one in three.
If one in three, beyond a doubt
Each gets three kisses from without.
If three in one, then is that one
Thrice kissed internally.*

*Four circles to the kissing come.
The smaller are the benter.
The bend is just the inverse of
The distance from the center.
Though their intrigue left Euclid dumb
There's now no need for rule of thumb.
Since zero bend's a dead straight line
And concave bends have minus sign,
The sum of the squares of all four bends
Is half the square of their sum.*

1 What are Soddy Circles?

When given three mutually externally tangent circles, how many circles are tangent to all three? This question is actually a specific case of Apollonius' problem back in Ancient Greece, which asks the same question for any three given circles in a plane. This article isn't about Apollonius, though, so let's stay with our mutually tangent circles. The answer is that there are exactly two circles, and these are named the Inner and Outer Soddy Circles. These two circles can be seen in Figure 1.

1.1 Brief History Lesson

Though these circles are called 'Soddy Circles', English Radiochemist Frederick Soddy did not discover them. As stated, these circles are a special case of Apollonius's problem, allegedly solved

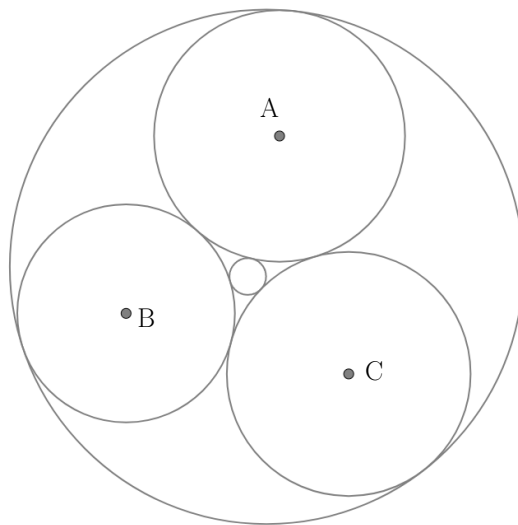


Figure 1: Inner and Outer Soddy Circles to Circles A , B , and C

in Apollonius of Perga's work *Tangencies* (English translation). The more formal statement of his problem is: Given three things (points, lines, or circles), construct a circle that passes through each point and is tangent to each line or circle. This text has been lost, though, and all that survives are references to the problems. Future mathematicians picked up the problem statement and went at solving it. This specific case of Apollonius's problem was picked up by René Descartes, who solved it in 1643 in a letter correspondence with Princess Elizabeth, the daughter of King Frederick of Bohemia. His formula gives the solution for the radii of both Soddy Circles, but the reasoning for his conclusion was not clear in the letter.

Descartes' Circle Theorem was rediscovered by multiple future mathematicians, one of whom being Frederick Soddy in 1936, who chose to publish his findings in the form of a poem, *The Kiss Precise*. This move by Soddy popularized the theorem in his name, which allowed Soddy to put his name on the circles. Soddy is actually better known for his Nobel Prize-winning work in chemistry as well as his work in Nuclear Physics and Economics. Descartes' Circle Theorem and its extensions were by no means discovered for the first time by Soddy. This is a common phenomenon in mathematics; it is very hard to find a problem that hasn't been explored before. This is not to say that Soddy just copied the theorem, he was a skilled mathematician and scientist and his contributions to these fields cannot be overlooked.

2 Existence / Construction

So how do we know there exist only two circles, and how do we know any exist at all? The tool we will use to prove the existence of the Soddy Circles is geometric inversion. This problem lends itself nicely to inversive geometry, due to the key property that inversion maintains tangency with respect to the reference circle.

2.1 Inversion

Inversion is the transformation of a figure with respect to a circle, which we call the reference circle. Let the reference circle have center O and radius r . Then, for any point P in the plane, we define the *inverse* of P with respect to the reference circle to be P' , such that P' is on the ray $\overrightarrow{OP}$ and

$$OP \cdot OP' = r^2.$$

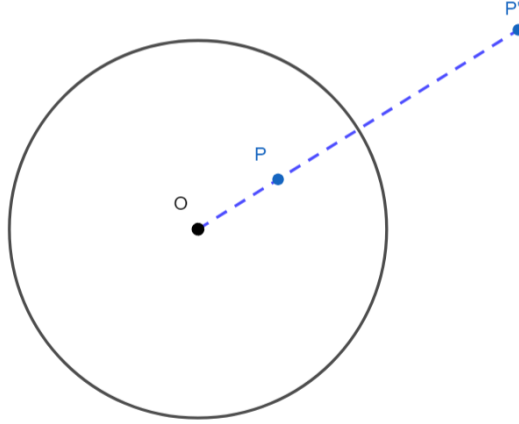


Figure 2: Visual representation of inversion of a point

From this definition we can see that the inverse of any point on the circumference of the reference circle is the same point. Considering Point S on the reference circle, we know the inverse S' is on $\overrightarrow{OS}$ and satisfies

$$OS \cdot OS' = r^2.$$

Since S is on the reference circle, $OS = r$. So, we have

$$\begin{aligned} OS \cdot OS' &= r^2 \\ r \cdot OS' &= r^2 \\ OS' &= r \end{aligned}$$

From this we know $S' = S$ since the only point on $\overrightarrow{OS}$ that is r away from O is point S . So, the inverse of any point on the circle is itself. Another way to say this is that the points on the reference circle are invariant under inversion. Similarly, we can see that the inverse of any point inside the reference circle lies outside the circle, and the inverse of any point outside the circle lies inside the circle.

2.1.1 Circles and Lines

With our definition of inversion, an interesting question to ask is: What is the inverse of the center of inversion? Let's try to find it using the definition! Let O' be the image of O under

inversion about the circle centered at O with radius r . Then,

$$OO \cdot OO' = r^2$$

$$0 \cdot OO' = r^2$$

$$0 = r^2$$

There is seemingly no solution if we use the definition of inversion (given $r > 0$). It would be quite inconvenient to not be able to invert any figure that passes through the center of inversion. For any point P , we can express $OP' = \frac{r^2}{OP}$ where P' is the image of P . Then, we can make a limiting argument, assuming OO' exists, to get

$$\begin{aligned} \lim_{P' \rightarrow O'} OP' &= \lim_{P \rightarrow O} \frac{r^2}{OP} \\ OO' &= \lim_{P \rightarrow O} \frac{r^2}{OP} \\ OO' &= r^2 \lim_{P \rightarrow O} \frac{1}{OP} \end{aligned}$$

Validating the first step in this limiting argument likely requires the condition that inversion is a bijection, or that each point has exactly one inverse and that inverse is unique to that point. This is because of the limits we introduce to both sides, since we are saying that P corresponds to P' for every point as we approach O . Inversion is defined so that it feels slightly obvious that this is true, since there is always only one point on $\overrightarrow{OP}$ of any given positive length. This won't be proven here, but the proof would involve geometric continuity.

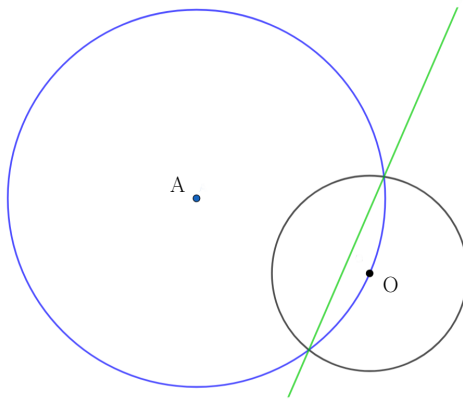


Figure 3: Line inverse of a circle passing through O

Since the length $OO = 0$, we know $\lim_{P \rightarrow O} \frac{1}{OP}$ is undefined (or, infinity). This means that OO' is also undefined. If we interpret this as the line segment OO' has infinite length, then we can say that O' is a *point at infinity*.

The basic figures that are used in inversive geometry are circles and lines (or, *clines*). Circles can invert to lines and vice versa. We have just shown how a circle can invert to a line: the inverse of a circle passing through the center of inversion is a line, since it must pass through a point at infinity! Proving that the inverted figure is exactly a line is more geometrically involved, so it will

remain unproven here (try it yourself!). When a circle inverts to a line, these lines are parallel to the tangent lines of the inverted circles at the center of inversion. In Figure 3, the green line is parallel to the tangent line to the blue circle through O .

Lines are sometimes considered to be degenerate circles with infinite radius, so they can be viewed as special cases of circles and be treated as such. With this interpretation, we can say that all circles are transformed to circles under geometric inversion.

2.1.2 Invariance of Orthogonal Circles

We know that the inverse of a circle intersecting the reference circle still intersects the reference circle, since the points of intersection lie on the reference circle and are thus invariant. When is the inverse of a circle invariant?

Along with points on the reference circle, orthogonal circles are preserved under inversion. Consider two circles, A and B , that intersect at two points. Circles A and B are called *orthogonal* if the tangent lines to A and B at each intersection point are perpendicular. This idea is important for our proof later. The most straightforward proof of this argument uses power of a point.

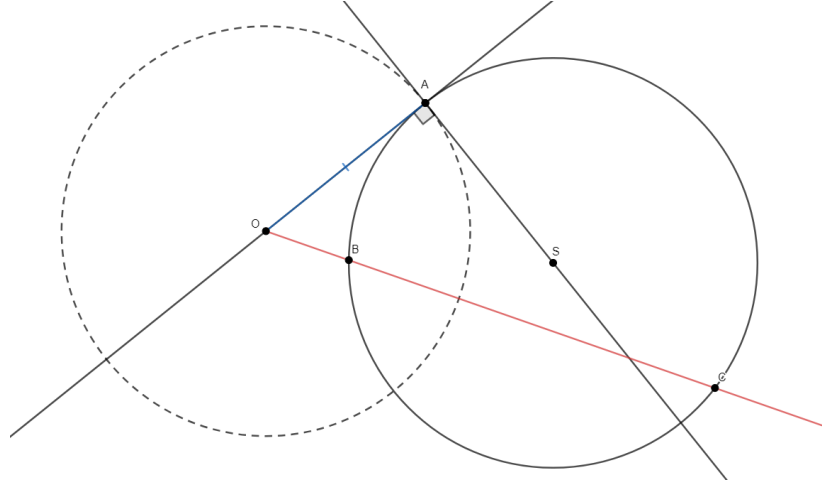


Figure 4: Orthogonal Circles O and S

Refer to Figure 4. Since the circles are orthogonal, the radius OA to the intersection point A is a tangent segment to circle S . We also consider an arbitrary ray $\overrightarrow{OB}$ through point B on circle S , and let point C be the other point that $\overrightarrow{OB}$ intersects S . We know OC is a secant, so we can use power of a point for point O with respect to circle S .

$$OA^2 = OB \cdot OC$$

Since A is on circle O , we know $OA = r$.

$$OB \cdot OC = r^2$$

Since this satisfies the definition of inversion, we have shown that B and C are inverses of each other with respect to circle O . So, any circle orthogonal to the reference circle will map onto itself under geometric inversion. The exception to this rule is if the orthogonal circle passes through O , in which then the circle inverts to a line.

2.1.3 Involutions, Bijections, and Other Inverses

When inverting a figure A to A' , we often call A' the *image* of A under inversion. Inversion is an *involution*, which means that the $(A')' = A$, or the inverse of the inverse of figure A is A . As mentioned before, we can see this pointwise.

Let P be the point we are inverting about circle O . Then, P' is the image of P and P'' is the image of P' . Note that, by definition, P' is on ray $\overrightarrow{OP}$, and P'' is on $\overrightarrow{OP'} = \overrightarrow{OP}$. We want to show that $P'' = P$. From the definition of inversion, we have

$$\begin{aligned} OP \cdot OP' &= r^2 \\ OP' \cdot OP'' &= r^2 \end{aligned}$$

Thus, we can express

$$OP = \frac{r^2}{OP'} = OP''$$

So, as long as OP' is defined, both P and P'' are the same distance from O . Since P and P'' lie on the same ray centered at O , we must have $P = P''$ since there is exactly one point on the ray satisfying $OP = x$ for any positive real x . This isn't a fully rigorous proof, but you should see the idea now that inversion is an involution.

In fact, knowing that inversion is an involution is enough to know that it is also a bijection (though showing inversion is a bijection directly is also possible). Consider a function $f : A \rightarrow A$ such that, for all $x \in A$, $f(f(x)) = x$, and try to show that this function must be bijective. To do this, try to individually show that this function must be injective and surjective from the given condition.

We can see the relation between geometric inversion and our notion of functional inverses or additive/multiplicative inverses. For example, the function $f(x) = \frac{1}{x}$ returns the multiplicative inverse of x , based on the relation $x \cdot \frac{1}{x} = 1$. We can also see the exception of 0 to this function corresponding to our exception of the center of inversion.

We have that these other notions of inverses are also involutions, since $f(f(x)) = x$ wherever $f(x)$ is defined. Taking the additive inverse as another example, we have the additive inverse of a is the unique value b such that $a + b = 0$. From this, we also know the inverse of b is a , since $a + b = 0$.

2.1.4 Inversion Exercises

Here are some exercises to test your new inversion skills. The last few are quite difficult, and you can find detailed solutions from Chapter 8 of Evan Chen's *Euclidean Geometry in Mathematical Olympiads* for the last three.

1. Find the inverse of the point $(4, 5)$ about the unit circle.
2. Find the inverse of the line $y = 4$ to the circle $x^2 + y^2 = 16$.
3. Circle O has radius 2. Circle B has radius 1 and is contained in circle O . What are the possible images of circle B under inversion about circle O ?

4. Circle A has diameter 1. Circles B , C , and D with diameter $\frac{1}{3}$ and are placed inside A , stacked directly vertically on another (in that order) such that every circle is tangent to two others. Perform an inversion with reference circle A , and then one with reference circle C . How can you relate these images, and what does that tell you about inversion as a transformation?
5. Circles A and B are externally tangent and have the same radius value. What is the image of circle A under inversion about circle B ? What about B about A ?
6. Consider an inversion centered at O . A and B are points outside the reference circle with images A' and B' . What can you figure out about triangles $\triangle OAB$ and $\triangle OA'B'$?
7. Given $\triangle ABC$, investigate what an inversion about the incircle does to points A , B , and C . Can you generalize this?
8. Congruent circles A , B , and C intersect at four points. Find the number of circles that exist tangent to any two of A , B , and C (and prove it!)
9. Try to prove Ptolemy's Theorem with inversion. Recall that Ptolemy's states that: If four coplanar points A , B , C , and D are concyclic, then

$$AB \cdot CD + BC \cdot AD = AC \cdot BD.$$

Try to prove the converse as well!

10. Let p be the semiperimeter of $\triangle ABC$. Points E and F are taken on line AB such that $CE = CF = p$. Prove the circumcircle of $\triangle EFC$ is tangent to the excircle of $\triangle ABC$ corresponding to side AB .
11. (2019 AIME II) Triangle ABC has side lengths $AB = 7$, $BC = 8$, and $CA = 9$. Circle ω_1 passes through B and is tangent to line AC at A . Circle ω_2 passes through C and is tangent to line AB at A . Let K be the intersection of circles ω_1 and ω_2 not equal to A . Then $AK = \frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.
12. (EGMO) Let $\triangle ABC$ have orthocenter H and altitudes $\overline{AD}$, $\overline{BE}$, and $\overline{CF}$. Perform an inversion around C with radius $\sqrt{CH \cdot CF}$. Where do the six points go?
13. (EGMO) Let A, B, C be three collinear points (in that order) and construct three semicircles $\Gamma_{AC}, \Gamma_{AB}, \omega_0$, on the same side of AC , with diameters $\overline{AC}$, $\overline{AB}$, and $\overline{BC}$ respectively. For each positive integer k , let ω_k be the circle tangent to Γ_{AC}, Γ_{AB} , and ω_{k-1} . Let n be a positive integer. Prove that the distance from the center of ω_n to $\overline{AC}$ is n times its diameter.
14. (USAMO 2009/5) Trapezoid $ABCD$, with $\overline{AB} \parallel \overline{CD}$, is inscribed in circle ω and point G lies within $\triangle BCD$. Rays $\overrightarrow{AG}$ and $\overrightarrow{BG}$ meet ω again at points P and Q respectively. Let the line through G parallel to $\overline{AB}$ intersect $\overline{BD}$ and $\overline{BC}$ at points R and S , respectively. Prove that quadrilateral $PQRS$ is cyclic if and only if $\overline{BG}$ bisects $\angle CBD$.

2.2 Existence of Soddy Circles

After that brief overview of inversion, we are ready to prove the existence of the Soddy Circles. First, we have to show that it is possible to have three mutually externally tangent circles. Soddy Circles and centers are often referred to with respect to a triangle, and we will soon see why. Given

an arbitrary triangle ABC , we are able to construct three mutually externally tangent circles centered at the vertices of $\triangle ABC$. Do you see how?

Consider the incircle of $\triangle ABC$. Let the points of tangency between the incircle and $\triangle ABC$ be A_1, B_1, C_1 on BC, AC , and AB respectively, as in Figure 5. You may recognize this setup if you've seen a proof for Heron's formula. If two tangent segments to a circle originate from the same external point, then those two segments are congruent. From this, we have $AB_1 = AC_1$, $BA_1 = BC_1$, and $CA_1 = CB_1$. Defining $a = AB_1$, $b = BC_1$, and $c = CA_1$, we can construct circles centered at A, B , and C with radii a, b , and c respectively. These circles will be mutually tangent at A_1, B_1 , and C_1 , since we have guaranteed that each pair of circles intersecting at these points are tangent to each other.

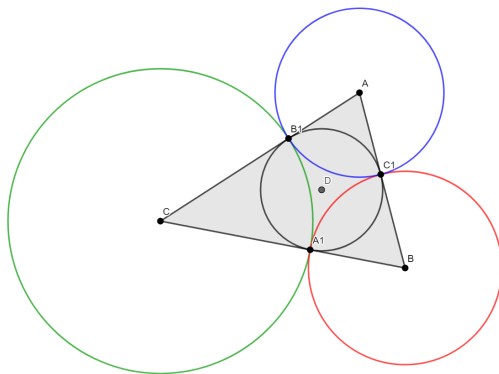


Figure 5: Construction of three mutually tangent circles

Now that we have constructed three mutually externally tangent circles, we can prove the existence of the Soddy Circles.

For the clever inversion, we choose the center of inversion to be A_1 (or any of the tangency points). We choose a radius such that the reference circle is orthogonal to circle A , so it is invariant under inversion about our reference circle.

Since both circles B and C pass through the center of inversion, A_1 , their images under inversion are lines. Both of their images are parallel to the tangent line to circles B and C through A_1 , so both lines are perpendicular to BC . We also know that, since both circles are tangent to circle A prior to inversion, they will also be tangent to A after inversion at the images of B_1 and C_1 .

We have determined that the images of circles B and C under inversion are two lines tangent to circle A and perpendicular to BC . Let the points of tangency to circle A be T_1 and T_2 . We know T_1T_2 is parallel to BC since the lines are perpendicular to BC and tangent to circle A , so T_1T_2 (which passes through A) completes a rectangle between BC and the inverted lines.

Now, our Soddy Circles must be tangent to both of the inverted lines and circle A , since that's what we want pre-inversion. There are only two circles that satisfy this, as seen in Figure 6. These circles cannot be on the opposite side of either line as circle A , because then it cannot be tangent to A . So, there is a fixed radius for these circles as they are restricted between both lines. Then, only two circles that are contained within the lines are also tangent to circle A . So, we have found

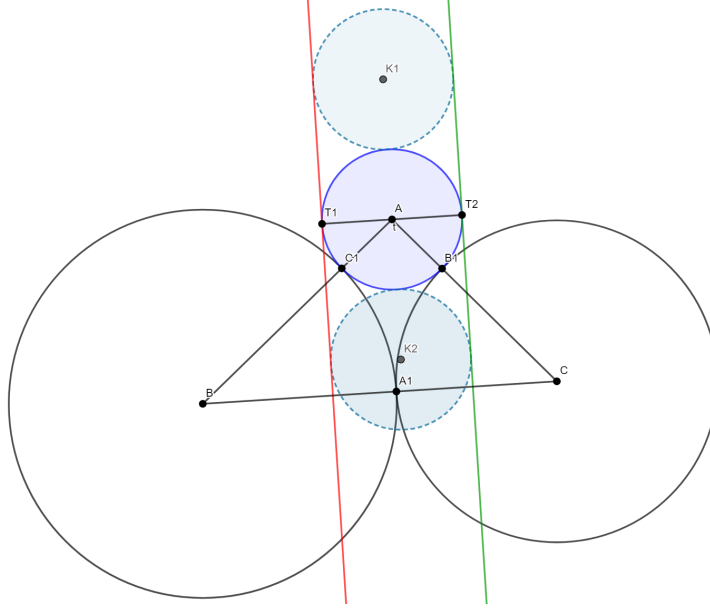


Figure 6: Images of Circles B and C and Soddy Circles

the images of the Soddy Circles under inversion. The circle K_1 is the Inner Soddy Circle, since it is farther away from the center of inversion, and circle K_2 is the Outer Soddy Circle. This approach actually provides a construction for both Soddy Circles, since we can perform the same inversion on this figure to get the desired inner and Outer Soddy Circles, and the process of inversion can be constructed.

2.3 Construction

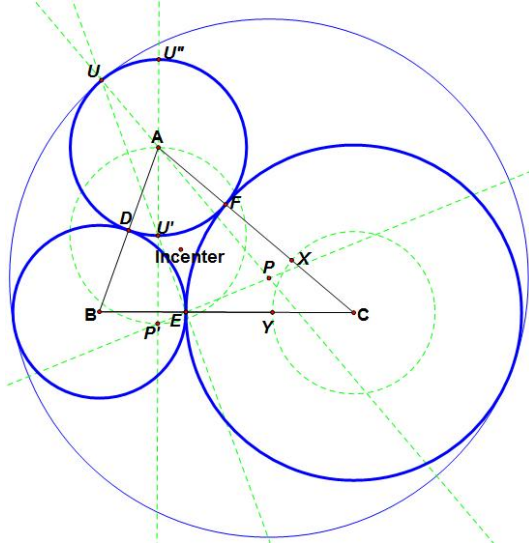
There is also a way to construct the inner and Outer Soddy Circles without inversion. These constructions are fully credited to Krista Floer, who's article on Soddy Circle Construction is cited in References.

2.3.1 Outer Soddy Circle

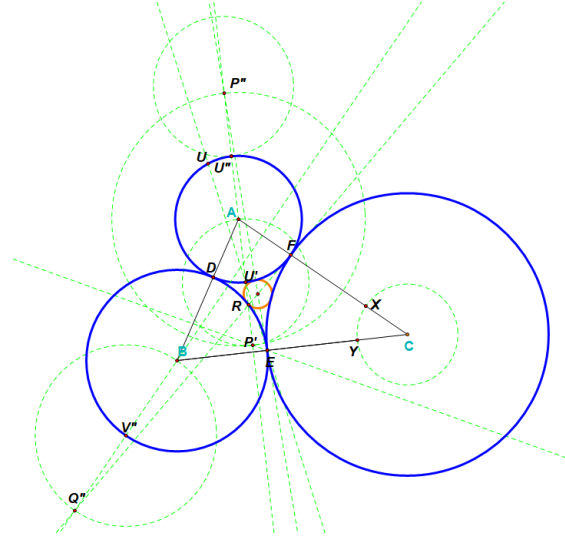
The Outer Soddy Circle is the easier of the two to construct without inversion. Given $\triangle ABC$ with AB as the shortest side (this is required), the Outer Soddy Circle can be constructed as follows (Refer to Figure 7a):

1. Construct point X on side AC so that $AX = AB$.
2. Construct point Y on side BC so that $CY = CX$.
3. Construct the midpoint E of segment BY .
4. Construct point D on side AB so that $BD = BE$.
5. Construct a circle (circle A) with center A and radius AD .
6. Construct a point U' as the intersection of circle A and the line through A perpendicular to BC . This point is the intersection point closest to side BC .

7. Construct EU' . This segment intersects circle A at U .
8. Construct P' on line AU' such that $U'P' = AU'$ and $P' \neq A$.
9. Let P be intersection point of AU and EP' .
10. Construct a circle centered at P with radius UP . This is the outer Soddy Circle.



(a) Outer Soddy Circle Construction



(b) Inner Soddy Circle Construction

Figure 7: Soddy Circle Constructions (Krista Floer)

2.3.2 Inner Soddy Circle

Following the same steps as the Outer Soddy Circle, we have $\triangle ABC$ with AB as the shortest side. Refer to Figure 7b as you read the construction process.

1. Follow steps 1-5 from the Outer Soddy Circle Construction. Note that this is the construction the three tangent circles centered at the triangle vertices.
2. Construct a point U'' as the intersection of circle A and the line through A perpendicular to BC . This point is the intersection point farthest from side BC .
3. Construct point U as the other point of intersection between EU' and circle A .
4. Construct P' on line AU' such that $U'P' = AU'$ and $P' \neq A$.
5. Construct a circle centered at A with radius AP' . Point P'' is the intersection of this circle and line AU' .
6. Construct Point F as the point of tangency between circles A and C .
7. Construct point V'' as the intersection of the line through B perpendicular to AC . The intersection point farthest from side AC is V'' .

8. Construct a circle with center V'' and radius $V''B$. The intersection point of this circle and line $V''B$ is point Q'' .
9. Construct lines FQ'' and EP'' . Let the intersection of these two lines be S . This is the center of the inner Soddy circle.
10. Construct point R as the intersection of circle B and line FQ'' .
11. Construct a circle centered at S with radius RS . This is the inner Soddy Circle.

2.4 Uniqueness

How do we know that the Soddy Circles of a triangle are unique? That is, do we know that there are only two Soddy Circles for any given triangle? Well, we know that for any three mutually externally tangent circles, there exists exactly two circles tangent to both (which is what we proved with inversion). So, as long as a triangle uniquely determines three mutually externally tangent circles centered at its vertices, then the Soddy Circles are unique.

Our mutually tangent circles were formed based on the tangent points of the incircle to the triangle. If there was another triplet of circles, then there would have to be another corresponding incircle to the triangle. However, this cannot be true since the incircle of a triangle is unique. The incenter must be equidistant from all three sides of the triangle, so it is formed by the intersection of the angle bisectors of each angle in the triangle. The angle bisectors of a triangle are concurrent at a single point, the incenter. The incenter is unique, so the incircle must also be unique. Thus, Soddy Circles/centers are unique for any given triangle.

3 Descartes' Circle Theorem

Descartes' Circle Theorem was stated by Descartes in 1643 but left unproved. Since then, proofs have been discovered and the theorem has been extended. For any four mutually tangent circles with radii r_1, r_2, r_3 , and r_4 , Descartes' Circle Theorem states that

$$2 \left(\frac{1}{r_1^2} + \frac{1}{r_2^2} + \frac{1}{r_3^2} + \frac{1}{r_4^2} \right) = \left(\frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3} + \frac{1}{r_4} \right)^2.$$

This statement can be simplified by taking $b_i = \frac{1}{r_i}$, getting

$$2(b_1^2 + b_2^2 + b_3^2 + b_4^2) = (b_1 + b_2 + b_3 + b_4)^2.$$

The quantity b_i is sometimes called the curvature or bend of a circle. We can use this equation to solve for the radii of the Inner and Outer Soddy Circles when given three mutually externally tangent circles. Note that, whenever solving, we will get a positive and a negative solution. Two solutions make sense, since there are exactly two circles tangent to all three circles.

The negative solution corresponds to the radius of the Outer Soddy Circle, for which we say it has negative radius/curvature since the other three circles are internally tangent to it. Similarly, the Inner Soddy Circle's radius corresponds to the positive solution of the quadratic, since the Inner Soddy Circle is externally tangent to all three circles.

3.1 Proof

Descartes' Circle Theorem can be proved using only Heron's formula and algebraic manipulation. We consider the line segments from the vertices of the triangles to the inner/outer Soddy centers. First, we know the sides of the triangle to be $r_1 + r_2$, $r_2 + r_3$, and $r_1 + r_3$, given these are

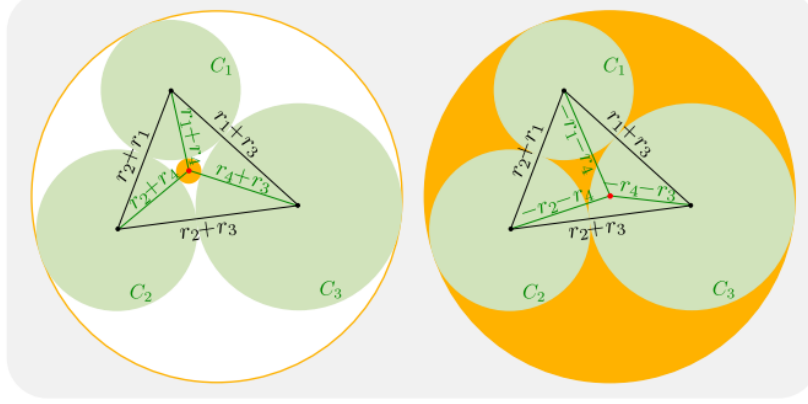


Figure 8: Labeled Diagrams for proof (Paul Levrie)

radii of the mutually externally tangent circles. Let side AB have length $r_1 + r_2$, side BC have length $r_2 + r_3$, and side AC have length $r_1 + r_3$.

For the inner Soddy center, we know the length of these segments to be $r_1 + r_4$, $r_2 + r_4$, and $r_3 + r_4$, where r_4 is the radius of the Inner Soddy Circle. Let the inner Soddy Center be O_1 , then $AO_1 = r_1 + r_4$, $BO_1 = r_2 + r_4$, and $CO_1 = r_3 + r_4$.

Since, when solving for radius using Descartes' Circle Theorem, the radius of the Outer Soddy Circle is considered to be negative, we consider $-R_4$ to be the radius of the Outer Soddy Circle with center O_2 . Then, we have that the segments have lengths $AO_2 = -R_4 - r_1$, $BO_2 = -R_4 - r_2$, and $CO_2 = -R_4 - r_3$. We will show the two cases to be equivalent when considering Heron's formula.

Heron's formula is a formula for triangle area. For $\triangle ABC$ with side lengths a, b, c , we define the semiperimeter to be $s = \frac{a+b+c}{2}$. Then,

$$[\triangle ABC] = \sqrt{s(s-a)(s-b)(s-c)}.$$

We know the side lengths of all of the triangles formed by the segments we constructed, so we can use Heron's formula.

First, we can find the area of the large triangle. The semiperimeter is

$$s = \frac{2r_1 + 2r_2 + 2r_3}{2} = r_1 + r_2 + r_3,$$

so the area via Heron's formula can be expressed as

$$[\triangle ABC] = \sqrt{(r_1 + r_2 + r_3)r_1r_2r_3}.$$

We know that, in both cases, the sum of the areas of the three triangles formed using the segments to the Soddy center is the large triangle.

For the inner Soddy center, we have three triangles. Then we have

$$\begin{aligned}s_{ABO_1} &= \frac{2r_1 + 2r_2 + 2r_4}{2} = r_1 + r_2 + r_4 \\s_{ACO_1} &= \frac{2r_1 + 2r_3 + 2r_4}{2} = r_1 + r_3 + r_4 \\s_{BCO_1} &= \frac{2r_2 + 2r_3 + 2r_4}{2} = r_2 + r_3 + r_4\end{aligned}$$

Similarly to $\triangle ABC$, we have the area of each subtriangle to be

$$\begin{aligned}[ABO_1] &= \sqrt{(r_1 + r_2 + r_4)r_1r_2r_4} \\[ACO_1] &= \sqrt{(r_1 + r_3 + r_4)r_1r_3r_4} \\[BCO_1] &= \sqrt{(r_2 + r_3 + r_4)r_2r_3r_4}\end{aligned}$$

Now, we can solve for the areas since $[ABC] = [ABO_1] + [ACO_1] + [BCO_1]$:

$$\sqrt{(r_1 + r_2 + r_3)r_1r_2r_3} = \sqrt{(r_1 + r_2 + r_4)r_1r_2r_4} + \sqrt{(r_1 + r_3 + r_4)r_1r_3r_4} + \sqrt{(r_2 + r_3 + r_4)r_2r_3r_4}.$$

From this step, only algebraic manipulation will get us Descartes' Circle Theorem. That won't be shown here, though, because it's very long.

We can perform the same steps for the Outer Soddy Circle, except all the semiperimeters and triangle sides will be negative. We get the same result, though, since there are four terms in the product for Heron's formula, so the factors of -1 will cancel out. So, we manipulate the above equation to get

$$2\left(\frac{1}{r_1^2} + \frac{1}{r_2^2} + \frac{1}{r_3^2} + \frac{1}{r_4^2}\right) = \left(\frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3} + \frac{1}{r_4}\right)^2,$$

which gives us the solution for both Soddy radii.

3.2 Degenerate Cases

As mentioned before, lines can be considered circles of infinite radius. It also makes sense, with our definition of curvature, that a line has 0 curvature. So, we have a few degenerate cases when some of the circles are lines.

3.2.1 One line

Since the curvature of a line is 0, this reduces Descartes' Theorem to

$$2(b_1^2 + b_2^2 + b_3^2) = (b_1 + b_2 + b_3)^2.$$

3.2.2 Two lines

Descartes' Theorem reduces to

$$2(b_1^2 + b_2^2) = (b_1 + b_2)^2.$$

In this case, we can solve for the radii and it isn't ugly. We can expand, getting

$$\begin{aligned} 2b_1^2 + 2b_2^2 &= b_1^2 + 2b_1b_2 + b_2^2 \\ b_1^2 + b_2^2 &= 2b_1b_2 \\ b_1^2 - 2b_1b_2 + b_2^2 &= 0 \\ (b_1 - b_2)^2 &= 0 \end{aligned}$$

Which we can see the only solution is $b_1 = b_2$, which is forced. We can see the relation to our inversion proof, where the the Soddy Circles under inversion had to be tangent to two lines and one circle, and they were both congruent to the circle.

3.2.3 Three lines

We have

$$2b_1^2 = b_1^2.$$

The only solution for this is when $b_1^2 = 0$, or $b_1 = 0$. This might not make sense, since we can have circles tangent to three different lines, but our lines are specifically degenerate circles. Since we require these 'circles' to be tangent, they must be parallel lines which intersect at infinity (remember how we inverted circles to lines). There cannot be a circle that is tangent to three distinct parallel lines, unless it is a parallel line (a degenerate circle) itself. This idea might seem confusing, but remember that all of the figures we are dealing with must be circles, even when considering lines as the degenerate case.

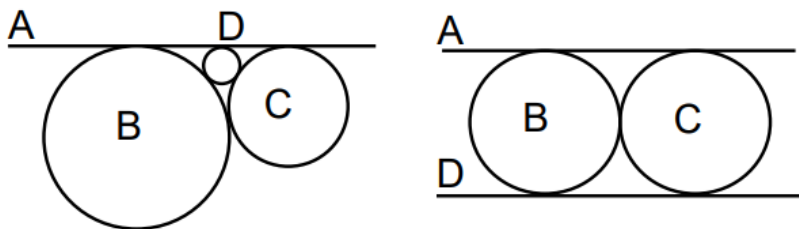


Figure 9: Two interesting degenerate cases of Descartes' Circle Theorem

4 Soddy Sequences

We can repeat the process of constructing the Inner Soddy Circle for any three mutually externally tangent circles to get a sequence of smaller and smaller Inner Soddy Circles. It is an interesting question to ask if, when repeating this infinitely many times, does the radius of the Soddy Circle converge? Well, every time we construct a new Soddy Circle, the area in which the next Soddy Circle can go (given we use the $(n - 1)^{\text{th}}$ Soddy Circle) is strictly decreasing. Since area must be positive, we know that the area that the next Soddy Circle can go approaches 0, so the radius of the n^{th} Soddy Circle approaches 0 as well.

4.1 Circle Packing

A cool application of this is circle packing, which is the study of figures of circles in a plane that all touch each other. When we take infinitely many Soddy Circles, the figure becomes a fractal, like in Figure 10.

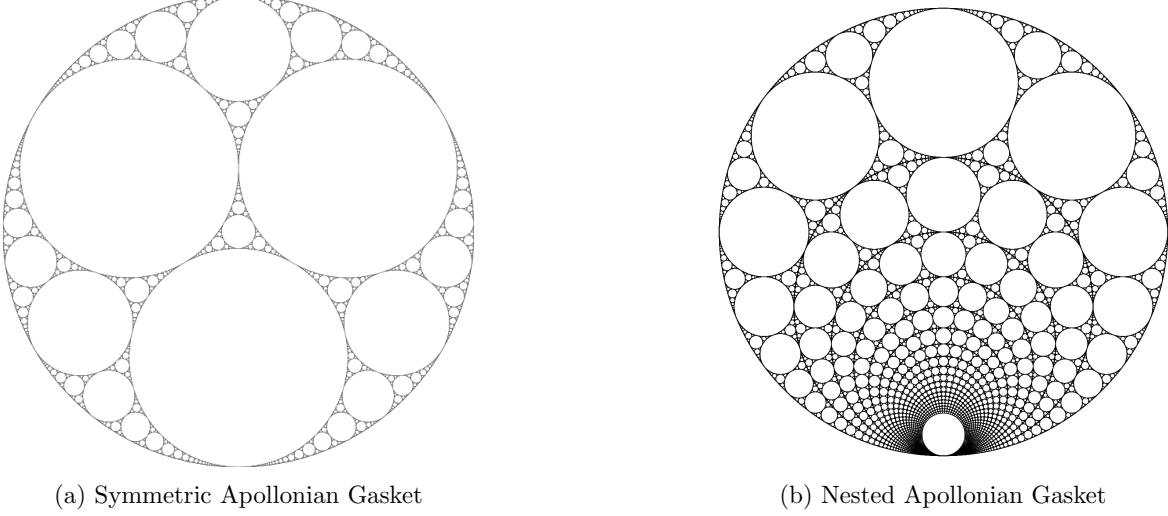


Figure 10: Fractals formed with infinitely many Soddy Circles

These packings are generated by an initial quadruple of circle curvatures, where three circles are mutually tangent within a larger one (like the Outer Soddy Circle). Then, the Inner Soddy Circle construction is applied indefinitely to get packed figures. Packings can also just be generated by triples, since we have shown that three mutually tangent circles determine the outer tangent one, but the outer circle is often included because packings only utilize the inner Soddy construction. These packings are often called Apollonian Gaskets after Apollonius. An interesting property of Apollonian Gaskets is that, if the initial quadruple of curvatures are all integers, then every circle in the gasket will also have an integer curvature. The proof of this utilizes Descartes' Circle Theorem, which we will use with curvatures:

$$2(b_1^2 + b_2^2 + b_3^2 + b_4^2) = (b_1 + b_2 + b_3 + b_4)^2.$$

Where $b_i = \frac{1}{r_i}$ for each circle. Let b_4 be the curvature we are solving for. Also, let $b_1 + b_2 + b_3 = n$ and $b_1b_2 + b_1b_3 + b_2b_3 = m$. Then, we have

$$\begin{aligned} 2(b_1^2 + b_2^2 + b_3^2 + b_4^2) &= (b_1 + b_2 + b_3 + b_4)^2 \\ 2((b_1 + b_2 + b_3)^2 - 2(b_1b_2 + b_1b_3 + b_2b_3) + b_4^2) &= (b_1 + b_2 + b_3 + b_4)^2 \\ 2(n^2 - 2m + b_4^2) &= (n + b_4)^2 \\ 2n^2 - 4m + 2b_4^2 &= n^2 + 2nb_4 + b_4^2 \\ b_4^2 - (2n)b_4 + n^2 - 4m &= 0 \end{aligned}$$

Now, we have a quadratic in b_4 , which we can solve with the quadratic formula:

$$\begin{aligned}
b_4 &= \frac{2n \pm \sqrt{4n^2 - 4(n^2 - 4m)}}{2} \\
&= \frac{2n \pm \sqrt{16m}}{2} \\
&= n \pm 2\sqrt{m} \\
&= b_1 + b_2 + b_3 \pm \sqrt{b_1b_2 + b_1b_3 + b_2b_3}
\end{aligned}$$

We have two solutions for b_4 . These are the curvatures of the inner and Outer Soddy Circles, so one is negative (outer) and the other is positive (inner). Let the Inner Soddy Circle have curvature $B = b_1 + b_2 + b_3 + \sqrt{b_1b_2 + b_1b_3 + b_2b_3}$ and the Outer Soddy Circle have curvature $B' = b_1 + b_2 + b_3 - \sqrt{b_1b_2 + b_1b_3 + b_2b_3}$. Then, $B + B' = 2(b_1 + b_2 + b_3)$, and we get

$$B = 2(b_1 + b_2 + b_3) - B'.$$

With the precondition that the starting quadruple (b_1, b_2, b_3, B') are all integers, we have shown that B will always be an integer. Due to this property, gaskets with integer starting quadruples, called Integral Apollonian Gaskets, are often studied.

4.2 Gasket Symmetry

We can construct these fractals to have as many symmetries as we want by changing the number of starting circles and having them be congruent. However, symmetry becomes much more restrictive when we only look at integral Apollonian circle packings. Integral packings can either have no symmetry, D_1 symmetry, or D_2 symmetry, where D refers to the dihedral symmetry group.

An integral Apollonian packing has no symmetry if the starting three curvatures (within the fourth circle) are all different. This makes sense, because there cannot be any rotational or reflectional symmetry if the three largest circles do not share values.

D_n symmetry refers to a group of transformations of rotations and reflections. For a figure to be D_n symmetric, it should be symmetric about n lines through the origin and by rotation of all multiples of $\frac{2\pi}{n}$ about the origin. Basically, it must have n different rotational symmetries (including 2π) and n different reflectional symmetries. Refer to figure 11 to see examples of each symmetry.

There are many integral packings with D_1 symmetry. The only requirement for this is that any two of the largest 5 circles have the same curvature. This gives us one line of reflectional symmetry about a diameter and no rotational symmetry in these packings.

There exists only one packing with D_2 symmetry: the packing with the first 5 curvatures $(-1, 2, 2, 3, 3)$. The -1 refers to the outer circle, but then there are two pairs of the same integer, so there are two lines of reflectional symmetry, and a rotational symmetry by π .

No packings exist with D_3 symmetry; This is because it would require three starting circles with the same curvatures. It can be shown that the ratio of the curvatures of these congruent circles to the circle they are inscribed in is irrational, so there exists no integral apollonian gasket with D_3 symmetry.

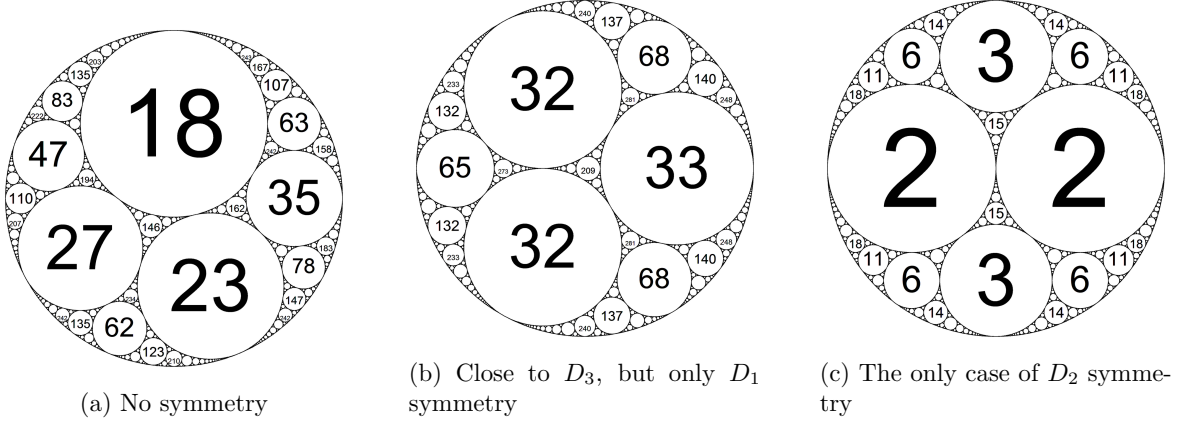


Figure 11: Integral Apollonian Gaskets

4.3 Interesting Case

There is an interesting case of circle packing in the degenerate case of Descartes' Circle Theorem with two lines and two circles. The lines have curvature 0, and the pattern repeats infinitely within the lines, creating a stripe figure, as seen in Figure 12. This case corresponds to the starting quadruple $(0, 0, 1, 1)$. Each value of curvature repeats infinitely, but there remains only D_1 symmetry being the line equidistant from both parallel lines. We can show that every square of an

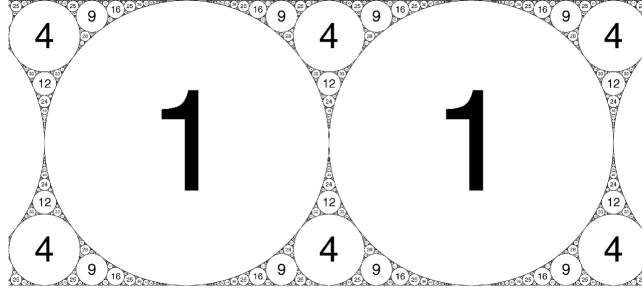


Figure 12: Strip Circle Packing

integer appears in the sequence of curvatures for this figure. Considering all of the circles tangent to one of the lines and a circle of radius 1, we have $B = 2(a + b + c) - B'$. For the first of these circles, it is tangent to two circles of curvature 1, so we have $B = 2(1 + 1 + 0) + 0 = 4$. We consider one of the lines to be the Outer Soddy Circle, the other figure tangent to the three circles we are considering. Then, the next circle in this sequence is tangent to a line, a circle with curvature 1, and a circle of curvature 4. This gets us $B = 2(1 + 4 + 0) - 1 = 9$. We consider the circle of curvature 1 to be the outer circle, since it is the other circle tangent to each of the circles we are considering. Now, we can prove that all squares appear in the sequence.

I claim that the circles with curvatures $(n + 1)^2$ and $(n - 1)^2$ will be the circles tangent to the circles with curvatures 0, 1, and n^2 .

The circle with curvature $(n - 1)^2$ will be the outer Soddy case, since a smaller curvature corresponds to a larger circle. We will show that, plugging in the values from the claim, we get an

identity that is always true:

$$\begin{aligned}
B &= 2(1 + 0 + n^2) - (n - 1)^2 \\
&= 2 + 2n^2 - n^2 + 2n - 1 \\
&= n^2 + 2n + 1 \\
&= (n + 1)^2.
\end{aligned}$$

We have shown that $B = (n + 1)^2$ given the conditions, which validates the claim. So, if we have a circle with curvature $(n - 1)^2$ that is tangent to circles of curvature 0, 1, and n^2 , we have shown that the other tangent circle to these three circles has curvature $(n + 1)^2$. We have already shown the base case of this hypothesis, that a circle of curvature $4 = 2^2$ is tangent to the circles of curvature 0, 1, and 1, with the other solution being $(2 - 2)^2 = 0$. Then, we can use curvatures 0, 1, and 4 as the next case, which will get us a new circle of curvature 9, using the fact that we have already shown a circle of curvature 1 is the other solution to these constraints.

Researchers have questioned the patterns in the integers that appear in the sequences formed by integral Apollonian gaskets. In fact, questions about integral Apollonian gaskets are still being studied today. In the summer of 2023, a group of researchers proved the *Local-Global Conjecture* false. In any given primitive integral Apollonian circle packing (a packing is primitive if the starting quadruple are all relatively prime), each curvature value must belong to either six or eight residue classes modulo 24. The local-global conjecture stated that every integer part of either of these residue classes must appear as a curvature, and this was proven false with many counterexamples.

5 Multidimensional Soddy Hyperspheres

The Kiss Precise (Continued)

Frederick Soddy, 1936

*To spy out spherical affairs
 An oscular surveyor
 Might find the task laborious,
 The sphere is much the gayer,
 And now besides the pair of pairs
 A fifth sphere in the kissing shares.
 Yet, signs and zero as before,
 For each to kiss the other four,
 The square of the sum of all five bends
 Is thrice the sum of their squares.*

The Kiss Precise (Generalized)

Thorold Gosset, 1937

*And let us not confine our cares
 To simple circles, planes and spheres,
 But rise to hyper flats and bends*

*Where kissing multiple appears,
 In n -ic space the kissing pairs
 Are hyperspheres, and Truth declares -
 As $n + 2$ such osculate
 Each with an $n + 1$ fold mate
 The square of the sum of all the bends
 Is n times the sum of their squares.*

In Soddy's original poem, he detailed his findings that Descartes' Circle Theorem generalizes to spheres, in which five spheres can be mutually externally tangent and Descartes' Theorem becomes

$$3(b_1^2 + b_2^2 + b_3^2 + b_4^2 + b_5^2) = (b_1 + b_2 + b_3 + b_4 + b_5)^2.$$

Expressing this in summation notation, it becomes

$$3 \cdot \sum_{i=1}^5 b_i^2 = \left(\sum_{i=1}^5 b_i \right)^2.$$

In 1937, Thorold Gosset extended Soddy's poem with his own stanza, detailing how the *Soddy-Gosset* theorem states that Descartes' Circle theorem can be extended to n -dimensions, with it being possible for $n + 2$ hyperspheres to be mutually tangent. Generalized to n -dimensions, Descartes' Circle theorem becomes

$$n \cdot \sum_{i=1}^{n+2} b_i^2 = \left(\sum_{i=1}^{n+2} b_i \right)^2.$$

The proof for this, as one can assume, will not use the simple geometry we used for Descartes' Circle Theorem on a 2-dimensional plane. Instead, a better method of generalizing to higher dimensions is to use matrix equations, and that's one way to prove this generalization.

5.1 Soddy's Hexlet

On January 9th, 1937, Frederick Soddy published his findings regarding 'The Bowl of Integers and the Hexlet', which are 3-dimensional sphere packings that use Descartes' Circle Theorem. The theorem for Soddy's Hexlet is stated as follows: Given two spheres A and B , both externally tangent to each other and internally tangent to a circle C , we can construct a 'chain' of spheres tangent to A , B , C , and two other spheres in the chain. Once the chain is completed, there will always be 6 spheres in the chain.

Soddy presented a proof for this in his publication by applying Descartes' Circle Theorem. First, define δ as the curvature of the first sphere in the chain (which can vary as long as tangencies hold).

Using the Generalized Descartes' Circle Theorem to 3 dimensions, we have two real solutions for b_5 , the fifth sphere, in terms of δ and the curvatures of A , B , and C . Instead of two solutions corresponding to inner and outer solutions, in this case, since C is already the outer sphere, both solutions will be within C and tangent to each sphere.

Let the sum of the curvatures of A , B , and C be σ . Then, let the two values of b_5 be ϕ and ϕ' . Similarly to how we found that the solutions for b_4 sum to $2(b_1 + b_2 + b_3)$ in section 4.1, Soddy

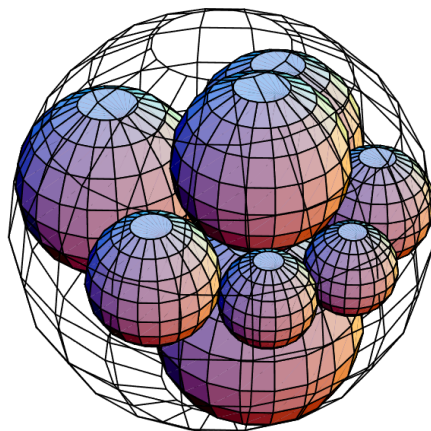


Figure 13: Soddy's Hexlet (Wolfram MathWorld)

found that, in 3 dimensions, the two solutions for curvatures also sum to a constant value. That is, we have the relation

$$\phi + \phi' = \sigma + \delta$$

We have guaranteed spheres with curvatures ϕ and ϕ' are tangent to the sphere with curvature δ by Descartes' Circle Theorem. So, ϕ, ϕ' , and δ are curvatures of consecutive spheres in the chain, and we can express this relationship recursively. Let ϕ_N be a the curvature of a sphere in the chain. Then, as Soddy found,

$$\phi_{N+1} + \phi_{N-1} = \sigma + \phi_N$$

From this, the curvatures of all spheres can be found (given any starting 2). Amazingly, this relation is periodic with period 6 (you can verify this yourself!), meaning every completed chain will consist of 6 spheres. This is why this figure is coined Soddy's Hexlet.

This theorem is actually a specific case of Kollros' theorem, which states that, for any ring of p spheres, there exists a ring of q spheres tangent to each of the p spheres such that

$$\frac{1}{p} + \frac{1}{q} = \frac{1}{2}$$

In the case of the hexlet, we start with A, B , and C , so $p = 3$. We can easily see that $q = 6$ is the solution from Kollros' theorem.

5.2 Soddy's Bowl of Integers

In addition to the Hexlet, Soddy introduced the Bowl of Integers in his article. Beginning with a unit sphere, place two spheres of radius $\frac{1}{2}$ in it such that they are externally tangent to each other and internally tangent to the unit sphere. This configuration is called the 'Bowl of Integers', due to the property that the curvature of all spheres that can be packed in it are integers. This property bares a striking resemblance to the integral Apollonian circle packings, and that's because both rely on the Descartes' Circle Theorem. Since this is quite similar to what we did before, try to prove the relation yourself! The first few curvatures are $-1, 2, 5, 6, 9, 11$, and 14 .

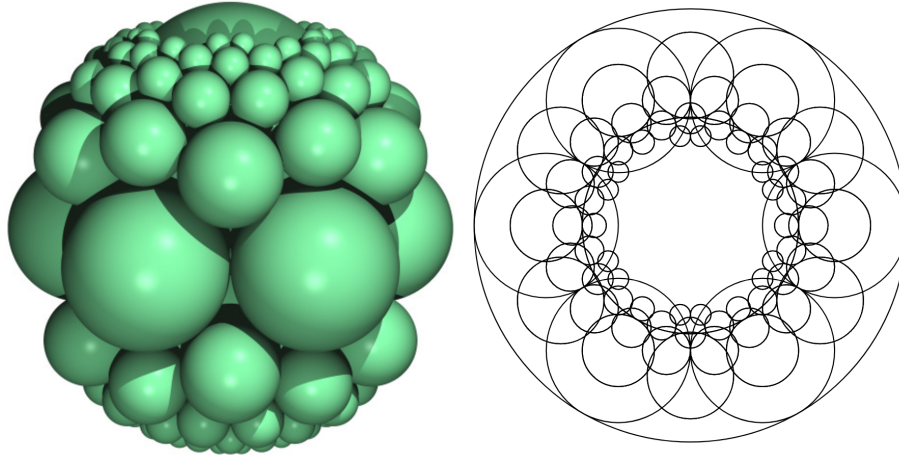


Figure 14: Bowl of Integers and 2 dimensional cross section

6 Non-Euclidean Spaces

The Kiss Precise (Further Generalized)

Fred Lunnon

*How frightfully pedestrian
My predecessors were
To pose in space Euclidean
Each fraternising sphere!
Let Gauss' k squared be positive
When space becomes elliptic,
And conversely turn negative
For spaces hyperbolic:
Squared sum of bends is sum times n
Of twice k squared plus squares of bends.*

Fred Lunnon extended Soddy's poem once more, describing a new generalization of Descartes' Theorem to non-Euclidean spaces, such as spherical, hyperbolic, and elliptical geometry.

6.1 Euclidean vs. Non-euclidean

Euclidean geometry is geometry on a flat plane, based on the axioms of Euclid in his book *Elements*. Non-Euclidean geometries are not conducted on planes, but rather on different 3D surfaces such as the surface of a sphere, ellipsoid, or hyperboloid. The main difference on these surfaces are in the notions of parallelism and perpendicularity of lines, which change fundamental properties that we are used to in Euclidean geometry. This article won't go in depth on non-Euclidean geometry, but Descartes' Circle Theorem generalizes quite nicely in these cases.

6.2 Descartes' Circle Theorem on Spheres and Hyperbola

In both spherical and hyperbolic geometry, curvature is defined differently than we have done so far in Euclidean geometry. In spherical geometry, a curvature $b_s = \cot(r_s)$ is defined in terms of r_i , named the *oriented intrinsic radius* of a circle, which is the radius of the circle on the surface of the sphere. Similarly, in hyperbolic geometry, a curvature $b_h = \coth(r_h)$, using the oriented intrinsic radius of a circle on a hyperbolic surface. Note that these two intrinsic radii have different definitions in relation to Euclidean geometry since we are relating different surfaces.

So, what does Descartes' Circle Theorem become? In spherical geometry, for three mutually tangent circles (curvatures b_1, b_2, b_3), the curvatures of the two circles tangent to all three (b_4) can be solved using the relation

$$(b_1 + b_2 + b_3 + b_4)^2 = 2(b_1^2 + b_2^2 + b_3^2 + b_4^2) + 4.$$

Notice that, this is exactly Descartes' Circle Theorem in Euclidean geometry with an added constant term! further, in hyperbolic geometry, we can solve for mutually tangent circle curvatures with the equation

$$(b_1 + b_2 + b_3 + b_4)^2 = 2(b_1^2 + b_2^2 + b_3^2 + b_4^2) - 4.$$

Interestingly, it is very similar to spherical geometry. In both cases, the fact we proved earlier still holds: for both solutions of b_4 (let those be B and B'), we have the relationship

$$B + B' = 2(b_1 + b_2 + b_3).$$

Which means we can create Soddy sequences and integral Apollonian circle packings on spherical and hyperbolic surfaces as well!

6.3 Putting it All Together

Combining the multidimensional and non-Euclidean generalizations of Descartes' Circle Theorem, we can get one Ultimate Descartes' Circle Theorem that applies in n -space and on different surfaces. In general n -space, we can guarantee $n + 2$ mutually tangent hyperspheres. Let the curvature of the i^{th} hypersphere be b_i . Then, we have

$$\left(\sum_{i=1}^{n+2} b_i\right)^2 = n \cdot \left(\sum_{i=1}^{n+2} b_i^2\right) + 2nv^2$$

Where v is the curvature of the space. For example, for hyperbolic geometry, $v^2 = -1$, in elliptical geometry, $v^2 = 1$, and in Euclidean geometry, $v^2 = 0$.

7 Conclusion

Taking a deep dive into mutually tangent circles in the lens of inversion allowed for a very straightforward proof of the existence of Soddy Circles. An associated result, Descartes' Circle Theorem, related the radii of these circles, and as it turns out the theorem generalizes far, far beyond two dimensional Euclidean geometry. Though there may not be much practical use to knowing one can pack infinitely many circles of integer curvature into a circle on a hyperbolic surface, Frederick Soddy and Descartes' Theorem are great examples of when mathematicians, driven purely by curiosity, aided by the findings of their predecessors, are able to explore problems

to their fullest extent. There appears to be a fundamental harmony present that allows us to generalize mutually tangent circles to the furthest extent with a simple equation. Should it just be chopped up to the ‘infinite symmetries’ of a circle? I don’t believe so, but questioning the structure that our mathematics has been built on is vital to advancing it. Is there some underlying basis that presents such a satisfying result? If there is, push it to its limits. If there isn’t, keep looking.

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Fourier Transforms

David Jiang

1 Introduction

Fourier analysis is the approximation of periodic functions by summing simpler, period trigonometric functions. The reason we want to do this is that trigonometric functions are simpler periodic functions to work with compared with constructing some new periodic function. Additionally, trigonometric functions also show up in the natural world, from representing sound waves to heat distribution. By representing these functions mathematically, we can create easier methods to perform certain calculations and operations to manipulate these periodic functions to get what we want.

2 Fourier Series

We define a Fourier series as follows. For a period function $f(x)$, its Fourier series is defined as

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx,$$

with

$$a_k = \frac{2}{\pi} \int_0^{\pi} f(x) \cos kx \, dx \text{ and } b_k = \frac{2}{\pi} \int_0^{\pi} f(x) \sin kx \, dx,$$

and a_0 is some constant term.

To find the coefficients of any Fourier series, we want to isolate each of a_k , b_k , so we get some expression in terms of $f(x)$. To do this, we can consider multiplying by either $\sin mx$ or $\cos mx$ and taking the integral from 0 to π because we know that

$$\begin{aligned} \int_0^{\pi} \sin nx \sin mx \, dx &= 0 \text{ if } m \neq n, \\ \int_0^{\pi} \cos nx \cos mx \, dx &= 0 \text{ if } m \neq n, \text{ and} \\ \int_0^{\pi} \sin nx \cos mx \, dx &= 0. \end{aligned}$$

However, this may seem somewhat unmotivated if we were just multiplying and taking the integral. One alternative thought process is that the sines and cosines can be represented as vectors, and we project $f(x)$ onto the trigonometric functions. What's nice about vectors is that they lead us to orthogonality, and when two vectors are orthogonal, their dot product is 0. Now we just need to consider how to take the dot product of a function.

2.1 Linear Motivation

Consider two functions $f(x)$ and $g(x)$. Then, we take the two vectors with infinite dimensions

$$f : \langle f(x), f(x + \Delta x), f(x + 2\Delta x), \dots \rangle \text{ and } g : \langle g(x), g(x + \Delta x), g(x + 2\Delta x), \dots \rangle$$

with Δx approaching 0.

Note that when we dot f and g , the components of f and g multiply with each other and add to get the final value. This is analogous to finding the area under the curve $h(x) = f(x)g(x)$, so

$$f \cdot g = \int h(x) dx = \int f(x)g(x) dx.$$

We can show that $\sin nx$ and $\sin mx$ are orthogonal to each other since the integral of their products are equal to 0, and similarly for $\cos nx$ and $\cos mx$, and $\sin nx$ and $\cos mx$. Assuming we know that these functions are all orthogonal to each other, multiplying by the trig functions and taking the integral seems much less random now.

3 Square Function Example

To get a better understanding of Fourier series, let's try approximate the piece wise square function $F(x)$, with one period defined as

$$F(x) = \begin{cases} 1 & \text{if } 0 \leq x < \pi \\ -1 & \text{if } \pi \leq x < 2\pi \end{cases}$$

from 0 to 2π using Fourier series.

Firstly, let's graph a section of this function to get a sense of what we're working with.



Since our function is odd, we only need to consider the sine portion of the Fourier series. Then,

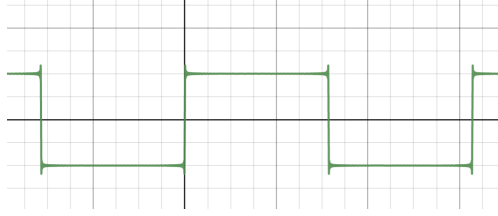
$$b_n = \frac{2}{\pi} \int_0^\pi \sin(nx) dx = \frac{2}{\pi} \left[\frac{-\cos nx}{n} \right]_0^\pi = \frac{-2 \cos \pi n}{\pi n}.$$

Note that all even $b_n = 2k$ will become 0 since $\cos 2kx = 0$ for all $k \in \mathbb{N}$.

Since our function is centered at 0, $a_0 = 0$ and we don't have to include it in our expression. Therefore, we can write our series as

$$F(x) = \sum_{n=1}^{\infty} \frac{2 - 2 \cos \pi n}{\pi n} \sin nx.$$

As a reality check, we can graph this on Desmos and sure enough, we get a graph very similar to the square function above.



4 Exponential Fourier Series

4.1 Functions with periods of 1

Recall that for any periodic function $f(x)$, we can write it as

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos(nx) + b_n \sin(nx).$$

Note that the period of this function is 2π . For convenience, let's let the period of $f(x)$ instead be 1 so that we can easily change it in the future (this is especially useful for when we take the transform). Then, if we let $0 \leq t \leq 1$, we can rewrite $f(x)$ in terms of t .

$$f(t) = a_0 + \sum_{n=1}^{\infty} a_n \cos(2\pi nt) + b_n \sin(2\pi nt)$$

The sine and cosine sum also reminds us of something else. Recall Euler's Formula. For all real θ ,

$$e^{i\theta} = \cos \theta + i \sin \theta.$$

One can prove this by relating the power-series expansions of e^x , the sine function, and the cosine function. More specifically,

$$\begin{aligned} e^{i\theta} &= 1 + i\theta + \frac{(i\theta)^2}{2!} + \frac{(i\theta)^3}{3!} + \dots \\ &= 1 + i\theta - \frac{\theta^2}{2!} - \frac{i\theta^3}{3!} + \dots \\ &= \left(1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \dots\right) + i\left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \dots\right) \\ &= \cos \theta + i \sin \theta. \end{aligned}$$

Thus, we get our identity.

One thing to note is that this proof relies on the fact that imaginary numbers act the same way as real numbers do when exponential. However, this can be proven, but for the sake of simplicity, we will take this for granted. This also gives some interesting properties about sine and cosine functions.

For all real θ , we can write $\cos \theta$ as

$$\cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2}$$

and we can write $\sin \theta$ as

$$\sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}.$$

Now, we can rewrite $f(t)$ in terms of e .

$$\begin{aligned} f(t) &= a_0 + \sum_{n=1}^{\infty} a_n \cos(2\pi nt) + b_n \sin(2\pi nt) \\ &= a_0 + \sum_{n=1}^{\infty} a_n \frac{e^{2\pi int} + e^{-2\pi int}}{2} + b_n \frac{e^{2\pi int} - e^{-2\pi int}}{2i} \\ &= a_0 + \sum_{n=1}^{\infty} \frac{a_n}{2} e^{2\pi int} + \frac{a_n}{2} e^{-2\pi int} + \frac{b_n}{2} e^{2\pi int} - \frac{b_n}{2i} e^{-2\pi int} \\ &= a_0 + \sum_{n=1}^{\infty} \left(\frac{a_n}{2} - \frac{b_n}{2} i \right) e^{2\pi int} + \left(\frac{a_n}{2} + \frac{b_n}{2} i \right) e^{-2\pi int} \end{aligned}$$

From here, we notice that we can let $c_n = \frac{a_n}{2} - \frac{b_n}{2}i$ and $c_{-n} = \frac{a_n}{2} + \frac{b_n}{2}i$. Additionally, notice that if we let $n = 0$ and sum the two coefficients together, $2c_0 = a_0$. In fact, textbooks generally have that the constant term of a trigonometric Fourier series as $\frac{a_0}{2}$, but for the sake of clarity, we let the constant term be a_0 . However, now we can completely ignore the constant term, and instead in our Fourier series write $f(t)$ as

$$f(t) = \sum_{n=-\infty}^{\infty} c_n e^{2\pi int}.$$

4.2 Motivation for coefficients

Rather than the linear motivation for finding the coefficients of a Fourier series by showing that two trigonometric functions are orthogonal, we will find an alternative motivation for multiplying by a function and taking the integral.

Suppose we wish to find the value of c_k . We are given

$$f(t) = \dots + c_1 e^{2\pi it} + c_2 e^{4\pi it} + \dots + c_k e^{2\pi ikt} + \dots.$$

A common first step would be to isolate the c_k term by multiplying by $e^{-2\pi ikt}$. This gives us

$$f(t)e^{-2\pi ikt} = c_k + \sum_{n=-\infty, n \neq k}^{\infty} c_n e^{2\pi i(n-k)t}.$$

Now we wish to apply some operation such that all the other c_n terms become 0. Recall that $f(t)$ is periodic with period 1, and so is $e^{2\pi i(n-k)t}$. Additionally, e has a nice integral, so we may try

integrating both sides from 0 to 1 with respect to t . For some generic n with $n \neq k$,

$$\begin{aligned} \int_0^1 c_n e^{2\pi i(n-k)t} dt &= c_n \int_0^1 e^{2\pi i(n-k)t} dt \\ &= c_n \left[\frac{1}{2\pi i n - k} e^{2\pi i(n-k)t} \right]_0^1 \\ &= c_n (e^{2\pi i(n-k)} - e^0) \\ &= c_n(1 - 1) \\ &= 0. \end{aligned}$$

With $\int_0^1 c_k dt$, we simply get c_k . Thus,

$$c_k = \int_0^1 f(t) e^{-2\pi i k t} dt.$$

For the sake of notation, we define $\hat{f}(n) = c_n$ for all valid n . The Fourier coefficient the n th term in the Fourier series of $f(t)$ is

$$\hat{f}(n) = \int_0^1 f(t) e^{-2\pi i n t} dt.$$

One can show that regardless of the start and end of the interval used for integration, so long as the interval is of length 1, the Fourier coefficient still holds true. More specifically,

$$\hat{f}(n) = \int_h^{h+1} f(t) e^{-2\pi i n t} dt$$

for $h \in \mathbb{R}$. A commonly used form is

$$\hat{f}(n) = \int_{-1/2}^{1/2} f(t) e^{-2\pi i n t} dt.$$

4.3 Functions with other periods

Let's say some periodic function $f(t)$ has a period of T . We can let $g(t) = f(Tt)$ so that $g(t)$ has period 1. If we let $s = Tt$, then

$$f(s) = g(t) = \sum_{n=-\infty}^{\infty} c_n e^{2\pi i n t} = \sum_{n=-\infty}^{\infty} c_n e^{2\pi i n s / T}.$$

We can apply a similar method as the previous section to find that

$$\hat{f}(n) = \frac{1}{T} \int_{-T/2}^{T/2} f(s) e^{-2\pi i n s / T} ds.$$

Note that this is essentially just the original Fourier coefficient except scaled down by the period T . We can now replace s with t , and our new Fourier series is essentially the same as before.

Now that we have a more generalized form of the Fourier series, let's generalize it even further and consider when the period of the function is infinity.

5 Fourier Transform

The Fourier transform takes in any function, regardless if its periodic or not, and still expresses it as the sum of sine and cosine functions. To start, let's first just consider taking the limit of $\hat{f}(n)$ as T goes to infinity.

$$\lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} f(t) e^{-2\pi i n t / T} dt$$

However, notice that

$$\left| \int_{-T/2}^{T/2} f(t) e^{-2\pi i n t / T} dt \right| \leq \int_{-T/2}^{T/2} |f(t)| \left| e^{-2\pi i n t / T} \right| dt = \int_{-T/2}^{T/2} |f(t)| dt,$$

which is just an expression independent of n and T . As a result, the above limit will tend towards 0, which is not what we want. Therefore, we scale everything up by T . Additionally, note that n/T is discrete, as $n \in \mathbb{N}$. We let $s = n/T$ to include all real terms, and thus we get

$$\hat{f}(s) = \int_{-\infty}^{\infty} f(t) e^{-2\pi i s t} dt,$$

and we're done. Hence, the Fourier transform of a function $f(t)$ is defined to be

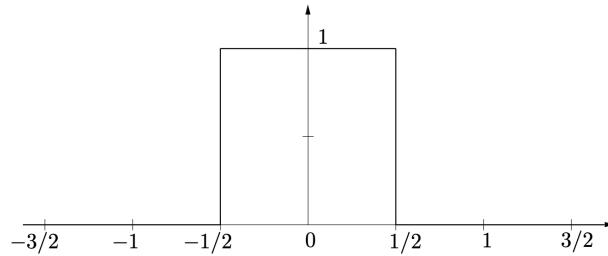
$$\hat{f}(s) = \int_{-\infty}^{\infty} f(t) e^{-2\pi i s t} dt.$$

Here's an example of how we can use the Fourier transform to express non periodic functions in terms of periodic functions.

The top hat function is defined as

$$\Pi(t) = \begin{cases} 1 & \text{if } |t| < 1/2 \\ 0 & \text{if } |t| \geq 1/2 \end{cases}$$

Perform the Fourier transform on $\Pi(t)$. As a hint, here is the graph of the top hat function.



If all goes well, the answer should be

$$\hat{\Pi}(s) = \frac{\sin \pi s}{\pi s}.$$

P-adic Numbers

Tanvir Ahmed

1 Introduction

This is a paper on p -adic numbers, a type of number that is of great relevance in number theory, and can also be thought of from an analytical perspective. All the sections up to section 5 focus on the number theoretic aspects of p -adics, while section 6 and on discuss them more analytically.

2 Bases

p -adic arithmetic works very similarly to arithmetic in different number bases, so we will first review bases. Typical numbers are written in base-10. The rightmost digit of a base-10 integer has a place value of 1, the digit directly to the left of that has a place value of 10, and so on. Furthermore, each of these digits are integers from 0 to 9. The value of a base-10 integer is then the sum of each digit multiplied by its place value. For example, $345 = 3 \cdot 100 + 4 \cdot 10 + 5 \cdot 1$. In general, if a number is written without a subscript for a base, it is written in base-10 unless it is clear otherwise. As a side note, base 2 is also called binary.

This can be generalized to other bases. In particular, if n is a positive integer, it can be uniquely expressed in base b as $a_k b^k + a_{k-1} b^{k-1} + \dots + a_1 b + a_0$ where $0 \leq a_i < b$ for $i = 0, 1, \dots, k$. While we will not formally prove the uniqueness of this representation, Section 2.1 goes over the general process of finding the base b representation of an integer. For ease of notation, n can be written here in base b as $\overline{a_k a_{k-1} \dots a_1 a_0}_b$. The overline typically isn't needed when expressing numbers whose digits aren't variables.

Notice how numbers in base b are analogous to numbers in base-10. Each digit is an integer between 0 and $b - 1$ inclusive and each digit's place value corresponds to some power of b . In base 10, this is the case as well because 10, 100, 1000, etc. are all powers of 10.

2.1 Arithmetic

Usual arithmetic for numbers in different bases is analogous to that of base 10. Here are some examples:

$$\begin{array}{r}
 \begin{array}{r}
 \begin{array}{r}
 1 \quad 1 \\
 2 \quad 4 \quad 5_7 \\
 + \quad 1 \quad 6 \quad 6_7 \\
 \hline
 4 \quad 4 \quad 4_7
 \end{array}
 \end{array}
 \qquad
 \begin{array}{r}
 \begin{array}{r}
 1 \quad 14 \\
 \cancel{2} \quad \cancel{4} \quad 6_7 \\
 - \quad 1 \quad 6 \quad 5_7 \\
 \hline
 0 \quad 5 \quad 1_7
 \end{array}
 \end{array}
 \qquad
 \begin{array}{r}
 \begin{array}{r}
 \begin{array}{r}
 \quad 2 \quad 4 \quad 5_7 \\
 \times \quad 1 \quad 6 \quad 5_7 \\
 \hline
 1 \quad 6 \quad 2 \quad 4_7 \\
 1 \quad 1 \quad 5 \quad 2 \quad 0_7 \\
 + \quad 2 \quad 4 \quad 5 \quad 0 \quad 0_7 \\
 \hline
 4 \quad 1 \quad 2 \quad 4 \quad 4_7
 \end{array}
 \end{array}
 \end{array}
 \end{array}$$

Notice how all remainders, carries, etc. are happening in the same base that the overall operation is taking place in.

2.2 Conversion Between Bases

Finally, we will look at conversions between bases. Let's say we want to convert the number 789 from base 10 to base 2. We will first find the last digit. In any base-2 number, if the last digit is 0, the whole number is even, and otherwise it is odd. This is because all other place values are divisible by 2. Therefore, simply by dividing 789 by 2 and finding the remainder, we get the last digit of 789 in binary, which is 1. However, if we keep the quotient of this division (which is 394) and divide by 2 again, we can find the second to last digit (which is 0). Here is the full process of converting 789 to binary.

$$\begin{aligned} 789 &= 2 \cdot 394 + 1 \implies \mathbf{1} \\ 394 &= 2 \cdot 197 + 0 \implies \mathbf{0} \\ 197 &= 2 \cdot 98 + 1 \implies \mathbf{1} \\ 98 &= 2 \cdot 49 + 0 \implies \mathbf{0} \\ 49 &= 2 \cdot 24 + 1 \implies \mathbf{1} \\ 24 &= 2 \cdot 12 + 0 \implies \mathbf{0} \\ 12 &= 2 \cdot 6 + 0 \implies \mathbf{0} \\ 6 &= 2 \cdot 3 + 0 \implies \mathbf{0} \\ 3 &= 2 \cdot 1 + 1 \implies \mathbf{1} \end{aligned}$$

At this point, we stop dividing because we have reached a quotient of 1, which is less than 2, so we can no longer divide it. We have to read this string of digits from the bottom to the top because we said the first division would give us the last digit, the second division would give us the second to last digit, and so on. Therefore, we have $789 = 100010101_2$. Using a similar process, we can convert numbers between any bases.

3 10-adics

The general idea of 10-adic numbers is taking a base-10 number and letting there be infinite digits going on to the left. Recall that when we use decimals, we let there be infinite digits going on to the right. If we have infinite digits going on to the left however, the place values of the digits can get infinitely large, and the number is no longer finite. However, we will ignore this problem for now and make some meaning out of these numbers with infinite digits to the left.

Some examples of 10-adic integers are $\dots 111$, $\dots 1212$, and $\dots 34345$. Notice that all these 10-adic integers have a sequence of digits which repeats infinitely to the left. We can write 10-adic integers with an infinitely repeating sequence of digits more concisely by putting a bar over the repeated part, just as we would for repeating decimals. Using this notation, the 3 previously mentioned 10-adic integers would be written as $\dots 1\overline{1}$, $\dots 12\overline{12}$, and $\dots 34\overline{34}5$, or $\dots \overline{1}$, $\dots \overline{12}$, and $\dots \overline{34}5$.

3.1 Arithmetic

Just like with regular base-10 integers, we can add 10-adic integers. When adding, we would just have to add digits infinitely to the left. For example, adding the two 10-adic integers $\dots\bar{1}$ and $\dots\bar{1}2$, we get:

$$\begin{array}{r} \dots \quad 1 \quad 1 \quad 1 \quad 1 \\ + \quad \dots \quad 1 \quad 2 \quad 1 \quad 2 \\ \hline \dots \quad 2 \quad 3 \quad 2 \quad 3 \end{array}$$

We could imagine doing this addition infinitely to the left, and we would always get pairs of the digits 2 and 3, so the result of this addition is the 10-adic integer $\dots\bar{2}3$. We can also perform multiplication as normal:

$$\begin{array}{r} \dots \quad 1 \quad 2 \quad 1 \quad 2 \quad 1 \quad 2 \\ \times \quad \dots \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \\ \hline \dots \quad 1 \quad 2 \quad 1 \quad 2 \quad 1 \quad 2 \\ \dots \quad 1 \quad 2 \quad 1 \quad 2 \quad 1 \quad 2 \quad 0 \\ \dots \quad 1 \quad 2 \quad 1 \quad 2 \quad 1 \quad 2 \quad 0 \quad 0 \\ + \quad \vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots \quad 0 \quad 0 \quad 0 \\ \hline \dots \quad 3 \quad 0 \quad 9 \quad 8 \quad 6 \quad 5 \quad 3 \quad 2 \end{array}$$

3.2 Negatives

Interestingly, in the 10-adics, negative signs are not needed. First of all, when we take a negative number x , we are really only finding the number y such that $x + y = 0$. Suppose we were trying to find the negative of 1 in the 10-adics. If the negative of 1 is $\dots\bar{a}_3\bar{a}_2\bar{a}_1\bar{a}_0$, then we know that $\dots\bar{a}_3\bar{a}_2\bar{a}_1\bar{a}_0 + 1 = 0$. Using this information, we can find $\dots\bar{a}_3\bar{a}_2\bar{a}_1\bar{a}_0$ digit by digit as follows:

$$\begin{array}{r} \dots \quad a_4 \quad a_3 \quad a_2 \quad a_1 \quad a_0 \\ + \quad \dots \quad 0 \quad 0 \quad 0 \quad 0 \quad 1 \\ \hline \dots \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \end{array}$$

For the last digit of the sum to be 0, a_0 must be 9. Putting $a_0 = 9$ into our addition gives:

$$\begin{array}{r} \dots \quad a_4 \quad a_3 \quad a_2 \quad a_1 \quad 9 \\ + \quad \dots \quad 0 \quad 0 \quad 0 \quad 0 \quad 1 \\ \hline \dots \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \end{array}$$

Once again, we must have $a_1 = 9$ due to the carry. Repeating this process, we get that all the digits of $\dots\bar{a}_3\bar{a}_2\bar{a}_1\bar{a}_0$ are 9, so $-1 = \dots999$ in the 10-adics. We can apply this procedure to any 10-adic number we want to get its negative. For example, you can try finding the negatives of $\dots\bar{2}369$ and $\dots\bar{2}598$. First, we will introduce a method that calculates negatives of 10-adics much easier.

To find the negative of a 10-adic integer $\dots\bar{a}_3\bar{a}_2\bar{a}_1\bar{a}_0$, first find $9 - a_i$ for each digit a_i . Since a_i is a digit and is in the range 0 to 9 inclusive, $9 - a_i$ will also be in that range, so it will also be a digit. Therefore, the number $\dots(9 - a_3)(9 - a_2)(9 - a_1)(9 - a_0)$ is a valid 10-adic (this is known as the 9's complement of our original number). When you add $\dots\bar{a}_3\bar{a}_2\bar{a}_1\bar{a}_0$ and $\dots(9 - a_3)(9 - a_2)(9 - a_1)(9 - a_0)$ digit, you get a 9 for every digit, so the sum of those two 10-adics is $\dots999 = -1$ as we have seen.

In other words:

$$\begin{aligned} \dots \overline{a_3 a_2 a_1 a_0} + \dots \overline{(9 - a_3)(9 - a_2)(9 - a_1)(9 - a_0)} &= -1 \\ \implies -(\dots \overline{a_3 a_2 a_1 a_0}) &= \dots \overline{(9 - a_3)(9 - a_2)(9 - a_1)(9 - a_0)} + 1 \end{aligned}$$

Thus, we have found the negative of $\dots \overline{a_3 a_2 a_1 a_0}$. It is 1 plus the 9's complement of $\dots \overline{a_3 a_2 a_1 a_0}$. Here's how we can use this method to find the negatives of the 2 examples mentioned earlier, $\dots \overline{2369}$ and $\dots \overline{2598}$.

For the first one, the 9's complement is $\dots \overline{7630}$. Adding 1 to this, we get $\dots \overline{7631}$.

For the second one, the 9's complement is $\dots \overline{7401}$. Adding 1 to this, we get $\dots \overline{7402}$.

Now that we can find the negative of any 10-adic integer, we can subtract any two 10-adic integers. For example, if we wanted to do $\dots \overline{2369} - (\dots \overline{2598})$, it would be equivalent to $\dots \overline{2369} + (-\dots \overline{2598})$. As we calculated earlier, $-\dots \overline{2598}$ is $\dots \overline{7402}$. Thus, our problem becomes $\dots \overline{2369} + (\dots \overline{7402})$, which becomes:

$$\begin{array}{r} \begin{array}{cccccccccccccccc} & & & 1 & 1 & & & 1 & 1 & 1 & 1 & & & 1 & 1 \\ \dots & 6 & 2 & 3 & 6 & 2 & 3 & 6 & 2 & 3 & 6 & 2 & 3 & 6 & 9 \end{array} \\ + \begin{array}{cccccccccccccccc} \dots & 7 & 4 & 7 & 4 & 7 & 4 & 7 & 4 & 7 & 4 & 7 & 4 & 9 & 8 \end{array} \\ \hline \dots & 3 & 7 & 1 & 0 & 9 & 8 & 3 & 7 & 1 & 0 & 9 & 8 & 6 & 7 \end{array}$$

Notice that we can now see a repetition in the sum up to 6 digits. Therefore, the desired answer to our original subtraction problem is $\dots \overline{37109867}$

3.3 Multiplicative Inverses

Some 10-adic integers also have a multiplicative inverse within the 10-adic integers. For example 1 and -1 ($= \dots 999$) are obviously multiplicative inverses of themselves. To explore this idea of multiplicative inverses, let us find the multiplicative inverse of 3 in the 10-adics.

First of all, multiplicative inverses are numbers such that when multiplied, result in 1. Therefore, we want to find $\dots \overline{a_3 a_2 a_1 a_0}$ such that $3(\dots \overline{a_3 a_2 a_1 a_0}) = 1$. We can now set up the multiplication as follows:

$$\begin{array}{r} \dots \quad a_3 \quad a_2 \quad a_1 \quad a_0 \\ \times \quad \dots \quad 0 \quad 0 \quad 0 \quad 3 \\ \hline \dots \quad 0 \quad 0 \quad 0 \quad 1 \end{array}$$

For the last digit of the product to be 1, $3a_0$ must have a remainder of 1 mod 10, which implies that $a_0 = 7$. With this value of a_0 , our multiplication then becomes:

$$\begin{array}{r} \begin{array}{cccc} & & & 2 \\ \dots & a_3 & a_2 & a_1 & 7 \end{array} \\ \times \quad \dots \quad 0 \quad 0 \quad 0 \quad 3 \\ \hline \dots \quad 0 \quad 0 \quad 0 \quad 1 \end{array}$$

This means $3a_1$ has a remainder of 8 mod 10, so $a_1 = 6$. When we plug this into the multiplication again, the next carry is going to be another 2, so a_2 will also be 6, and every digit besides a_0 will be 6. In other words, we get:

$$\begin{array}{r} \dots \quad 6 \quad 6 \quad 6 \quad 7 \\ \times \quad \dots \quad 0 \quad 0 \quad 0 \quad 3 \\ \hline \dots \quad 0 \quad 0 \quad 0 \quad 1 \end{array}$$

The digits $\dots 6667$ are very familiar. In fact, they are just the digits of $\frac{2}{3}$ expressed as a decimal, rounded. However, if we negative the 10-adic integer $\dots 6667$, we get $\dots 3333$, which are exactly the digits of $\frac{1}{3}$ in its decimal representation. This is not a coincidence, and we will investigate why in the next section.

3.4 (Non)existence of Multiplicative Inverses

Unlike in our above example, multiplicative inverses don't always exist in the 10-adics. For example, 10 doesn't have a 10-adic inverse. This is because no matter what you multiply by 10, there will always be a 0 in the ones place and not a 1.

In general, any 10-adic with a last digit of 0 will not have a multiplicative inverse. Furthermore, 10-adics with a last digit of 2 or 5 won't have multiplicative inverses. This is because when you multiply such a number by any other 10-adic number, the last digit of the product will retain either a factor of 2 or 5, and thus the last digit will not be 1. This is directly due to the fact that 2 and 5 are factors of 10.

Instead, what we need for multiplicative inverses is a last digit that has a multiplicative inverse mod 10. Multiplicative inverses are numbers that multiply to something congruent to 1 mod 10, such as the 3 and 7 from the example in the previous section. If in the 10-adic number $\dots a_3 a_2 a_1 a_0$, a_0 has a multiplicative inverse mod 10, then we can have the multiplicative inverse of the entire 10-adic number end in the multiplicative inverse of a_0 , allowing the product to have a final digit of 1.

As it turns out, a 10-adic number having a last digit that has a multiplicative inverse mod 10 is sufficient for it to have a 10-adic inverse, which we summarize in the following theorem:

Theorem 3.1. If in the 10-adic integer $a = \dots a_3 a_2 a_1 a_0$, a_0 has an inverse mod 10, $\dots a_3 a_2 a_1 a_0$ has an inverse in the 10-adics.

Proof. As discussed previously, the last digit of the inverse of a must be the inverse of a_0 mod 10, which we will denote as a_0^{-1} . Let the rest of the digits be denoted by b_i , making the inverse $b = \dots b_3 b_2 b_1 a_0^{-1}$. We now have to solve for the b_i 's. Here is the column multiplication for ab (b is the inverse):

$$\begin{array}{rcccccc} & & c_3 & c_2 & c_1 & & \\ & \dots & b_3 & b_2 & b_1 & a_0^{-1} & \\ \times & \dots & a_3 & a_2 & a_1 & a_0 & \\ \hline & \dots & 0 & 0 & 0 & 1 & \end{array}$$

Notice that all the carries in this multiplication are denoted by c_i . The first carry, c_1 is already determined by the values of a_0 and a_0^{-1} , so we can definitely use it to find b_1 .

For the second digit of the product to be 0, we must have $a_0 b_1 + a_1 a_0^{-1} + c_1 \equiv 0 \pmod{10}$. Solving for b_1 , we get $b_1 \equiv -a_0^{-1}(a_1 a_0^{-1} + c_1)$. a_0^{-1} exists by assumption, and we are also assuming we already know a_1 and c_1 . Now that we know b_1 , we can calculate c_2 . With a similar calculation as the one for b_1 , we can find that $b_2 \equiv -a_0^{-1}(a_2 a_0^{-1} + a_1 b_1 + c_2) \pmod{10}$. We can keep on doing this for each b_i , showing that the inverse of a does indeed exist. In fact, we have

$$b_i \equiv -a_0^{-1} \left(a_i a_0^{-1} + c_i + \sum_{k=1}^{i-1} a_k b_{i-k} \right) \pmod{10}.$$

□

4 Duality between 10-adics and Decimals

The main idea of this section is that we can relate 10-adics and decimals to each other and sometimes do algebra with the two by considering both as representing fractions. In other words, both (repeating) 10-adics and decimals both represent specific rational numbers.

4.1 Geometric Series

One formula that would be really useful if we could apply to the 10-adics is the infinite geometric series formula, which says:

$$1 + x + x^2 + \cdots = \sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$$

For real numbers, this only holds if $|x| < 1$. Otherwise, the sum doesn't converge. For example, this formula would suggest that $1 + 2 + 2^2 + \cdots = \frac{1}{1-2} = -1$, but this is ridiculous because the terms in the sum get infinitely large, and can't possibly converge to a finite value, much less a negative value. However, we do have the following proposition:

Proposition 4.1. If x is a number in any algebraic number system with commutative addition and multiplication, and the sum $1 + x + x^2 + \cdots$ converges to some value in that number system, then it converges to $\frac{1}{1-x}$.

Proof. Suppose that $1 + x + x^2 + \cdots$ converges to S . Then, we have that

$$\begin{aligned} S &= 1 + x + x^2 + \cdots = 1 + x(1 + x + x^2 + \cdots) = 1 + Sx \\ \implies S(1-x) &= 1 \\ \implies S &= \frac{1}{1-x} \end{aligned}$$

□

"Convergence" and "algebraic number system" here are used quite loosely, but since we are only applying this to the 10-adics (and p -adics later), this will suffice.

Now, here is the main claim that will allow us to use infinite geometric series for 10-adics:

Theorem 4.1. For a 10-adic integer x , the infinite geometric series $1 + x + x^2 + \cdots$ converges if and only if 10 divides x .

Here, 10 dividing x is equivalent to x having a 0 as its ones digit. In general, divisibility of a 10-adic by a regular positive integer isn't well-defined unless if it's divisibility by 1, 2, 5, or 10 because those are the factors of 10. For example, it doesn't make sense to ask if a 10-adic integer is divisible by 3.

Proof. Rather than providing a rigorous proof of the statement, we will sketch one with some particular examples.

Let's start with the backwards direction. Suppose x is 90, which is divisible by 10. Then, the geometric series $1 + x + x^2 + \cdots$ is $1 + 90 + 90^2 + 90^3 + \cdots = 1 + 90 + 810 + 729000 + \cdots$. Writing

this using column addition, we get:

$$\begin{array}{r}
 9 \ 0 \\
 8 \ 1 \ 0 \ 0 \\
 7 \ 2 \ 9 \ 0 \ 0 \ 0 \\
 + \ \vdots \ \vdots \ \vdots \ \vdots \ \vdots \ \vdots \\
 \hline
 \end{array}$$

The last digit of the final sum is pretty clearly 0 because every term has 0 as its last digit. The second to last digit is pretty clearly 9 because after adding the 9 from 90, the second to last digit of every other number is 0. In fact, for every digit of the final sum, you only have to sum a finite number of digits from the summands in the geometric series. Thus, we can calculate each digit of the final sum if we wanted to, so it must converge.

For the forwards direction, we will do a proof by contrapositive. Suppose that x is a 10-adic integer that isn't divisible by 10, say 2. Then, the geometric series we are interested in is $1 + 2 + 2^2 + \dots$, which is $1 + 2 + 4 + 8 + 16 + 32 + 64 + \dots$. Let's consider only the last digit of this series. To do this, we only need to consider the last digits of each of the terms in the geometric series, so the last digit of $1 + 2 + 4 + 8 + 16 + 32 + 64 + 128 + 256 + \dots$ (if it exists) is the same as that of $1 + 2 + 4 + 8 + 6 + 2 + 4 + 8 + 6 \dots$.

Notice that these last digits repeat. In this case, the repetition is 2, 4, 8, 6. In general, these last digits will always repeat because of Fermat's Little Theorem. Now, if we take the last digits of the partial sums of the series $1 + 2 + 4 + 8 + 6 + \dots$, it never becomes constant, because the sequence 1, 2, 4, $\dots$ never becomes 0. Instead, the last digit only repeats between a couple of values. Therefore, the last digit of the series doesn't converge to one value, so the entire series also can't converge. $\square$

With this, we can calculate things like $1 + 90 + 90^2 + \dots$. We just get $\frac{1}{1-90} = -\frac{1}{89}$. What this means is that $89(1 + 90 + 90^2 + \dots) = -1$. Conversely, we can also use this to calculate the reciprocal of 89 in the 10-adics. It is just $-(1 + 90 + 90^2 + \dots)$. Like we said before, all digits of $1 + 90 + 90^2 + \dots$ can be calculated with enough work, so $\frac{1}{89}$ can be calculated as well.

In the last section, we also saw that $\frac{1}{3} = \dots 6667 = -(\dots 333)$. Notice that

$$\begin{aligned}
 \dots 333 &= 3(\dots 111) \\
 &= 3(1 + 10 + 10^2 + \dots) \\
 &= 3\left(\frac{1}{1-10}\right) \\
 &= 3\left(-\frac{1}{9}\right) \\
 &= -\frac{1}{3}.
 \end{aligned}$$

Therefore, $\dots 6667 = -(\dots 333) = -(-\frac{1}{3}) = \frac{1}{3}$, which is the same as what we saw previously.

4.2 Calculating Multiplicative Inverses

Using the geometric series formula, we can calculate the reciprocal of any finite 10-adic number. We saw in the previous section how $\frac{1}{89} = -\frac{1}{1-90} = -(1 + 90 + 90^2 + \dots)$. However, we can't apply

4.4 Euler's Doubly Infinite Sum

Calculating fractions in the 10-adics like we did for $\frac{1}{7}$ is quite time-consuming, but there is a faster way. We have seen that repeating 10-adics can be written as fractions, and we know that repeating decimals can also be written as fractions. Therefore, it makes some sense to be able to add 10-adics with decimals; we can just add the corresponding fractions.

Before we start actually adding 10-adics to decimals, let us use the following notation: let $\lambda(x)$ be the fraction corresponding to a repeating 10-adic number x and $\mu(x)$ the fraction corresponding to a repeating decimal. Then, we have the following result:

Theorem 4.2. If $x \neq 0$ is an integer divisible by 10, then

$$\mu\left(\sum_{n=1}^{\infty} \frac{1}{x^n}\right) + \lambda\left(\sum_{n=0}^{\infty} x^n\right) = \mu\left(\cdots + \frac{1}{x^2} + \frac{1}{x}\right) + \lambda(1 + x + x^2 + \cdots) = 0$$

Proof. Since x is an integer divisible by 10, $|x| \geq 10$, so $|\frac{1}{x}| < 1$, allowing us to use the geometric series formula for the first sum, giving:

$$\mu\left(\sum_{n=1}^{\infty} \frac{1}{x^n}\right) = \mu\left(\frac{1}{1 - \frac{1}{x}} - 1\right) = \mu\left(\frac{x}{x-1} - 1\right) = \frac{1}{x-1}.$$

Furthermore, since x is divisible by 10, we can use the geometric series formula from Section 4.1 to obtain

$$\lambda\left(\sum_{n=0}^{\infty} x^n\right) = \lambda\left(\frac{1}{1-x}\right) = \frac{1}{1-x}.$$

Adding the two together, we get

$$\frac{1}{x-1} + \frac{1}{1-x} = \frac{1}{x-1} - \frac{1}{x-1} = 0$$

□

The way that Theorem 4.4 is written is mostly just a way to formalize the following, known as Euler's Doubly Infinite Identity:

$$\sum_{n=-\infty}^{\infty} x^n = \cdots + \frac{1}{x^2} + \frac{1}{x} + 1 + x + x^2 + \cdots = 0$$

One way you can "prove" this is as follows:

$$\begin{aligned} S &= \cdots + \frac{1}{x^2} + \frac{1}{x} + 1 + x + x^2 + \cdots \\ S &= \cdots + x \cdot \frac{1}{x^2} + x \cdot \frac{1}{x} + x \cdot 1 + x \cdot x + x \cdot x^2 + \cdots \\ S &= x(\cdots + \frac{1}{x^2} + \frac{1}{x} + 1 + x + x^2 + \cdots) = Sx \\ 0 &= Sx - S = S(x-1) \implies S = 0 \end{aligned}$$

However, this makes many assumptions about convergence. For example, it would imply that the sum $\cdots + \frac{1}{2^2} + \frac{1}{2} + 1 + 2 + 2^2 + \cdots = 0$, even though it obviously doesn't even converge. Furthermore,

one has to define what it means to add a 10-adic and decimal together when writing Theorem 4.4 and what it means for the sum to be 0. The statement of Theorem 4.4, however, avoids this.

On the other hand, Theorem 4.4 motivates us to define numbers with both a 10-adic part and a decimal part. Such a number would take the form $\dots a_3 a_2 a_1 a_0 . b_1 b_2 b_3 \dots$. In this light, Theorem 4.4 is saying that $\dots \overline{a_n a_{n-1} \dots a_1 a_0} . \overline{a_n a_{n-1} \dots a_1 a_0} \dots = 0$ where the overline indicates repetition and the $a_n a_{n-1} \dots a_1 a_0$ is the repeated sequence.

As an example, take $\overline{3434.3434}$. This is equal to $34(\dots \overline{0101.0101} \dots) = 34(\dots + \frac{1}{100^2} + \frac{1}{100} + 1 + 100 + 100^2) = 0$. Therefore, $\dots \overline{3434} = -0.\overline{3434} \dots = -\frac{34}{99}$. In a similar fashion, we can calculate $\frac{1}{7}$ as we teased earlier in the section. In particular, we know $\frac{1}{7} = 0.\overline{142857}$ as a decimal, and that $\overline{142857.142857} = 0$, so $\overline{142857} = -0.142857 = -\frac{1}{7}$. Therefore, $\frac{1}{7} = -\overline{142857} = \overline{2857143}$

5 Orders, Primitive Roots, and Decimal Expansions

Now that we know that repeating 10-adics, repeating decimals, and fractions are all related to each other, the question arises as to how we can tell how long the repeating part of a decimal or 10-adic representation of a fraction is. To do that, we need to use the notion of order mod n .

5.1 Orders and Primitive Roots

Mod n for any positive integer n , if you continually add any number to itself, you eventually get $0 \bmod n$, the additive identity. For example, $6 + 6 + 6 + 6 \equiv 0 \bmod 8$. If we instead consider continually multiplying a number itself mod n , the situation is slightly different and a little more interesting. To parallel the case where we continually added a number to itself and got the additive identity, we would expect repeated multiplication to eventually give us $1 \bmod n$, the multiplicative identity, and usually it does. For example, consider $3 \bmod 7$:

$$\begin{aligned} 3^2 &\equiv 3 \cdot 3 \equiv 2 \bmod 7 \\ 3^3 &\equiv 2 \cdot 3 \equiv 6 \bmod 7 \\ 3^4 &\equiv 6 \cdot 3 \equiv 4 \bmod 7 \\ 3^5 &\equiv 4 \cdot 3 \equiv 5 \bmod 7 \\ 3^6 &\equiv 5 \cdot 3 \equiv 1 \bmod 7 \end{aligned}$$

Since 3^6 is the lowest power of 3 congruent to $1 \bmod 7$, we say that the order of $3 \bmod 7$ is 6. The formal definition of an order is as follows:

Definition 5.1. If $\gcd(a, n) = 1$, then the order of $a \bmod n$ is the smallest positive integer k such that $a^k \equiv 1 \bmod n$

The reason we must impose the relatively prime condition in the definition is because numbers fail to have an order otherwise. For example, consider $2 \bmod 10$:

$$\begin{aligned} 2^2 &\equiv 2 \cdot 2 \equiv 4 \bmod 10 \\ 2^3 &\equiv 4 \cdot 2 \equiv 8 \bmod 10 \\ 2^4 &\equiv 8 \cdot 2 \equiv 6 \bmod 10 \\ 2^5 &\equiv 6 \cdot 2 \equiv 2 \bmod 10 \end{aligned}$$

At this point, we have come back to $2 \bmod 10$, and multiplying by 2 anymore will just keep us in this loop. The problem here is that 2 and 10 aren't relatively prime. Therefore, the factor of 2 in 2 is never lost mod 10, and since 1 does not have a factor of 2, no power of 2 can be 1 mod 10. On the other hand, if two positive integers a and n are relatively prime, then a does have an order mod n , which is due to Euler's Theorem, which states:

Theorem 5.1. If $\gcd(a, n) = 1$, then $a^{\phi(n)} \equiv 1 \pmod{n}$ where $\phi(n)$ is the Euler Totient Function, defined by the number of positive integers less than or equal to n relatively prime to n .

Euler's Theorem also leads to the following proposition:

Proposition 5.1. If $\gcd(a, n) = 1$ and m is the order of $a \bmod n$, then $m | \phi(n)$.

Proof. we have $a^m \equiv a^{\phi(n)} \equiv 1 \pmod{n}$. We can write $\phi(n)$ in terms of a quotient remainder when dividing by m . In particular, we can let $\phi(n) = mk + r$ where $0 \leq r < m$. Therefore, $a^{\phi(n)} \equiv a^{mk+r} \equiv (a^m)^k a^r \equiv a^r \equiv 1 \pmod{n}$. If $r > 0$, then m is no longer the smallest positive integer such that $a^m \equiv 1 \pmod{n}$, so m would not be the order of a in that case. Therefore, we must have $r = 0$, and thus $\phi(n) = mk \implies m | \phi(n)$ $\square$

When the order of $a \bmod n$ and $\phi(n)$ are equal, then we say that a is a primitive root mod n . One example of a primitive root is $2 \bmod 11$. Note that $\phi(11) = 11 - 1 = 10$ because 11 is a prime.

$$\begin{aligned} 2^1 &\equiv 2 \pmod{11} \\ 2^2 &\equiv 4 \pmod{11} \\ 2^3 &\equiv 8 \pmod{11} \\ 2^4 &\equiv 5 \pmod{11} \\ 2^5 &\equiv 10 \pmod{11} \\ 2^6 &\equiv 9 \pmod{11} \\ 2^7 &\equiv 7 \pmod{11} \\ 2^8 &\equiv 3 \pmod{11} \\ 2^9 &\equiv 6 \pmod{11} \\ 2^{10} &\equiv 1 \pmod{11} \end{aligned}$$

Notice how each number from 1 to 10 is included in the list of powers of 2 mod 11. This should make sense because if two powers of 2, say 2^a and 2^b were congruent such that $1 \leq a < b \leq 10$ and $2^a \equiv 2^b \pmod{11}$, then we would have $2^{b-a} \equiv 1 \pmod{11}$, and since $0 < b - a < 10$, we would have that 10 is not actually the order of 2 mod 11. Therefore, primitive roots, if they exist, can be thought of as generating all the numbers mod n that are relatively prime to n by multiplication.

5.2 Repeats in Decimal Expansions

So, why are orders and primitive roots important to repeating 10-adics and decimals? It is because of the following theorem:

Theorem 5.2. If $\gcd(n, 10) = 1$ and m is the order of 10 mod n , then the repeating part of the decimal expansion of $\frac{1}{n}$ has m digits.

Notice how the condition of relatively primeness is needed here to ensure that 10 has an order mod n .

Proof. We will demonstrate a particular example rather than providing a rigorous proof. Consider $\frac{1}{7}$. We can obtain its decimal expansion via the long division below:

$$\begin{array}{r} 0.\overline{142857} \\ 7 \overline{)1.000000} \\ \underline{7} \\ 30 \\ \underline{28} \\ 20 \\ \underline{14} \\ 60 \\ \underline{56} \\ 40 \\ \underline{35} \\ 50 \\ \underline{49} \\ 1 \end{array}$$

What we want to focus on are the results of each subtraction step, which includes 3, 2, 6, 4, 5, and 1 in this case. We obtain the first difference 7, by subtracting 10 by the largest multiple of the 7 less than 10, which is $10\%7$ (% indicates remainder). Similarly, the next number 2, is the remainder when 3 is multiplied by 10 and then divided by 7, which is $((10\%7) \cdot 10)\%7$. By properties of modular arithmetic, we can write this as $10^2\%7$. Similarly, the next remainder is $10^3\%7$, and so on.

When the remainder becomes 1 again, the decimal starts to repeat because we restart the same division problem as the one we started with. By definition, the number of remainders we go through before we get to 1 is the order of 10 mod 7, and each remainder corresponds to one of the digits in the repeating part of the decimal expansion of $\frac{1}{7}$ $\square$

Since we now know how many digits are in the repeating parts of the decimal representation of $\frac{1}{7}$, we can also find that information for the 10-adic representation. In particular, if $0.\overline{142857} = \frac{1}{7}$, then $\overline{142857} = -\frac{1}{7}$, and $\frac{1}{7} = \overline{2857143}$. Negating $\overline{142857}$ did not have any effect on the number of digits on the repeating part of the number; it only added an extra digit not part of the repeating part. Therefore, 5.2 is also true for 10-adics, not just decimals.

5.3 Some Example Problems

The following 2 problems demonstrate how useful n -adics can be. Here, 2-adics will be used rather than 10-adics, but they utilize the same concepts we've been discussing.

Problem 1. (AMC 12B 2022 #23) Let $x_0, x_1, x_2, \dots$ be a sequence of numbers, where each x_k is either 0 or 1. For each positive integer n , define

$$S_n = \sum_{k=0}^{n-1} x_k 2^k$$

Suppose $7S_n \equiv 1 \pmod{2^n}$ for all $n \geq 1$. What is the value of the sum

$$x_{2019} + 2x_{2020} + 4x_{2021} + 8x_{2022}?$$

Solution. Observe that the S_n 's define a 2-adic integer because in binary, the S_n 's essentially define infinitely long string of binary digits going off to the left. Let's call this 2-adic number S . In other words, S is the (2-adic) limit of S_n as n goes to ∞ . Furthermore, because S_n has the property that $7S_n \equiv 1 \pmod{2^n}$, S is the 2-adic inverse of $7 = 111_2$. In other words, $S = \frac{1}{111_2}$. Doing binary long division, we can find that $\frac{1}{111_2} = 0.\overline{001}_2$. This makes sense because the order of $2 \pmod{7}$ is 3, so by Theorem 5.2 applied to 2-adics, we should expect to have some factor of 3 as the number of digits in the repeating part. Now, converting to a 2-adic, we get $S = \frac{1}{111_2} = -\overline{001}_{1.2} = \overline{011}_{1.2}$. Now, we essentially have all the x_k 's embedded into S . In particular, x_0 is the last digit of S , x_1 is the second last, and so on. Finally, if $k > 0$, we have the following:

$$x_k = \begin{cases} 1 & k \equiv 1 \pmod{3} \\ 1 & k \equiv 2 \pmod{3} \\ 0 & k \equiv 0 \pmod{3} \end{cases}$$

This is simply from reading off the repeated digits. Finally, we have $(x_{2019}, x_{2020}, x_{2021}, x_{2022}) = (0, 1, 1, 0)$, giving an answer of $2 + 4 = \boxed{6}$.

Problem 2. (AIME II 2023 #15) For each positive integer n let a_n be the least positive integer multiple of 23 such that $a_n \equiv 1 \pmod{2^n}$. Find the number of positive integers n less than or equal to 1000 that satisfy $a_n = a_{n+1}$.

Solution. Since each of the a_i 's are multiples of 23, we can let $a_i = 23b_i$ for $b_i \in \mathbb{N}$. Since b_i is the inverse of $23 \pmod{2^i}$, like with the previous problem, we can say that the b_i 's converge to the 2-adic inverse of 23. Call this limit b , so $b = \frac{1}{23} = \frac{1}{10111_2}$ in the 2-adics. We proceed by first doing binary long division to find $\frac{1}{10111_2} = 0.\overline{00001011001}_2$. Therefore, as a 2-adic, $\frac{1}{10111_2} = -\overline{00001011001}_{1.2} = \overline{01111010011}_{1.2}$. Now, each b_i is embedded in this 2-adic number as the i rightmost digits. For example, $b_5 = 00111_2$.

If we have the condition that $a_n = a_{n+1}$ as in the problem statement, then $b_n = b_{n+1}$. This can only happen if the $n + 1$ st digit of b is 0. In other words, we are looking for the number of zeros in b up to and including the 1001st place (1001 because $n \leq 1000 \implies n + 1 \leq 1001$). Each repeated segment of b has a total of 11 digits, 4 of them being 0. One can check that there are 90 such full-length repeated segments before the 10001st place, and 9 left after that. The 90 repeated segments give $90 \cdot 4$ zeros, and the 9 remaining digits give 3 more zeros (counting through the repeated segment from the right). Thus, we have a total of $90 \cdot 4 + 3 = \boxed{363}$.

6 P-adics and Convergence

6.1 Why the 10-adics Go Wrong

Consider taking 5, squaring it, and doing so repeatedly. Here are the first few results of this process:

$$\begin{aligned}5^2 &= 25 \\25^2 &= 625 \\625^2 &= 390625 \\390625^2 &= 152587890625\end{aligned}$$

Notice how for the most part, each time we square a number, the resulting number has most of the digits of the original number at the end. The only exception is 390625 where its square has 890625. Therefore, we will proceed with only the part that matches with 390625, and keep on proceeding in this way:

$$\begin{aligned}90625^2 &= 8212890625 \\8212890625^2 &= 67451572418212890625 \\67451572418212890625^2 &= 4549714621689417981542646884918212890625\end{aligned}$$

As we keep on squaring, we get a number n^2 which has more and more common digits with n . In a weird sense, we are getting a number which is equal to its own square (which is not 0 or 1). In the real numbers, this doesn't make sense. The problem here is that the 10 isn't a prime number, so the 10-adics contain so-called zero divisors. In particular, zero divisors are two numbers, neither of which are 0 but whose product is 0.

If we have a solution to the equation $n^2 = n$ as is sort of happening above, we would have $n^2 - n = n(n - 1) = 0$. However, since $n \neq 0, 1$, neither n or $n - 1$ is 0, so n and $n - 1$ are zero divisors. For those interested, below is a rigorous justification that such an n exists in the 10-adics

We will prove that for every positive integer k , there exists a positive integer n_k such that $n_k(n_k - 1) \equiv 0 \pmod{10^k}$. We can let $n_k = a \cdot 5^k$ such that $a \equiv (5^k)^{-1} \pmod{2^k}$. Therefore, $n_k - 1 = a \cdot 5^k - 1 \equiv (5^k)^{-1} \cdot 5^k - 1 \equiv 1 - 1 \equiv 0 \pmod{2^k}$. Therefore, n_k is a number such that its last k digits are the same as the last k digits of its square. Therefore, when taking the limit of n_k as k goes to infinity, we get a 10-adic number which squares to itself. The sequence we were generating above is the same as this sequence n_k except that it skips a few terms.

6.2 More Definitions/Theorems

p -adics are defined analogously to how the 10-adics are defined, except that the digits are between 0 and $p - 1$ inclusive and that p is a prime. We will denote them by $\mathbb{Z}_p$. Addition and multiplication also work in the same column-by-column way as for 10-adics. Now, many things of the things we've proven for 10-adics hold for p -adics as well, since 10 is not really a special number. Here is a list of some those theorems but applied to p -adics:

Theorem 6.1. If in the p -adic number $a = \dots a_3a_2a_1a_0$, a_0 has an inverse mod p , then a has an inverse in the p -adic integers. In other words, if $a_0 \neq 0$, then a has an inverse in the p -adic integers.

Theorem 6.2. For a p -adic integer x , the infinite geometric series $1 + x + x^2 + \dots$ converges to a p -adic integer if and only if its last digit is 0.

Theorem 6.3. For a p -adic integer x , let $\lambda(x)$ be the fraction corresponding to the repeating p -adic number x and $\mu(x)$ the fraction corresponding to the real number x , written in base p .

If $x \neq 0$ is an integer divisible by p , then

$$\mu\left(\sum_{n=1}^{\infty} \frac{1}{x^n}\right) + \lambda\left(\sum_{n=0}^{\infty} x^n\right) = \mu\left(\dots + \frac{1}{x^2} + \frac{1}{x}\right) + \lambda(1 + x + x^2 + \dots) = 0$$

Theorem 6.4. If $\gcd(n, p) = 1$ and m is the order of p mod n , then the repeating part of the expansion of $\frac{1}{n}$ in base p and in the p -adics has m digits.

6.3 P-adic Field

Of particular interest is Theorem 6.2 because it guarantees that there exists an inverse to a p -adic integer as long as the last digit is not 0, whereas in the 10-adics, inverses don't exist even if the last digit is 2 or 5. Ideally, we would like to have p -adic inverses for all elements, and to do so, we have to introduce the field of p -adics.

In particular, the p -adic field $\mathbb{Q}_p$ consists of all numbers with a p -adic part to the left of the decimal point and a finite part after the decimal point. For example, this would include numbers like $\overline{356}.84576_{11}$ in the 11-adics. The addition of the decimal point effectively allows us to multiply by negative powers of p . This also gives us inverses for p -adics that had 0's at the end. These 0's represent some power of p , say p^k so by just multiplying by p^{-k} , we have a number that does have an inverse. For example, $(2^4)^{-1} = 10000_2^{-1} = 0.0001_2 = 2^{-4}$.

Overall, this makes $\mathbb{Q}_p$ a field, which is a set of an algebraic structure with addition and multiplication such that addition and multiplication obey the usual commutative associative, and distributive laws, and such that there exist additive and multiplicative inverses. A more precise definition of a field can be found here: [https://en.wikipedia.org/wiki/Field_\(mathematics\)](https://en.wikipedia.org/wiki/Field_(mathematics)).

Another consequence of $\mathbb{Q}_p$ being a field is that it doesn't have zero divisors. Suppose that $x, y \neq 0$ are p -adic numbers. When multiplying them, the last nonzero digit of their product will be the product of the last nonzero digits of x and y mod p . Since p is a prime, this cannot be 0.

While all rational numbers are embedded within this field, this field also contains irrational numbers; a number would be irrational in the p -adics if it never repeats. For example, $\dots 56295141.3_{11}$ where the digits are the (base 10) digits of π reversed, is irrational. This is because every rational number in the p -adics does repeat.

6.4 P-adic Absolute Value

Just like there is an absolute value for the real numbers, there is a corresponding absolute value for each p -adic field. In particular, the p -adic absolute value of x is denoted by $|x|_p$ and is defined to be $\frac{1}{p^k}$ where k is the place value of the rightmost nonzero digit of x . For example, $|1011000|_2 = \frac{1}{2^3}$ and $|10100.11001|_2 = \frac{1}{2^{-5}} = 2^5$. In other words, if $x = p^k y$ where y is a p -adic whose units digit is nonzero and has nothing to the right of the decimal point, then $|x|_p = \frac{1}{p^k}$. Also the p -adic absolute

value of 0 is defined to be 0. Using this absolute value, we can define the p -adic metric between two p -adic numbers x and y as the absolute value of their differences, or $|x - y|_p$. In general terms, a metric is a notion of distance defined a set called a metric space. More details can be found here: https://en.wikipedia.org/wiki/Metric_space. With this, we have the following proposition:

Proposition 6.1. The following are true for a p -adic absolute value where x and y are p -adic numbers:

- $|x - y|_p \geq 0$ with equality if and only if $x = y$
- $|x - y|_p = |y - x|_p$
- $|x + y|_p \leq \max(|x|_p, |y|_p) \leq |x|_p + |y|_p$
- $|xy|_p = |x|_p |y|_p$

Proof.

- The first statement is trivial because the $|\cdot|_p$ is always positive by the way it is defined.
- Suppose $x - y = p^k z$ where z has a nonzero units digit and has nothing to the right of the decimal point. Then, $|x - y|_p = \frac{1}{p^k}$. Then, when taking the negative of z to $-z$ will also have a nonzero units digit, so $|y - x|_p = |-(x - y)|_p = |p^k(-z)|_p = \frac{1}{p^k} = |x - y|_p$.
- Suppose $x = p^k z$ and $y = p^j w$ where z and w have nonzero units digits and nothing to the right of the decimal point, so $|x|_p = \frac{1}{p^k}$ and $|y|_p = \frac{1}{p^j}$. Suppose $k \leq j$, so $|x|_p \geq |y|_p$ and $\max(|x|_p, |y|_p) = |x|_p$. When adding x and y , we get

$$x + y = p^k z + p^j w = p^k (z + p^{j-k} w).$$

Since $j - k \leq 0$, $z + p^{j-k} w$ has nothing to the right of the decimal point. If $z + p^{j-k} w$ has a zeros at the end of it then, $|x + y|_p = \frac{1}{p^{k+a}} \leq \frac{1}{p^k} = \max(|x|_p, |y|_p)$.

- Suppose $x = p^k z$ and $y = p^j w$ where z and w have nonzero units digits and nothing to the right of the decimal point. Then, their product, $xy = p^{k+j} zw$. zw also has a nonzero units digit, so $|xy|_p = \frac{1}{p^{k+j}} = |x|_p |y|_p$.

□

Notice that two numbers are close together by the p -adic metric when they have many digits in common, because then their difference will have many 0's at the end of it, resulting in a small absolute value. For example, $|22232323 - 32332323|_5 = |(222_5 - 323_5) \cdot 5^5|_5 = \frac{1}{5^5}$, which is pretty small.

6.5 Convergence

With an absolute value, we can define convergence for sequences of p -adic numbers. In particular, if $a_0, a_1, \dots$ is a sequence of p -adic numbers, we say that the sequence converges to the p -adic number a and write $\lim_{n \rightarrow \infty} a_n$ if $\forall \epsilon > 0$, we can find an $N > 0$ such that for all $n > N$, $|a_n - a| < \epsilon$. In other words, the terms in the sequence gets arbitrarily close to a p -adically.

We've already seen one example of p -adic convergence with Theorem 4.1. In this case, we saw the convergence of a series rather than a sum. The difference is that the sequence of partial sums was what was converging, where the N th partial of sum of the series $\sum_{n=0}^{\infty} a_n$ is $\sum_{n=0}^N a_n$. In other words, we had the following:

$$\sum_{n=0}^{\infty} \frac{1}{x^n} = \lim_{N \rightarrow \infty} \sum_{n=0}^N \frac{1}{x^n} = \frac{1}{1-x}$$

where x was divisible by 10 in the 10-adics (and analogously, by p in the p -adics).

This is where p -adic series differ very interestingly from real series. For example, consider the harmonic series for the real numbers, which is:

$$\sum_{n=1}^{\infty} \frac{1}{n} = 1 + \frac{1}{2} + \frac{1}{3} + \dots$$

It can be shown that this doesn't converge, by the following argument:

$$\begin{aligned} & 1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) + \left(\frac{1}{5} + \dots + \frac{1}{8}\right) + \left(\frac{1}{9} + \dots + \frac{1}{16}\right) + \dots \\ & \geq 1 + \frac{1}{2} + \frac{2}{4} + \frac{4}{8} + \frac{8}{16} + \dots \\ & = 1 + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \dots \rightarrow \infty \end{aligned}$$

If we compare this with a series that does converge, such as the geometric series, we notice that for both series, the terms get smaller and smaller; in particular, their absolute values go to 0. In fact, for any convergent series in the real numbers, the absolute values of the sequence must converge to 0, which can be summarized in the following proposition:

Proposition 6.2. If $a_0, a_1, \dots, a_n$ is a sequence of real numbers such that $\sum_{n=0}^{\infty} a_n = a$ for some $a \in \mathbb{R}$, then the limit $\lim_{n \rightarrow \infty} |a_n| = 0$.

Proof. The convergence of the series $\sum a_n$ means that the limit of the partial sums converges. In particular, we have:

$$a = \lim_{N \rightarrow \infty} \sum_{n=0}^N a_n = \lim_{N \rightarrow \infty} \left(a_N + \sum_{n=0}^{N-1} a_n \right) = \lim_{N \rightarrow \infty} a_N + \lim_{N \rightarrow \infty} \sum_{n=0}^{N-1} a_n = \lim_{N \rightarrow \infty} a_N + a.$$

Reading off the far left and right hand sides of this equation and subtracting a from both sides, we get $\lim_{N \rightarrow \infty} a_N = 0$. Then, taking the absolute value on both sides, we get $\lim_{N \rightarrow \infty} |a_N| = |0| = 0$, the desired result. $\square$

However, as the harmonic series shows, the converse to Proposition 6.2 is not true; not all sequences whose absolute values converge to 0 have a convergent sequence. However, the converse is true for p -adics. The justification for this is essentially the same as that of the justification for the backwards direction of Theorem 4.1, but here is the statement of the theorem anyway.

Theorem 6.5. Let $a_0, a_1, \dots$ be a sequence of p -adic numbers. Then, $\lim_{n \rightarrow \infty} a_n = 0$ if and only if $\sum_{n=0}^{\infty} a_n$ converges.

The if and only if part comes from the fact that Proposition 6.2 is also true for p -adics.

6.6 Exponential Function

In the real numbers, one very important series is the one corresponding to the exponential function e^x , also known as $\exp(x)$. We will see if this makes sense for p -adics. However, to avoid ambiguity, since e is not necessarily defined in the p -adics, we will only refer to the exponential function as $\exp(x)$. It is defined as follows:

$$e^x = \exp(x) = \sum_{n=0}^{\infty} \frac{x^n}{n!}.$$

For real numbers x , this series always converges, but that is not so for the p -adics. For example, consider $\exp(2)$ in the 3-adics. We get the sum

$$\sum_{n=0}^{\infty} \frac{2^n}{n!} = 1 + 2 + \frac{2^2}{2!} + \frac{2^3}{3!} + \dots$$

No term of the sum has a positive power of 3; in fact, many of them contain negative powers of 3. This correlates to the fact that each term doesn't have any 0's in the 3-adics, which means that when considering the partial sums, ones digits (and basically all the other digits) keep changing, and thus this sum can't converge by Theorem 6.5 (the p -adic absolute values are always greater than or equal to 1)

On the other hand, if we consider $\exp(3)$ in the 3-adics, it is different. We get the sum:

$$\sum_{n=0}^{\infty} \frac{3^n}{n!} = 1 + 3 + \frac{3^2}{2!} + \frac{3^3}{3!} + \dots$$

Now, all the terms (besides the 1) have a positive power of 3, and in fact each successive term has a higher power of 3. We can count the number of powers of 3's in the term $\frac{3^n}{n!}$ by the difference in the powers of 3 in the numerator and denominator. The number of powers of 3 in the numerator is n , and the number of powers of 3 in the denominator is well known to be

$$\left\lfloor \frac{n}{3} \right\rfloor + \left\lfloor \frac{n}{3^2} \right\rfloor + \left\lfloor \frac{n}{3^3} \right\rfloor + \dots \leq \frac{n}{3} + \frac{n}{3^2} + \frac{n}{3^3} + \dots = \frac{n}{2}$$

Overall, the difference is greater than or equal to $n - \frac{n}{2} = \frac{n}{2}$, which is increasing.

Therefore, $\exp(3)$ converges in the 3-adics, and with a similar analysis, $\exp(6)$, $\exp(9)$, etc also exist in the 3-adics. In general, for an odd prime p and an integer n , $\exp(n)$ in the p -adics exists if and only if $p|n$. The reason that $p = 2$ is an exception is that when finding the number of factors of 2 in $n!$, we get

$$\left\lfloor \frac{n}{2} \right\rfloor + \left\lfloor \frac{n}{2^2} \right\rfloor + \left\lfloor \frac{n}{2^3} \right\rfloor + \dots \leq \frac{n}{2} + \frac{n}{2^2} + \frac{n}{2^3} + \dots = n$$

Therefore, the p -adic absolute values of terms in $\exp(2)$ (of the form $\frac{2^n}{n!}$) don't actually have a limit of 0, and thus don't converge.

6.7 Natural Logarithm

Similar to e^x , the natural log, or $\ln(1+x)$, also has a nice series expansion. In particular:

$$\ln(1+x) = \sum_{n=0}^{\infty} \frac{(-1)^{n+1} x^n}{n} = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$$

Just like for $\exp(x)$, for this to be defined p -adically, we need the terms to have increasing powers of p . For this to occur, we need x to be divisible by p . With this, we can define exponentiation of two p -adics as $a^b = \exp(b \ln a)$, if it exists. Many other functions that can be written as power series can also be defined on the p -adics, albeit with restrictions on when convergence occurs.

7 Ostrowski's Theorem

We will finish off with a proof of Ostrowski's Theorem, which essentially says that given an arbitrary absolute value function on the rational numbers, it is either the standard absolute value function for real numbers or a p -adic absolute value function. Let us first define what is meant by an absolute value, or norm, which we will define more generally in terms of a field:

Definition 7.1. $|\cdot|$ is a norm on field F if it satisfies the following properties:

- $|x| \geq 0$ for all $x \in F$, and equality holds only when $x = 0$.
- $|x + y| \leq |x| + |y|$ for all $x, y \in F$.
- $|xy| = |x||y|$ for all $x, y \in F$.

The first two conditions essentially define a metric, and the third condition gives us a multiplicative property. Furthermore, we define two norms to be equivalent if the following is true:

Definition 7.2. Two norms $|\cdot|_1$ and $|\cdot|_2$ on a field F are equivalent if there exists a positive real number a such that for all $x \in F$, $|x|_1 = |x|_2^a$.

Also, from here on, we will denote the usual norm for real numbers as $|\cdot|_\infty$, so $|x|_\infty = x$ if $x \geq 0$, and $|x|_\infty = -x$ if $x < 0$. Furthermore, we define the trivial norm $|\cdot|_T$ by $|x|_T = 0$ if $x = 0$ and $|x|_T = 1$ otherwise. It is easy to see that this is indeed a norm. We say a norm is nontrivial if it is not the trivial norm.

Given a rational number $\frac{a}{b}$ where $a, b \in \mathbb{Q}$, we can write both a and b in terms of primes and then reduce the powers of each prime, allowing us to write $\frac{a}{b}$ as $\pm p_1^{r_1} p_2^{r_2} \dots p_n^{r_n}$ where the p_i 's are primes and the r_i 's are integers (not necessarily positive). Given a norm $|\cdot|$, we can then write $|\frac{a}{b}|$ as $|\pm p_1^{r_1} p_2^{r_2} \dots p_n^{r_n}| = |p_1|^{r_1} |p_2|^{r_2} \dots |p_n|^{r_n}$. In other words, any norm on the rationals is defined only by its values on the primes. This will be helpful in now proving Ostrowski's Theorem:

Theorem 7.1. If $|\cdot|$ is a nontrivial norm on the rationals, then it is either equivalent to $|\cdot|_\infty$ or $|\cdot|_p$ for some prime p .

Proof. There are two major cases. The first case is when for all positive integers n , $|n| \leq 1$, in which case we claim that $|\cdot|$ is equivalent to $|\cdot|_p$ for some prime. The other case is when there exists a positive integer n such that $|n| > 1$, in which case we claim that $|\cdot|$ is equivalent to $|\cdot|_\infty$.

- **Case 1:** For all positive integers n , $|n| \leq 1$

Since $|\cdot|$ is nontrivial, there exists a positive integer n such that $|n| < 1$. Then, we can write n out in terms of its prime factors, giving us $|p_1|^{r_1} |p_2|^{r_2} \dots |p_n|^{r_n} < 1$ for distinct primes p_i and positive integers r_i . It follows that one of the p_i 's has an absolute value less than 1 under this norm, call it p . We now claim that p is the only prime with an absolute value strictly less than 1.

Suppose $q \neq p$ is also a prime such that $|q| < 1$. Then, we can find some integer k such that $|p|^k, |q|^k < \frac{1}{2}$ since $|p|, |q| < 1$. By the Euclidean Algorithm/Bezout's Lemma, we can also find integers a and b such that $ap^k + bq^k = \gcd(p^k, q^k) = 1$. Therefore, we have the inequality

$$1 = |1| = |ap^k + bq^k| \leq |a||p|^k + |b||q|^k < \frac{1}{2}|a| + \frac{1}{2}|b| \leq \frac{1}{2} + \frac{1}{2} = 1$$

Note that along the way we used the fact $|a|, |b| < 1$ even though they probably are negative. This is ok because in general, $|x| = |-x|$. Anyway, we see that we get the inequality $1 < 1$, which is absurd, so p is the only prime with absolute value less than 1. All that remains is to show that $|\cdot|$ is equivalent to $|\cdot|_p$. If we let $a = -\log_p |p|$, we get that

$$|p|_p^a = \left(\frac{1}{p}\right)^{-\log_p |p|} = p^{\log_p |p|} = |p|.$$

Also, it is obvious that $|q|_p^a = |q| = 1$ for any other prime q , so $|\cdot|$ and $|\cdot|_p$ are equivalent (remember, a norm on the rationals is defined by its values on the primes).

- **Case 2: There exists a positive integer n such that $|n| > 1$**

First of all, it follows that $n > 1$. Our goal is to show that for any positive integer b greater than 1, $|b| > 1$ as well. To do this, we can write n^k in base b , where k is an arbitrary positive integer for now. First of all, $|n^k| = |n|^k > 1$. Furthermore, we can write n^k in base b with m digits, giving us

$$n^k = c_0 + c_1b + c_2b^2 + \cdots + c_{m-1}b^{m-1}$$

where $0 \leq c_i < b$ and m is a positive integer. For each term $c_i b^i$, we can construct the following inequality: First of all, $c_i < b$, so $|c_i| \leq |1| + |1| + \cdots + |1| = c_i < b$, so $|c_i| \leq b-1$. Furthermore, $|b^i| = |b|^i \leq \max\{1, |b|^{m-1}\}$ (this can be shown by considering $|b| > 1$ and $|b| < 1$). Also, we have the inequality $n^k \geq b^{m-1}$, which after solving for m , we get $m \leq 1 + k \log_b n$. Putting all this together, we can form the following inequality involving $|n|^k$:

$$\begin{aligned} |n|^k &= |c_0 + c_1b + c_2b^2 + \cdots + c_{m-1}b^{m-1}| \\ &\leq |c_0| + |c_1||b| + |c_2||b|^2 + \cdots + |c_{m-1}||b|^{m-1} \\ &\leq \underbrace{(b-1) \max\{1, |b|^{m-1}\} + (b-1) \max\{1, |b|^{m-1}\} + \cdots + (b-1) \max\{1, |b|^{m-1}\}}_{m \text{ times}} \\ &= m(b-1) \max\{1, |b|^{m-1}\} \\ &\leq (1 + k \log_b n)(b-1) \max\{1, |b|^{m-1}\} \\ &\leq (1 + k \log_b n)(b-1) \max\{1, |b|^{k \log_b n}\}. \end{aligned}$$

Now, taking the k th root on the far left and right sides, we get

$$|n| \leq (1 + k \log_b n)^{\frac{1}{k}} (b-1)^{\frac{1}{k}} \max\{1, |b|^{\log_b n}\}.$$

We can now take the limit as k goes to infinity. The only slightly challenging limit here is

that of $(1 + k \log_b n)^{\frac{1}{k}}$. We can write it as

$$\begin{aligned} & \lim_{k \rightarrow \infty} \left((1 + k \log_b n)^{\frac{1}{k \log_b n}} \right)^{\log_b n} \\ &= \lim_{x \rightarrow \infty} \left((x + 1)^{\frac{1}{x}} \right)^{\log_b n} \\ &= 1^{\log_b n} \\ &= 1. \end{aligned}$$

The limit $\lim_{x \rightarrow \infty} (x + 1)^{1/x}$ can be evaluated by taking the natural logarithm and then using L'hôpital's Rule.

Overall, we get $|n| \leq \max\{1, |b|^{\log_b n}\}$, which implies that $|b| > 1$ because otherwise, we would have $|b|^{\log_b n} \leq 1$, and $|n| \leq 1$. Therefore, we have proven that every integer greater than 1 has an absolute value greater than 1. We have also shown that $|n| \leq \max\{1, |b|^{\log_b n}\} = |b|^{\log_b n}$ because $|b| > 1$. This implies the following inequalities after rearranging with logarithm rules:

$$\begin{aligned} |n| &\leq |b|^{\log_b n} \\ \log_{|b|} |n| &\leq \log_b n \\ \frac{\log_b |n|}{\log_b |b|} &\leq \frac{\log_{|n|} n}{\log_{|n|} b} \\ \frac{\log_b |n|}{\log_b |b|} &\leq \frac{\log_b |n|}{\log_n |n|} \\ \log_n |n| &\leq \log_b |b| \end{aligned}$$

All we needed to derive this inequality was the fact that $|n| > 1$, but we also know that $|b| > 1$, so the opposite inequality is also true. In other words, $\log_b |b| \leq \log_n |n|$, which means $\log_n |n| = \log_b |b|$ for all integers n and b greater than 1. Finally, we can let a be this constant value of $\log_n |n|$, and then for all primes p , $|p|_\infty^a = p^{\log_p |p|} = |p|$. Since this is true for all primes, we have that $|\cdot|$ and $|\cdot|_\infty$ are equivalent.

□

Cantor's Infinities

Jacob Levin

1 Introduction

Suppose you have two sets of numbers: one with 5 numbers and the other with 10 numbers. Which set is larger? Obviously, the set with 10 numbers is larger. What if one set had 20 numbers and the other had 21? Well, once again, the set with 21 numbers is larger. How about the set of all natural numbers and the set of all real numbers? Well, now it is trickier. This question was so tricky in fact, that only 128 years ago was this question even taken into serious consideration and solved.

The concept of infinities terrified mathematicians for thousands of years. The ancient Greeks were known to fear infinities. One of Pythagoras' friends was thrown off of a boat for proving that $\sqrt{2}$ was an irrational number whose decimal representation would go on forever without a pattern, while Zeno's paradoxes attempted to describe infinitude as an unknown that would create more questions than answers. Even in the late 19th century, the father of infinities, Georg Cantor, fearing for his reputation, had to write his ideas and theorems about infinities in the appendix of his paper about number theory.

2 Definition

In order to deal with infinities, some definitions are needed. A set is a collection of unique elements that are grouped together (e.g. set $X = \{1, 2, 4, 8, 16\}$, set $Y = \{a, b, c, d, e, f, g, h\}$, set $Z = \{this, is, a, set\}$) which is unordered. An injection from set A to set B is when every element in set A can be assigned to a unique element in set B . A surjection from A to B is when every element in B can be assigned to a unique element in A (so a surjection from A to B is the same as an injection from B to A). A bijection is when every element in A can be mapped to a unique element in B such that there are no elements left over in B .

We also have to get comfortable with sets that have infinite elements. For example, any set X where $|X| = |\mathbb{N}|$ will be defined as a set with countably infinite elements (the absolute value symbol counts the number of elements in a set, or the cardinality of a set). We say that two sets, even if they are infinite, have the same cardinality if there exists a bijection between them. For example, $|\mathbb{N}| = |\text{even positive numbers}|$, since we can map every element in $\mathbb{N}$ to every even number through the function $f(x) = 2x$.

As we are finding out, even if an infinite set contains another infinite set, they can still have the same cardinality. For example, the set of integers is countably infinite. We can assign 1 to 0, 2 to 1, 3 to -1 , 4 to 2, 5 to -2 , and so on, such that $f(n) = (-1)^2 \lfloor n/2 \rfloor$. We have a bijection, since every element in $\mathbb{N}$ can be assigned to every unique element in $\mathbb{Z}$. We can use this similar method

of snaking back and forth to prove that the set of rational numbers is countable by snaking from the top left corner of this infinite grid:

Figure 1: An easy proof that rational numbers are countable.

	1	2	3	4	5	6	7	8	...
1	$\frac{1}{1}$	$\frac{1}{2}$	$\frac{1}{3}$	$\frac{1}{4}$	$\frac{1}{5}$	$\frac{1}{6}$	$\frac{1}{7}$	$\frac{1}{8}$	...
2	$\frac{2}{1}$	$\frac{2}{2}$	$\frac{2}{3}$	$\frac{2}{4}$	$\frac{2}{5}$	$\frac{2}{6}$	$\frac{2}{7}$	$\frac{2}{8}$	...
3	$\frac{3}{1}$	$\frac{3}{2}$	$\frac{3}{3}$	$\frac{3}{4}$	$\frac{3}{5}$	$\frac{3}{6}$	$\frac{3}{7}$	$\frac{3}{8}$	...
4	$\frac{4}{1}$	$\frac{4}{2}$	$\frac{4}{3}$	$\frac{4}{4}$	$\frac{4}{5}$	$\frac{4}{6}$	$\frac{4}{7}$	$\frac{4}{8}$	...
5	$\frac{5}{1}$	$\frac{5}{2}$	$\frac{5}{3}$	$\frac{5}{4}$	$\frac{5}{5}$	$\frac{5}{6}$	$\frac{5}{7}$	$\frac{5}{8}$	...
6	$\frac{6}{1}$	$\frac{6}{2}$	$\frac{6}{3}$	$\frac{6}{4}$	$\frac{6}{5}$	$\frac{6}{6}$	$\frac{6}{7}$	$\frac{6}{8}$	...
7	$\frac{7}{1}$	$\frac{7}{2}$	$\frac{7}{3}$	$\frac{7}{4}$	$\frac{7}{5}$	$\frac{7}{6}$	$\frac{7}{7}$	$\frac{7}{8}$	...
8	$\frac{8}{1}$	$\frac{8}{2}$	$\frac{8}{3}$	$\frac{8}{4}$	$\frac{8}{5}$	$\frac{8}{6}$	$\frac{8}{7}$	$\frac{8}{8}$	...
...	...	...	...	...	...	...	...	...	...

We can assign 1 to $\frac{1}{1}$, 2 to $\frac{2}{1}$, 3 to $\frac{1}{2}$, 4 to $\frac{3}{1}$, 5 to $\frac{2}{2}$, and so on. We can also prove that if you have a countably infinite number of baskets with a countably infinite number of apples in each basket, then the amount of apples is countably infinite. Suppose you label each basket 0, 1, 2, 3, and so on. In each basket, label the apples as 1, 2, 3, and so on. For each natural number, break it up into digits of even powers of ten and digits of odd powers of ten (for example, 12345 would be 135 and 24). The digits in odd powers of ten would describe the basket and the digits in even powers of ten would describe the apples. So the 27th apples in the 84th basket would be described by the number 8247 (if the numbered basket has less digits than the place it holds in the basket or vice versa, you can just fill in the rest of the even or odd digits with 0's). We have our bijection.

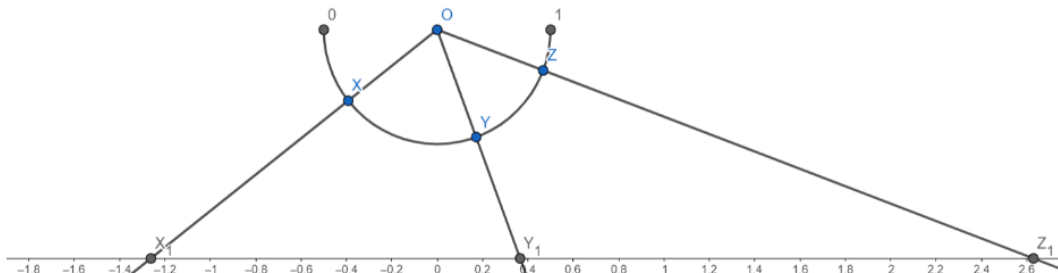
3 The Union Principle

From this last bijection, we have the Union Principle, which states that if $|A| = |B| = |\mathbb{N}|$, then $|A \cup B| = |\mathbb{N}|$. The Union Principle can be repeated multiple times, so this can be expanded to the following:

$$|A_1 \cup A_2 \cup A_3 \dots A_i| = |\mathbb{N}|$$

if $|A_i| = |\mathbb{N}|$ for any positive integer i .

Before we figure out the cardinality of the real numbers, for simplification, we will prove that the cardinality of the real numbers from 0 to 1 is equal to the cardinality of all real numbers. This can be accomplished through a simple, elegant image:



The semicircle is the segment from 0 to 1, while the line at the bottom represents all real numbers. As we can see, every real number on the semicircle can be mapped to every real number on the number line by drawing a line from the center of the semicircle to the semicircle, and then extending it to the number line.

4 Cardinality of Real Numbers

We can now find the cardinality of the real numbers. Since the cardinality of the real numbers and the real numbers from 0 and 1 are the same, we can compare the natural, countable numbers, to the real numbers from 0 to 1. We will conduct a proof by contradiction. Assume that it is possible to assign every natural number to any real number. We will write out a random list of real numbers, but for this argument, any list would work.

- 1 0.283401982491...
- 2 0.230498123809...
- 3 0.912358934576...
- 4 0.492750740595...
- 5 0.933850384947...
- ...

Now, if we can generate a new real number between 0 and 1 that is different from every number on this list, then we have proved that there exists a higher cardinality than the countables. If you take the first digit after the decimal point of the first number and change it to a different digit, the second digit of the second number and change it to a different digit, the third digit of the third number and change it to a different digit, and so on, we would have generated a new number which is different from the 1st (since it would have a different first digit), different from the 2nd, 3rd, and every number already on the list. Thus, we have proved that the cardinality of the reals is greater than the cardinality of the natural numbers.

But wait, a skeptical reader like yourself might wonder, some real numbers have two decimal representations. For example, 0.5 can also be written as 0.499... So when generating a new number, one might write the new number as 0.499..., even if 0.5 is a part of the list, and vice versa. So when generating a new number, we will have to exclude the digits 0 and 9 when selecting the new digits. So now we have proved that the cardinality of the reals is greater than the cardinality of the natural numbers. We will call this new cardinality uncountable infinity.

By the Union Principle, since all real numbers are either rational or irrational, and rational numbers are countably infinite, if the irrationals were countably infinite, then the reals would also

be countably infinite. And since we just proved that the reals are uncountable, then the irrationals are also uncountable. But we are not done. In fact, we have found just one out of an infinite number of new cardinalities of infinities.

5 Power Sets

A power set is the set of all subsets of a set. For example, the power set of the first 3 perfect squares starting from 1 is $\{\{\}, \{1\}, \{4\}, \{9\}, \{1, 4\}, \{1, 9\}, \{4, 9\}, \{1, 4, 9\}\}$. As we can see, there are $2^3 = 8$ elements. We can say that the power set of a set X is $P(X)$. We can prove that the size of the power set of a set with n elements is 2^n through induction:

Base Case: $n = 1$

$$|\{\{\}, \{1\}\}| = 2$$

Assume that the size of the power set of a set with $n = k$ elements is 2^n

Induction step: Prove that this holds for $n = k + 1$

Assume that the set is the natural numbers from 1 to $k + 1$. Each subset will either have or not have $k + 1$. So the size of this power set is the size of the power set of $n = k$ plus the size of the power set of the power set of $n = k$ but each subset has $k + 1$. Since each set has 2^k elements, the size of the power set of the natural numbers from 1 to $k + 1$ is $2^{(k+1)}$.

Now, let's conduct a similar proof to that which we did before with the reals. We can assign each element in an infinite set (we will use the naturals as an example) to a subset of that very set. Such a subset will be represented as an infinitely long binary number, where a 0 signifies an element not being present in the subset and a 1 signifying the presence of that element in a set. So for example, if we have the set $\{0, 1, 2, 3\}$, the subset $\{0, 2, 3\}$ would be represented as 1011. We can create such a chart:

1	1000101110111...
2	0110100100010...
3	0100011110001...
4	1010001110100...
5	1001100011011...
...	

We can do the same diagonal trick as with the reals, where we flip the 1st number of the subset assigned to 1 and flip it (a flip would change a 0 to a 1 and a 1 to a 0), flip the 2nd number of the subset assigned to 2, and so on. We have now generated a new subset that is unique from the rest. This means that the power set of a set has a higher cardinality than the set itself.

This can generate infinitely many infinite cardinalities, ranging from $\mathbb{N}$ to $P(\mathbb{N})$, which is just uncountable infinity, to $P(P(\mathbb{N}))$, to $P(P(P(\mathbb{N})))$, and so on.

The notation we will use to describe the sizes of infinities is the Hebrew letter ($\aleph$). Countable infinity is written as $\aleph_0$, the first uncountable infinity is written as $\aleph_1$, and the rest follows until

infinity.

6 Problems

1. Prove that if A is an infinitely large set, then $|A| \geq |\mathbb{N}|$.
2. Prove that there is a bijection between the number line and the coordinate plane (hint: don't forget to watch out for the fact that there are two decimal representations for some numbers, such as 0.5 being equal to 0.499...).
3. Prove that the cardinality of the reals is the same as the cardinality of complex numbers (Hint: use the result from problem 2).
4. Prove that there if there is both a injection and surjection from A to B , then $|A| = |B|$. (Schröder-Bernstein Theorem). Very difficult!

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The Layers of Cake-Cutting

Shaon Anwar

1 Introduction

A birthday is a day of celebration. Whether you’re enjoying one with friends, family, or a mix of both, birthdays provide a semblance of unity entwined with festivity. . . until the cake is pulled out.

It’s a tale as old as time. Everybody sights out their own desired slice of cake. A couple partygoers aim for the strawberry at the center, while another few desire the delicately swirled puffs of meringue frosting; nobody wants to walk away with a plain slice. Unfortunately, there often isn’t enough cake to get picky. Though quarreling over slice preference seems inevitable with every birthday celebration, mathematicians have been steadily working towards an algorithm that would finally facilitate a utopian party, where everyone leaves with what they want.

The aptly-named cake-cutting problem defines a scenario where a heterogenous, finite object is split among a number of people, each valuing their own pieces of the object. Because of the scenario’s applicability to the real world, from territorial disputes to birthday parties, the cake-cutting problem has been cracked at by mathematicians for generations. With modern algorithms to solve the problem having been largely derived through software programs, the efficiency of these algorithms has been steadily becoming more refined.

The cake-cutting problem hinges on the game theory concept of “envy-freeness.” That is, in a fair division situation, n players intend to obtain a piece of an object (denoted as C) such that each participant is satisfied that their share is as sufficient as any other player’s share. Given the subjectivity of how much a player values an element of the object, desirability is expressed with the value function, V_x , for player x . If player x favors their own share (C_x) over player y ’s share (C_y), then x is envy-free:

$$V_x(C_x) \geq V_x(C_y)$$

Conversely, if player x favors player y ’s share over their own, x is considered envious of y :

$$V_x(C_x) < V_x(C_y)$$

In an envy-free system, no player is envious of each other.

The modern mathematical concept of an envy-free cake-splitting division was defined in 1958 by polymath George Glamow and Marvin Stern. Since then, the problem’s prevalence in resource division launched it into prominence as an optimizable algorithm. By the 1960s, algorithms for fair division among 2 and 3 players had been established.

For $n = 2$, the simplistic “divide and choose” process is employed; player 1 begins by dividing the cake into two pieces that they value equally. Then, player 2 selects the piece that they value

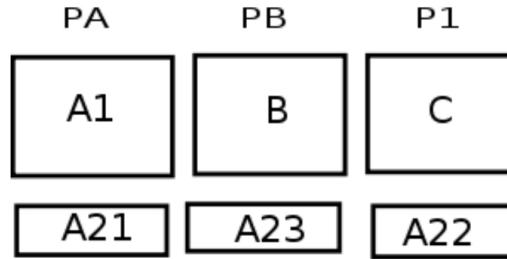
more (C_2), leaving the other piece (C_1) for player 1. Therefore:

$$V_1(C_1) = V_1(C_2) \text{ and } V_2(C_2) \geq V_2(C_1)$$

This satisfies the condition of envy-freeness, as both players value their piece as much or more than the other piece.

The Selfridge-Conway procedure, discovered in 1960 by mathematician John Selfridge, allows for envy-free division among 3 players. Notably, this algorithm assumes the cake can be split into “general pieces”, where a sum of pieces can constitute a player’s share. This contrasts from a “connected pieces” scenario, where n pieces must be available for n players, and each player must receive one contiguous piece (therefore meaning that $n - 1$ cuts can be performed). The Selfridge-Conway procedure has an upper bound of 5 cuts, and is as follows:

1. Akin to the “divide and choose” algorithm, player 1 begins by dividing the cake into three slices that they perceive as equal.
2. Player 2 trims off part of their most valued slice (C_A) so that player 2 values it equally as much as the next most valuable slice. The trimmings are put aside for later. Therefore, C_A is now divided into a trimmed slice (C_{A1}) and trimmings (C_{A2}).
3. Player 3 then chooses any slice.
4. If player 3 did not choose C_{A1} , player 2 must select it. Otherwise, player 2 takes any slice.
5. Player 1 takes the remaining slice, leaving only the trimmings. Now, the trimmings must be divided. Whichever player took the slice C_{A1} is denoted as player A , and the other player as player B .
6. Player B divides the trimmings into three equally-valued slices.
7. Player A chooses a slice of the trimmings (C_{A21}).
8. Player 1 chooses a slice of the trimmings (C_{A22}).
9. Player B chooses the last trimming slice (C_{A23}).



Player A has C_{A1} and C_{A21} . Player B has C_B and C_{A23} . Player 1 has C_C and C_{A22} . Player A is satisfied because:

- $V_a(C_{A1}) > V_A(C_B)$
- $V_a(C_{A1}) > V_A(C_C)$

- $V_a(C_{A21}) > V_A(C_{A23})$
- $V_a(C_{A21}) > V_A(C_{A22})$
- Therefore, $V_A(C_{A1} + C_{A21}) \geq V_A(C_B + C_{A23}), V_A(C_C + C_{A22})$

Player B is satisfied because:

- $V_B(C_B) \geq V_B(C_{A1})$
- $V_B(C_B) \geq V_B(C_C)$
- $V_B(C_{A23}) = V_B(C_{A22}) = V_B(C_{A21})$
- Therefore, $V_B(C_B + C_{A23}) \geq V_B(C_{A1} + C_{A21}), V_B(C_C + C_{A22})$.

Player C is satisfied because:

- $V_C(C_C) \geq V_C(C_{A1} + C_{A21})$
- $V_C(C_C) = V_C(C_B)$
- $V_C(C_{A22}) \geq V_C(C_{A23})$
- Therefore, $V_C(C_C + C_{A22}) \geq V_C(C_B + C_{A23}), V_C(C_{A1} + C_{A21})$.

This scenario fulfills envy-freeness; ultimately, each player is satisfied with their combination of piece and trimming.

That was a doozy.

Since the 1960s, the cake-splitting problem has continued to see development. In 1995, mathematician Alan Taylor and political scientist Steven Brams deduced a procedure that promised envy-freeness. However, the unbounded nature of the algorithm meant that for complicated scenarios, runtime could exceed billions of steps based on the number of players and the specificity of their preferences.

Further progress in 2016 by computer scientists Simon Mackenzie and Haris Aziz reduced the upper bound to only depend on the number of players. Using the trim-dividing premise of the Selfridge-Conway procedure in conjunction with novel concepts of player prioritization, Mackenzie and Aziz were able to decrease the number of players requiring cake with each subsequent step. Though the algorithm's hefty exponential runtime may render it unfit for your next birthday dispute, the cake-splitting problem's applicable zaniness has underscored one of the most fascinating slices of modern game theory.

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Galois and Quintic Formula

Patrick Was

1 Definitions

1.1 Groups

First, some notation. In order to make this more readable, define the symbol " $*$ " to mean some binary operation (meaning it is performed on two elements/numbers), such as addition, multiplication, subtraction, or division. Also, let G be some set. A **group** is a set of elements that satisfies the following properties:

- **Closure.** $\forall a, b \in G, a * b \in G$

For example, the set $\{1, 2\}$ is closed under multiplication, since $1 \cdot 2 = 2$ and $2 \cdot 1 = 2$, which are both in the set. However, it is not closed under addition, since $1 + 2 = 3$, which is not in the set.

- **Identity.** $\exists e \in G \mid \forall x \in G, e * x = x * e = x$

The set $\{0, 5\}$ is an obvious example of a set closed under addition. It is clear that $5 + 0 = 0 + 5 = 5$, and also that $0 + 0 = 0 + 0 = 0$. So, both elements have an identity element, that being 0.

- **Inverse.** $\forall x \in G, \exists y \in G \mid y * x = x * y = e$

This e is the same as from before, the identity element. The set $\{-1, 1\}$ may appear to satisfy this property, since generally -1 is the inverse of 1 , $-1 + 1 = 0$, and since there is no identity element 0 in the set, this set does not follow this property. $\{-1, 0, 1\}$, however, does.

- **Associativity.** $\forall x, y, z \in G, (x * y) * z = x * (y * z)$

This is analogous to associativity under the set of real numbers.

It is not the most well-defined to attempt associativity on an set with two elements, such as $\{-1, 1\}$. However, we do know that $-1, 1 \in \mathbb{R}$, and all elements of the real numbers satisfy associativity under multiplication, so this set must satisfy associativity under multiplication.

One of the elements can also be used twice in the demonstration.

An **abelian group** is one in which $\forall x, y \in G, x * y = y * x$. Basically, it is analogous to the commutative property when dealing with numbers. Any set with complex numbers will satisfy this property, since there it would just be the commutative property, which is a property that extends to that set.

A **cyclic group** is one in which $G = \{g^n \mid n \in \mathbb{Z}\}$ or $G = \{ng \mid n \in \mathbb{Z}\}$. Basically, this just means that every element of G can be formed by a single **generator element** through either multiplying by some constant or raising it to a power; in other words, it acts on itself somehow to form the other elements. G is typically defined more rigorously as a cyclic group in terms of the **subgroup** of the element generator, $\langle g \rangle$, if $\langle g \rangle = \{g^n \mid n \in \mathbb{Z}\} = G$, or if the group is abelian and

additive, then $\langle g \rangle = \{ng \mid n \in \mathbb{Z}\} = G$. This group is also typically written as C_n for its n elements.

An example of a cyclic group is $(\mathbb{Z}, +)$, which represents the additive group of integers. This is because every element can be generated using 1 and -1 simply by multiplying by some constant n . $(\mathbb{Z}_{11}^+, \times_{11})$, the set of positive integers mod 11 under mod 11 multiplication, is also a cyclic group with one of its group generators being 6. As n goes from 1-10, 6^n hits every element of the group: $\{6, 3, 7, 9, 10, 5, 8, 4, 2, 1\}$, in that order.

A **symmetric group** is a group of all permutations on a set, with the binary operation being the composition of functions. If the set contains n elements, we know that the group will contain $n!$ elements, since that encompasses all permutations of those elements. The idea of permutations will be expanded upon later. The symmetric group is also typically written as S_n for its n elements.

An **alternating group** is the set of all **even** permutations in S_n . In order to define what it means for a permutation to be even, it's important to know that a **transposition** is the permutation of 2 elements of a set. Therefore, every 2-cycle is a transposition. A permutation is even if one of its transposition representations consists of an even number of transpositions. In other words, if there was a sequence of transpositions that led to the identity element of the set, there would need to be an even number of transpositions in that sequence. This will be expanded upon later. This group is typically written as A_n for its n elements.

1.2 Fields

A **field** is a set F which is closed under the two binary operations of addition and multiplication such that

- F is an abelian group under addition.
- $F \setminus \{0\}$ is an abelian group under multiplication.

A field K is said to be a **field extension** of a field F if F is a subfield of K . This is denoted as K/F , though field extensions are more generally written in terms of one number at a time, written as $F(\alpha)$.

For example, consider the set of rational number, $\mathbb{Q}$. If we were trying to solve for all solutions to the polynomial $x^2 - 3$, we would not be able to under the field of $\mathbb{Q}$. We do know that the roots are $\pm\sqrt{3}$, so if we were to extend our field to include $\sqrt{3}$, we would be able to find the solutions. This could be written as $\mathbb{Q}(\sqrt{3})$, which is essentially equivalent to $\{a + b\sqrt{3} \mid a, b \in \mathbb{Q}\}$. Any algebraic combination involving rational numbers will therefore be in the field $\mathbb{Q}(\sqrt{3})$. The expression $x^2 - 3$ will therefore **split**, or factor, in $\mathbb{Q}(\sqrt{3})[x]$, which is the field of the coefficients of this particular quadratic. The **splitting field** is therefore the smallest field containing the solutions to the polynomial.

As a further example, we could express $\mathbb{Q}(\sqrt{3}, \sqrt{5}) = \{a + b\sqrt{3} + c\sqrt{5} + d\sqrt{15} \mid a, b, c, d \in \mathbb{Q}\}$, but we cannot have an element like $e\sqrt{10}$, since this would involve multiplying $\sqrt{5}$ by $\sqrt{2}$, which is not an element of our field.

A cool fun fact is that $\mathbb{R}(i)$ is equivalent to $\mathbb{C}$.

An **isomorphism** is a special mapping between two groups that sets up a one-to-one correspondence with the group that respects the binary operations pertaining to each group. A more rigorous definition is the following:

A group isomorphism from $(G, *)$ to (H, Δ) is a bijection from G to H such that for all x and y in G :

$$f(x * y) = f(x) \Delta f(y)$$

An **automorphism** is very similar, except that it is a mapping to itself with the same test but the two operations must be the same. Basically, it is an isomorphism from the group to itself.

1.3 Permutations

A **permutation** of a set S is a bijective function such that $S \rightarrow S$.

1.3.1 Cycle Notation

When permuting a set, **cycle notation** is typically used. It is a simple way to express what each element of the original set permutes to.

Consider the set $\{1, 2, 3, 4, 5\}$. If we wanted to compute every element to its following element, we could write it as the cycle $(1, 2, 3, 4, 5)$. This new set would look like $\{2, 3, 4, 5, 1\}$. We have 1 permute to 2, 2 permute to 3, and so on until 5 permutes to the beginning of the set, 1.

Using the same set, we could write the permutation that permutes all of the sets' elements to itself. This would be $(1)(2)(3)(4)(5)$. The parentheses denote the end of the cycle, so we see the set remains identical. Similarly, in $(1, 2)(4, 5)$, we have 1 permuting to 2 and 2 permuting to 1 as well as 4 permuting to 5 and 5 permuting to 4, but the remaining elements permute to themselves. This kind of permutation that contains **disjoint cycles**, since they do not affect each other. In this case, it is the product of two disjoint cycles.

1.3.2 Functional Cycle Notation

Since we know every permutation is a function, our symmetric group is a group composed of these functions. Since every element is a function, we can see what happens if we compose these functions with each other. In tune with how compositions of functions are typically denoted, $f \circ g$ will be equivalent to fg .

If we consider the two permutations from earlier, $(1, 2, 3, 4, 5)$ and $(1, 2)(4, 5)$, we can denote each as σ and τ respectively. We could find $\sigma\tau$: $(1, 3, 4, 5)$. In this case, 2 actually goes back to map to itself. It's also important to note that this group is not abelian, meaning that the elements will not commute. As an example, $\tau\sigma = (2, 3, 5)$, meaning 1 maps to itself and 4 maps to itself. We see that $\sigma\tau \neq \tau\sigma$.

We can also compose a function on itself. For example, $\sigma^2 = \sigma \circ \sigma$.

We also know that if permutations $\alpha\beta$ are two disjoint cycles, then $\alpha\beta$ do indeed commute. We can see that this is true by a quick example: since $(1, 2)$ and $(3, 4)$ do not affect each other in any

way, then $(1, 2)(3, 4) = (3, 4)(1, 2)$.

In addition, we know that since this is a group, there must be an identity element function. This would just be $(1)(2)(3)(4)(5)$, since every element is permuted to itself.

There must also be a permutation σ^{-1} for any element σ such that $\sigma^{-1}\sigma$ or $\sigma\sigma^{-1}$ equals to the identity element.

1.3.3 More on Alternating Groups

A **simple** alternating group is one that only consists of its entire group (the improper subgroup) and its trivial group of order one, or of one element as its **normal subgroups** (which are subgroups that do not change by any sort of conjugation; a subgroup H is normal iff $gHg^{-1} = H$).

There is a very special alternating group that is the foundation of Galois theory: A_5 . It is the smallest non-abelian simple group, which eventually paves the way for why there is no general quintic formula solvable by any binary operation.

1.4 Galois groups

A **galois group** is the set of automorphisms of L/K , denoted as either $\text{Gal}(L/K)$ or $\text{Aut}(L/K)$, for automorphism.

2 Quintic formula

Galois's theorem states that an algebraic equation is **solvable** iff its galois group is solvable (such that each of its normal subgroups are abelian).

Though rather difficult, it can be proved that A_5 is the smallest non-abelian simple group, and will therefore make its parent group, S_5 , not solvable either. Then, all that needs to be done is to form a connection between the quintic and the group S_5 .

The symmetric group S_5 does have some special isomorphisms, though, which means that it will share properties with that group. For instance, if we let $s_1s_2s_3, \dots, s_n$ be the elementary symmetric functions and let $x_1, x_2, x_3, \dots, x_n$ represent these symmetric functions, then $F(y_1, y_2, \dots, y_n)$ is a field extension of degree $n!$ of $F(s_1, s_2, \dots, s_n)$. This makes it and its galois group isomorphic to S_n , since they share the same number of elements.

If we let the group $E = \mathbb{Q}(y_1, y_2, y_3, y_4, y_5)$ and let

$$f(x) = \prod_{i=1}^5 (x - y_i) = x^5 - s_1x^4 + s_2x^3 - s_3x^2 + s_4x - s_5$$

where s_n is the roots taken n at a time, we know that $s_1, s_2, s_3, s_4, s_5 \in E$, so $f(x) \in E[x]$. If we let $F = \mathbb{Q}(s_1, s_2, s_3, s_4, s_5)$, we see that $f(x) \in F[x]$ and E is the splitting field over F of $f(x)$, since it contains all its roots. The galois group $G(E/F)$ is isomorphic to S_5 , and since that group is not solvable, neither is this one, making $f(x)$ not solvable.

The basic premise behind this is that if the roots aren't re-arrangeable if not given their specific values, then it cannot be possible to find a way to manipulate the roots such that they satisfy an algebraic equation. It would likely take an infinite number of nested roots to actually satisfy some polynomial.

3 Other Applications

There are a considerable number of geometric construction problems that are shown to be impossible by Galois theory, since these involve simple binary arithmetic operations. Some such examples include the impossibility of trisecting general angles or doubling the volume of cubes, as well as other n -dimensional cubes.

It is also very useful in RSA encryption, seeing as it leads the way for commutators and other properties in groups that make certain values hard to decode.

Magic Squares

Ivy Zhu

1 Introduction

An order n magic square is an n by n grid filled with integers such that the rows, columns, and the two largest diagonals all add up to one number: the magic constant. Shown below is an example of a order 3 magic square with 15 as its magic constant.

8	1	6
3	5	7
4	9	2

2 Types of Magic Squares

The magic square we described is known as the normal magic square. A normal n by n magic square is filled up with consecutive integers from 1 to n^2 . Based on the definition, it follows that the magic constant has to be $\frac{1+2+3+\dots+(n^2-1)+n^2}{n}$ or $\frac{n^2(n^2+1)}{2n}$, since the total sum of the numbers must be split up evenly between each row or column. However, there are many variations that open up new possibilities, such as the need for the numbers filling the grid to be consecutive, multiplying instead of adding, and more explained below.

2.1 Multiplicative Magic Squares

Multiplicative magic squares are ones that multiply instead of adding to a constant. If we took the left magic square shown below and raised 2 to the power of each number in the cells, we will get the multiplicative magic square on the right, where each row, column, and diagonal multiplies to 4096.

3	8	1
2	4	6
7	0	5

2^3	2^8	2^1
2^2	2^4	2^6
2^7	2^0	2^5

8	256	2
4	16	64
128	1	32

We can take any magic square and raise any number, x , to the power of each number inside it. In this case, we didn't necessarily need to use powers of 2, since multiplication across these rows,

columns, and diagonals involve adding up the exponents. We already know the exponents add up to a magic constant which means they multiply to the same power of x in the end.

2.2 Complementary Magic Square

When you already have a magic square, you can make a complementary magic square by subtracting each cell from $n^2 + 1$.

8	1	6
3	5	7
4	9	2

$10-8$	$10-1$	$10-6$
$10-3$	$10-5$	$10-7$
$10-4$	$10-9$	$10-2$

2	9	4
7	5	3
6	1	8

2.3 Associative Magic Square

In an associative magic squares, pairs of cells opposite the center sum to $n^2 + 1$. In other words, consider each cell as a point and the opposite as a reflection over the square in the center, in this case 13. In this square, 1 is across from 25, 15 is across from 11, 24 is across from 2, and so on.

1	15	24	8	17
23	7	16	5	14
20	4	13	22	6
12	21	10	19	3
9	18	2	11	25

2.4 Panmagic Square

In a panmagic square, all of the diagonals, when wrapped around, rows, and columns sum to the magic constant. For a diagonal to wrap around, it continues on a diagonal on the opposite side of the main diagonal. Let's take the square shown in the section above, which is also a panmagic square! In the diagram on the lower left, we start from blue with 1. Then we continue onto the diagonal with 4 squares starting with 14 and sum all the numbers up. Shown below are the wrapped diagonals with each color presenting a different diagonal. The magic constant in this case is 65 (sum from 1 to 25 is 325, $\frac{325}{5} = 65$)

1	15	24	8	17
23	7	16	5	14
20	4	13	22	6
12	21	10	19	3
9	18	2	11	25

1	15	24	8	17
23	7	16	5	14
20	4	13	22	6
12	21	10	19	3
9	18	2	11	25

3 Making Magic Squares

Using these magic squares from above, we can develop techniques for creating them.

A complementary magic square is a square is defined by taking an existing magic square and transforming it into another!

Additionally, we can take one magic square and transform it by rotating the square 90° around any corner. We can also draw a mid-line horizontally or vertically and reflect the numbers to the boxes opposite. These transforms preserve the value of numbers in each row, column, and diagonal and are still valid magic squares after. The following 4 magic squares were created from the leftmost square by preserving the numbers that are in each row or column. Notice that 16, 2, 3, and 13 all stay in a line.

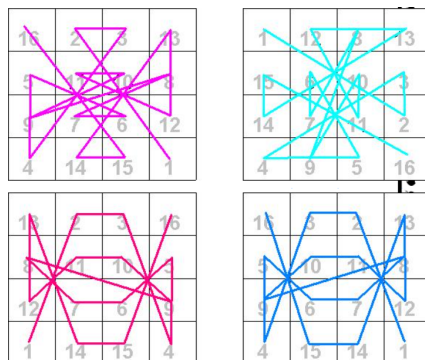
16	2	3	13
5	11	10	8
9	7	6	12
4	14	15	1

1	12	8	13
15	6	10	3
14	7	11	2
4	9	5	16

13	2	3	16
8	11	10	5
12	7	6	9
1	14	15	4

16	3	2	13
5	10	11	8
9	6	7	12
4	15	14	1

Note: When drawing a line through these magic squares from 1 to 16, they form a symmetrical pattern!



From the definition of panmagic squares, wrapped around diagonals all sum to a magic constant, it alludes to an approach that involves focus on the diagonals. One example of this called the Lozenge method, is shown below.

18	24	5	6	12
22	3	9	15	16
1	7	13	19	25
10	11	17	23	4
14	20	21	2	8

This starts with putting the numbers 1 through 5 in a wrap-around diagonal with odd numbers in a “diamond” in the middle, formed by the bolded lines, and even numbers outside. Then 6 through 10 are placed, 11 to 15, and so on.

A method using the idea of associative magic squares. Take an 8 by 8 grid, we fill in the grid with numbers 1 to 64 and draw these diagonals. Then we take the numbers along these diagonals that are opposite from each other, for example 1 and 64, and switch their positions. This involves the concept of associative magic squares with opposite numbers summing to $n^2 + 1$.

64	2	3	61	60	6	7	57
9	55	54	12	13	51	50	16
17	47	46	20	21	43	42	24
40	26	27	37	36	30	31	33
32	34	35	29	28	38	39	25
41	23	22	44	45	19	18	48
49	15	14	52	53	11	10	56
8	58	59	5	4	62	63	1

This diagram shows the end result after the swaps.

Some creative methods for producing order $4n + 2$, where m is an integer, normal magic squares such as the LUX method involve filling the magic squares with shapes. First, we start with an order $2n + 1$ magic square and $n + 1$ rows of Ls, 1 row of Us, and $n - 1$ rows of Xs. Then we would swap the adjacent L and U in the middle column which forms the letters shown below in the intersection of each group of four squares. Those letters refer to how each consecutive group of 4 integers will be arranged.

68	65	96	93	4	1	32	29	60	57
L	L	L	L	L	L	L	L	L	L
66	67	94	95	2	3	30	31	58	59
92	89	20	17	28	25	56	53	64	61
L	L	L	L	L	L	L	L	L	L
90	91	18	19	26	27	54	55	62	63
16	13	24	21	49	52	80	77	88	85
L	L	L	L	L	L	L	L	L	L
14	15	22	23	50	51	78	79	86	87
37	40	45	48	76	73	81	84	9	12
38	39	46	47	74	75	82	83	10	11
41	44	69	72	97	100	5	8	33	36
X	X	X	X	X	X	X	X	X	X
43	42	71	70	99	98	7	6	35	34

Let's try making a magic square with this method! For the sake of simplicity, we will consider a smaller case with an order 3 magic square which maps onto an order 6 normal magic square. We start with 0 and apply the L shape to get that the top middle group of 2 by 2. Then we move on to 1 to get the lower right group and so on.

7	0	5
2	4	6
3	8	1

L	L	L
L	U	L
U	X	U

Here is the final result you should get.

32	29	4	1	24	21
30	31	2	3	22	23
12	9	17	20	28	25
10	11	18	19	26	27
13	16	36	33	5	8
14	15	34	35	6	7

Taking from the idea of splitting large grids into smaller ones, the one below shows splitting a 6 by 6 grid into four quadrants of four groupings of consecutive numbers.

						165
8	1	6	26	19	24	84
3	5	7	21	23	25	84
4	9	2	22	27	20	84
35	28	33	17	10	15	138
30	32	34	12	14	16	138
31	36	29	13	18	11	138
111	111	111	111	111	111	57

Only the columns line up right now, but with a few swaps between the cells, we can eventually get the rows to work. We can see that the sum of the first three rows and last three rows are constant which motivates us to swap numbers from rows 1-3 to rows 4-6 in the same column. As shown below, the yellow boxes swap with the orange boxes. Notice how the swaps occur 3 rows apart in the same column for the reasoning aforementioned.

						111
35	1	6	26	19	24	111
3	32	7	21	23	25	111
31	9	2	22	27	20	111
8	28	33	17	10	15	111
30	5	34	12	14	16	111
4	36	29	13	18	11	111
111	111	111	111	111	111	111

This strategy also works on a 10 by 10 grid, with the swaps looking recognizable as the pattern above, but with some extra steps involved.

										505
92	99	1	8	15	67	74	51	58	40	505
98	80	7	14	16	73	55	57	64	41	505
4	81	88	20	22	54	56	63	70	47	505
85	87	19	21	3	60	62	69	71	41	505
86	93	25	2	9	61	68	75	52	34	505
17	24	76	83	90	42	49	26	33	65	505
23	5	82	89	91	48	30	32	39	66	505
79	6	13	95	97	29	31	38	45	72	505
10	12	94	96	78	35	37	44	45	53	505
11	18	100	77	84	36	43	50	27	59	505
505	505	505	505	505	505	505	505	505	505	505

As we can see throughout the examples provided, magic squares of the same order don't have to look similar! Even similar ones are interesting to think about, for example, the rotations of the order 4 magic square shown before. Try playing around with creating magic squares and thinking about why each strategy works!

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Cool Things You Need to Know About Triangles

Qinwen Zheng

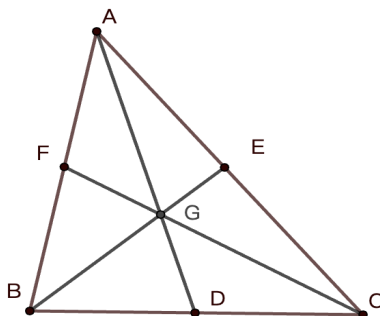
1 Why are triangles so important?

In geometry, polygons are everywhere. For instance, even though a circle is not a defined polygon, it can still be considered as a polygon that has infinite sides. Since any polygon can be formed by a number of triangles, the triangle becomes one of the most important shapes in geometry. (Think like this: all polygons can be cut to a bunch of triangles.) This article will introduce a series of triangle properties, including the centroid, incenter, orthocenter, and some famous theorems related to triangles, such as the Angle Bisector Theorem, Law of Cosines, and Stewart's Theorem.

2 Cevians

In order to talk about the properties of triangles and their applications, we have to know what a cevian is: A cevian is a line segment or ray that extends from one vertex of a triangle to the opposite side. Angle bisectors, altitudes, and medians of a triangle are all special types of cevians. One famous theorem related to cevians is Ceva's theorem, which states that in any $\triangle ABC$:

$$\frac{AF}{FB} \cdot \frac{BD}{DC} \cdot \frac{CE}{EA} = 1$$



2.1 Proof of Ceva's Theorem

Note: $[ABC]$ indicates the area of triangle ABC. Notice that when triangles share the same height, the ratio of their areas is simply the ratio of the lengths of their bases.

$$\frac{AF}{FB} = \frac{[AGF]}{[BGF]}$$

$$\frac{AF}{FB} = \frac{[ACF]}{[BCF]}$$

A quick proof of the baseball theorem in case you are wondering what that is:

Statement: If $\frac{a}{b} = \frac{c}{d}$, then $\frac{a}{b} = \frac{c}{d} = \frac{a-c}{b-d}$.

Proof: $\frac{a}{b} = \frac{c}{d} \rightarrow ad = bc$. Subtract cd from both sides: $ad - cd = bc - cd \rightarrow (a - c)d = (b - d)c \rightarrow \frac{a-c}{d-d} = \frac{c}{d}$, which also equals to $\frac{a}{b}$.

Applying baseball theorem, we get:

$$\frac{[ACF] - [AGF]}{[BCF] - [BGF]} = \frac{AF}{FB}$$

$$\frac{[ACG]}{[CBG]} = \frac{AF}{FB}$$

Applying the same method on $\frac{BD}{DC}$ and $\frac{CE}{EA}$, we obtain:

$$\frac{BD}{DC} = \frac{[AGB]}{[AGC]}, \frac{CE}{EA} = \frac{[BGC]}{[GBA]}$$

Multiply all three area ratios and obtain:

$$\frac{[ACG]}{[CBG]} \cdot \frac{[AGB]}{[AGC]} \cdot \frac{[BGC]}{[GBA]} = 1$$

3 Incenter, Centroid, and Orthocenter

Before talking about these terms, let's first start with the following definitions:

An **angle bisector** is a cevian which divides the given vertex's angle into two congruent angles.

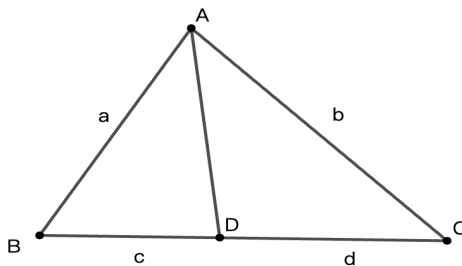
A **median** is a cevian which intersects at the midpoint of its opposite side.

An **altitude** is a cevian which is perpendicular to its opposite side.

3.1 Angle Bisector Theorem

The Angle Bisector theorem is commonly used in triangles, which can be used to find the relationships between segments of the triangle, given there is an angle bisector.

This theorem states that for any triangle ABC, angle bisector AD divides the opposite side in a ratio of AB:AC.

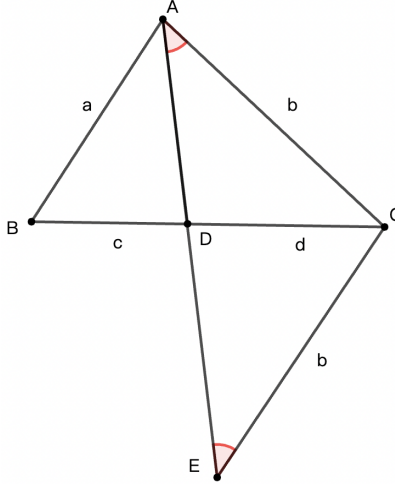


Written as an equation:

$$\frac{AB}{BC} = \frac{AC}{DC} \longrightarrow \frac{a}{c} = \frac{b}{d}$$

3.2 Proof of the Angle Bisector Theorem

There are many different ways to prove Angle Bisector theorem. Here is one simple method: Extend angle bisector AD until a point E ($E \neq A$) such that $CA = CE$.



Since $CA = CE$, $\triangle ACE$ is an isosceles triangle, meaning $\angle CAE \cong \angle CEA$. Since $\angle CAE \cong \angle DAB$, by the definition of an angle bisector, $\angle CEA \cong \angle DAB$, which proves $AB \parallel CE$. Because $AB \parallel CE$, $\angle ABD \cong \angle DCE$. Using AA congruence with $\angle ABD \cong \angle DCE$ and $\angle CAE \cong \angle DAB$, $\triangle ADB \sim \triangle EDC$. Using $\triangle ADB \sim \triangle EDC$:

$$\frac{c}{a} = \frac{d}{b} \longrightarrow \frac{a}{c} = \frac{b}{d}$$

This is equivalent to the angle bisector theorem statement at section 3.1: $\frac{a}{c} = \frac{b}{d}$.

3.3 Brief Introduction

An **incenter** is the point of intersection of three angle bisectors of a triangle, which is also the center of the triangle's incircle.

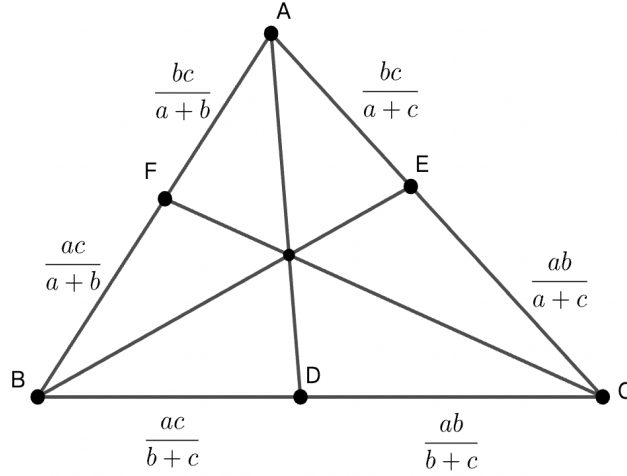
A **centroid** is the point of intersection of three medians of a triangle, which divides each median to a 1 : 2 ratio.

An **orthocenter** is the point of intersections of three altitudes of a triangle.

4 Incenter

I believe the incenter is more interesting to talk about, so I decided to write more about it in this article.

You may wonder why three angle bisectors intersect at the same point, here you go:



Here is the setup:

Let $BC = a$, $AC = b$, $AB = c$.

Let $\overline{AD}$ and $\overline{CF}$ to be angle bisectors. Since we do not know if $\overline{BE}$ is an angle bisector, we need to treat it as a cevian that pass through the intersection point of $\overline{AD}$ and $\overline{CF}$.

Apply angle bisector theorem on angle bisector $\overline{AD}$ and $\overline{CF}$, we can obtain:

$$BD : DC = \frac{ac}{b+c} : \frac{ab}{b+c} \text{ and } AF : FB = \frac{bc}{a+b} : \frac{ac}{a+b}.$$

Assume $\overline{BE}$ is an angle bisector, then angle bisector theorem will hold true, otherwise it will not.

If $\overline{BE}$ is an angle bisector:

$$AE : EC = \frac{bc}{a+c} : \frac{ab}{a+c}.$$

If this is valid, then Ceva's theorem needs to be hold true also:

$$\frac{AF}{FB} \cdot \frac{BD}{DC} \cdot \frac{CE}{EA} \stackrel{?}{=} 1$$

Plug in those side lengths obtained from applying angle bisector theorem:

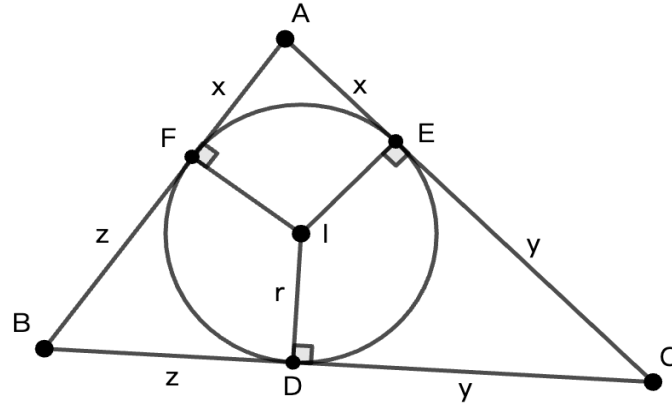
$$\begin{aligned} \frac{\frac{bc}{a+b}}{\frac{ac}{a+b}} \cdot \frac{\frac{ac}{b+c}}{\frac{ab}{b+c}} \cdot \frac{\frac{ab}{a+c}}{\frac{bc}{a+c}} &\stackrel{?}{=} 1 \\ \frac{bc}{ac} \cdot \frac{ac}{ab} \cdot \frac{ab}{bc} &\stackrel{?}{=} 1 \\ 1 &= 1 \end{aligned}$$

Therefore, if $\overline{BE}$ is not an angle bisector, then the ratio of $AE : EC$ is not $\frac{bc}{a+c} : \frac{ab}{a+c}$, which will make Ceva's theorem not holds true, causing a contradiction.

Conclusion: Three angle bisectors of a triangle are **concurrent**.

4.1 Interesting property of incircle

As mentioned in section 3.3, the incenter of a triangle is also the center of its incircle. The incircle of a triangle is the circle that is inscribed inside of a triangle. Here how an incircle looks like:



The incenter is indicated by the letter I . Points D, E, F are the feet of the segment dropped from I . r , the inradius of the triangle, which is the radius of the incircle, equals to the measure of any of ID, IE, and IF.

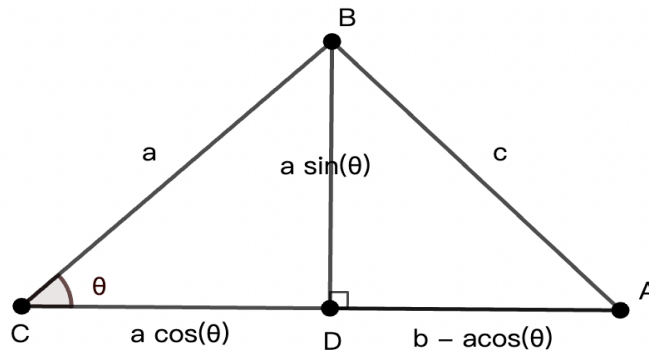
Points D,E,F divide each side by a certain ratio. As indicated in the diagram, $AF = AE$, $CE = CD$, $BD = BF$.

5 Law of Cosines

Law of cosines is an useful theorem which creates a relationship between three sides and one of three angles in a triangle. Using law of cosine, we are able to figure out any side length given one angle and two of the side lengths, or any angle measurement given three side lengths.

5.1 Proof for acute triangle

Consider a triangle ABC, and call angle ACB θ .



First, drop an altitude from B to $\overline{CA}$, call the foot D.

Since ABD is a right triangle, the length of $\overline{BD}$ is $a \sin(\theta)$ and the length of $\overline{CD}$ is $a \cos(\theta)$, which means the length of $\overline{DA}$ is $b - a \cos(\theta)$.

$\triangle BDA$ is also a right triangle, meaning we can apply Pythagorean theorem on it to obtain the following relationship: $(b - a \cos(\theta))^2 + (a \sin(\theta))^2 = c^2$.

Expand this, we have:

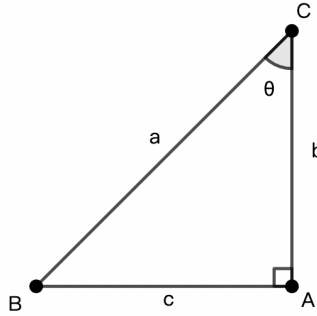
$$b^2 - 2ab \cos(\theta) + a^2 \cos^2(\theta) + a^2 \sin^2(\theta) = c^2$$

$$b^2 - 2ab \cos(\theta) + a^2 = c^2$$

Here we have the statement for law of cosine: $c^2 = a^2 + b^2 - 2ab \cos(\theta)$.

5.2 Proof for right triangle

Since law of cosine can be applied to all triangles and I have only shown the case for acute triangle, we need to know if it works for the case of right triangle.



Starting from a similar setup, law of cosine obviously holds true when θ is the right angle. Therefore, we only need to consider when θ is one of the other two angles.

Knowing that $b = a \cos(\theta)$ and $c = a \sin(\theta)$, we can derive from $2b^2 = 2b^2$:

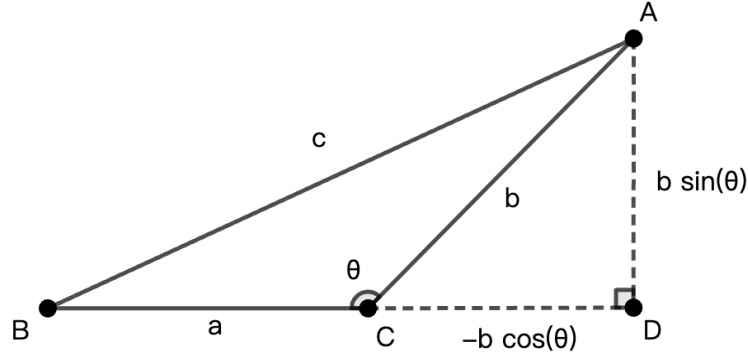
$$c^2 + 2b^2 = b^2 + a^2$$

$$c^2 = a^2 + b^2 - 2b^2$$

$$c^2 = a^2 + b^2 - 2ab \cos(\theta)$$

5.3 Proof for obtuse triangle

Also using the similar setup, we are going to prove if law of cosine holds for the obtuse angle:



First, extended side $\overline{BC}$ to point D and drop an altitude from point A to point D.

Knowing that $\cos(\pi - \theta) = -\cos(\theta)$, we know that the length of $\overline{CD}$ is $-b \cos(\theta)$. Also since $\sin(\theta) = \sin(\pi - \theta)$, the length of $\overline{AD}$ is $b \sin(\theta)$.

Applying Pythagorean theorem on triangle ABD, we have: $(a - b \cos(\theta))^2 + (b \sin(\theta))^2 = c^2$, expanding this:

$$a^2 - 2ab + b^2 \cos^2(\theta) + b^2 \sin^2(\theta) = c^2$$

$$a^2 - 2ab \cos(\theta) + b^2 = c^2$$

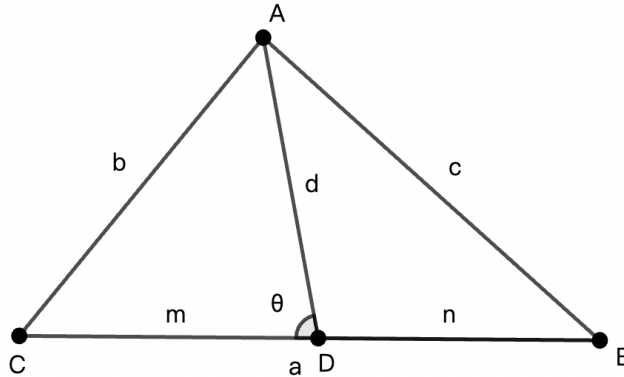
$$c^2 = a^2 + b^2 - 2ab \cos(\theta)$$

6 Stewart's Theorem

Stewart's Theorem, discovered by a Scottish mathematician named Matthew Stewart, shows a relationship between one cevian and three sides of a triangle.

Here is the theorem statement:

$$b^2n + c^2m = a(m + n) + d^2a$$



Note: a is the length of $\overline{CB}$.

6.1 Proof

Here is the setup:

$$AC = b, AB = c, CB = a, CD = m, DB = n, AD = d.$$

We will apply law of cosine twice on triangle ADC and triangle ADB to obtain the statement for Stewart's Theorem.

$$\text{On triangle ADC: } b^2 = d^2 + m^2 - 2dm \cos(\theta)$$

$$\text{On triangle ADB: } c^2 = d^2 + n^2 - 2dn \cos(\pi - \theta)$$

Multiply n on both sides of the first equation, multiply m on both sides of the second equation, we can obtain:

$$\begin{aligned} b^2 n &= d^2 n + m^2 n - 2dmn \cos(\theta) \\ c^2 m &= d^2 m + n^2 m - 2dmn \cos(\pi - \theta) \end{aligned}$$

As I mentioned earlier in this article, $\cos(\pi - \theta) = -\cos(\theta)$, we can do this substitution to the second equation:

$$c^2 m = d^2 m + n^2 m + 2dmn \cos(\theta)$$

Now adding two equations together, we get:

$$b^2 n + c^2 m = d^2(m + n) + n^2 m + m^2 n$$

Since $m + n = a$ and we can factor out mn from $n^2 m + m^2 n$, we get the following:

$$b^2 n + c^2 m = d^2 a + nma$$

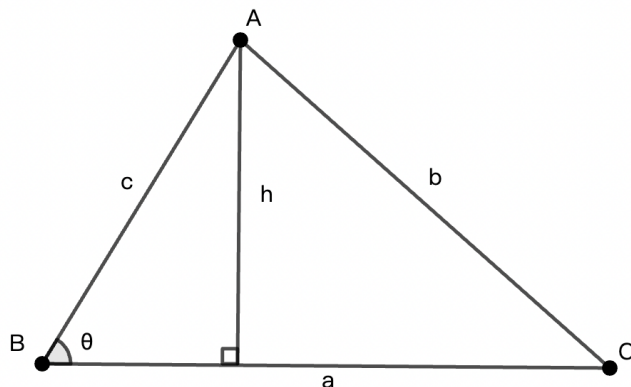
Therefore, the statement for Stewart's theorem is: $b^2 n + c^2 m = d^2 a + amn$.

7 Alternative formula for triangle area using trigonometry

The most common area formula we used in math classes is probably $A = \frac{bh}{2}$, but there are other ways to find the area of a triangle!

Here is one of them that uses trigonometry:

$A = \frac{1}{2}ab \sin(\theta)$, where a and b are lengths of two sides and θ is the measure of the angle formed by these two sides.



The proof of this formula is straightforward. Indicated in the diagram, h is also $c \sin(\theta)$, then we can use the basic $\frac{bh}{2}$ formula. After substitution we can get: $\frac{ac \sin(\theta)}{2}$.

7.1 The case of equilateral triangle

You may heard the area formula for a equilateral triangle before, which is $\frac{s^2 \sqrt{3}}{4}$, where s is the side length. This comes from the formula derived above. Since all three angles of an equilateral triangle are 60 degrees, substitute this back to the formula, we get: $\frac{s \cdot s \cdot \sin(\frac{\pi}{3})}{2} = \frac{s^2 \sqrt{3}}{4}$.

8 Problems

1. Given triangle ABC with $AB = 3$, $AC = 4$, $m\angle BAC = 30^\circ$, find BC.
2. Find the area of a regular hexagon with side length 2.
3. Given triangle ABC with $AB = 6$, $AC = 9$, $BC = 10$, find the length of angle bisector AD, which bisects the angle BAC. (If you are able to derive a formula for the length of this segment, then you discovered the second angle bisector theorem)
4. Try to prove this: show that the centroid divides a median into two segments of a ratio of 1 : 2.
5. Derive a triangle area formula using perimeter and its inradius. (Hint: Look back to section 4.1)
6. In triangle ABC, $AB = 6$, $BC = 7$, and $CA = 8$. Point D lies on $\overline{BC}$, and $\overline{AD}$ bisects $\angle BAC$. Point E lies on $\overline{AC}$, and $\overline{BE}$ bisects $\angle ABC$. The bisectors intersect at F. What is the ratio AF : FD? (2016 AMC 12A Problem 12)
7. The lengths of the sides of a triangle are consecutive integers, and the largest angle is twice the smallest angle. What is the cosine of the smallest angle? (1982 AHSME Problem 23)

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Rise of the Cosine

Elizabeth Burnell

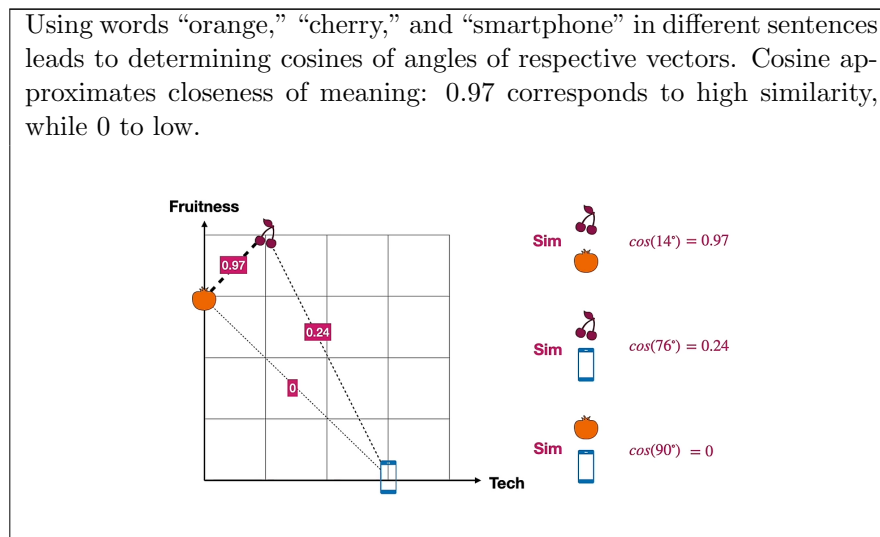
1 Introduction

The now classic paper “Attention Is All You Need” laid the conceptual foundation for current implementation of the models such as Chat Generative Pre-trained Transformer (ChatGPT) by employing similarity between words, sentences, and partial paragraphs, in vector calculations called together “attention,” which rely on the cosine¹ function. While reviewing Do-It-Yourself guides for building a model of this type, “generative pre-trained transformer”, the backbone of ChatGPT, I discovered that a very familiar trigonometric function—cosine—was enabling its success. Here, I want to explain the first part of this process (and hope to provide more in later editions) to provide motivation for trigonometry enthusiasts.

2 Steps to Build a Model

A preparatory step in building the model is embedding – representing each word (or root and ending) as a number, and then the whole a sentence as a vector, making it into an array of numbers. In large language models, this provides the link from human speech to the computer.

Although the Euclidean distance could be used as an indicator of relative proximity between two points, using the cosine function has important advantages.



¹To be fair, sin is also used but the correct term is “cosine similarity.”

In simple terms, softmax replaces cosine with its exponent, making all numbers positive. The maximum is amplified if they are further normalized to sum up to 1.

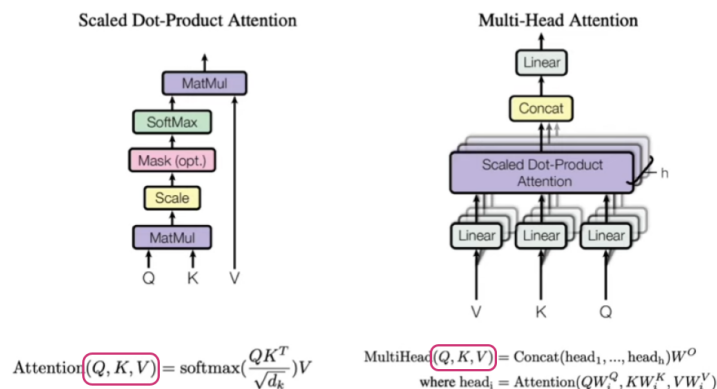
Softmax

$$x \longrightarrow e^x$$

$$\text{Orange} \longrightarrow \frac{e^1 \text{Orange} + e^{0.71} \text{Apple}}{e^1 + e^{0.71}} = 0.57 \text{Orange} + 0.43 \text{Apple}$$

$$\text{! Orange} \longrightarrow \frac{e^1 \text{Orange} + e^{-1} \text{Motorcycle}}{e^1 + e^{-1}} = 0.88 \text{Orange} + 0.12 \text{Motorcycle}$$

The original article by Google researchers uses a more sophisticated formula with a similar meaning:



First, the arrays of numbers built off sentences (embeddings) can be also conceptually likened to vectors in an n -dimensional space. n is the number of words coming together with the one that interests us. Thus, a vector measure is cohesive. The mathematical techniques of vector transformations in linear algebra are well understood, allowing for further work with data in such form. See illustration in Appendix for stretching the angle, and therefore lowering its cosine, between Apple the phone and apple a fruit, with vertex at the origin, by reshaping a square into a narrow and wide parallelogram using linear transformations.

The cosine function has an important advantage as it can be measured for vectors of different dimensions. Other measures such as Euclidean distance, it turned out, tended to increase with the number of dimensions, complicating the task of comparing words in paragraphs and texts of different lengths.

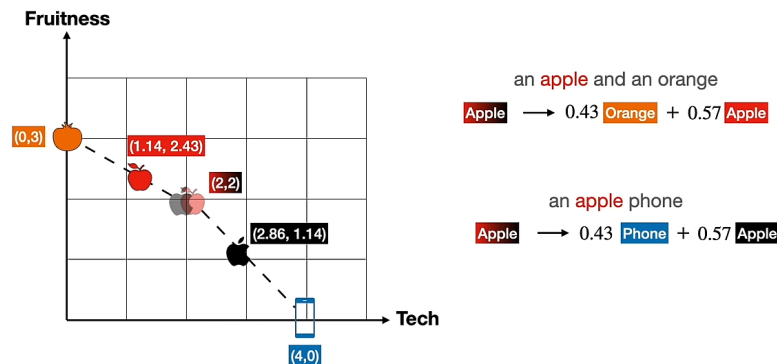
Linked to this property is the fact that cosine's range is $[-1; 1]$, which offers uniformly comparable bounds which make use of words in sentences, paragraphs, and texts of different lengths

comparable. The Euclidean distance is unbounded towards infinity in dimensions and could cause a risk, even for modern computers, of overflow (root calculations are also relatively more “costly” in terms of computing time). This fact can be seen in calculating a distance from the origin to point $(1, 1, 1, 1, \dots, 1)$ in n dimensions: $\sqrt{(1-0)^2 + \dots + (1-0)^2} = \sqrt{n}$, and for $n = 10,000$, corresponding to a smaller dictionary than the one used in ChatGPT, this distance is 100 making this point far away from a “unit circle.”

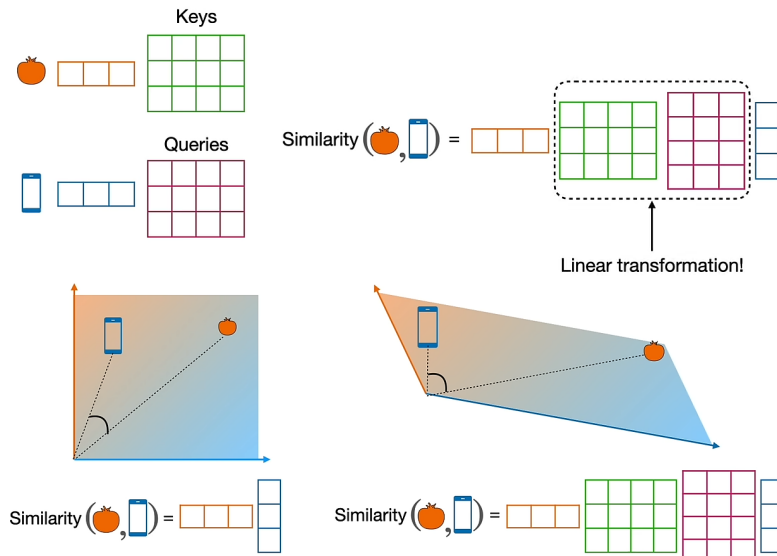
3 Final Adjustment

In the real word, the cosines are replaced with their exponents, as the exponent of a cosine will be a positive number. All of such values lie between $\frac{1}{e}$ (e to power -1) and e (e to power 1). The maximum of several cosines is amplified if exponents are further normalized to sum up to 1, completing the analysis of similarity and finding the likeliest match.

Appendix Illustration: cosine similarity calculation with further transformation



Keys and Queries Matrices

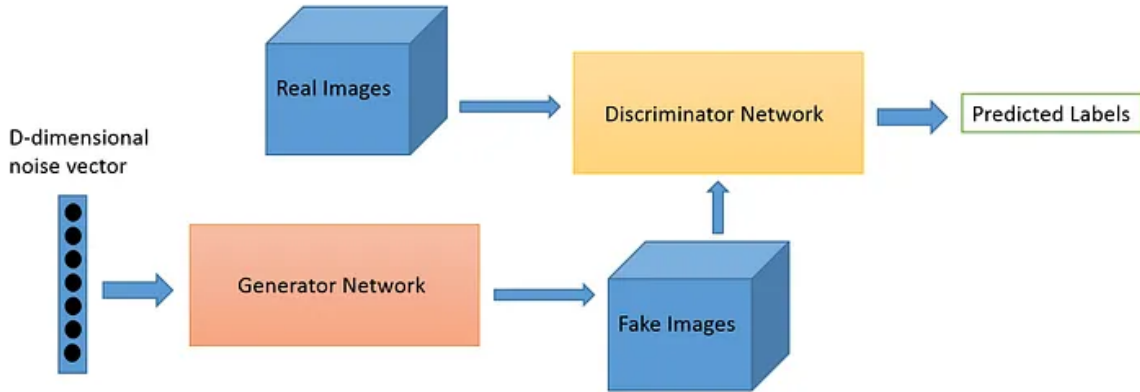


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GANs

Han Xiao



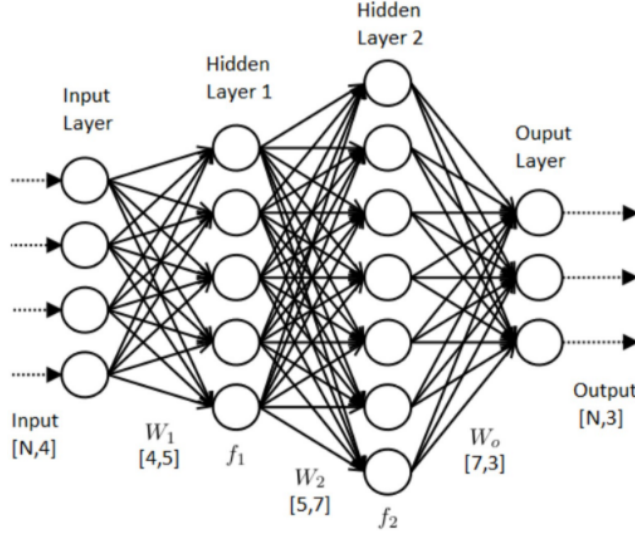
1 Introduction

Generative adversarial networks consist of two parts: a generator network and a discriminator network. The generator creates fake data using random noise chosen from a distribution with the same domain as the preexisting set of real data, while the discriminator acts as a simple binary classifier that tries to determine whether an input is generated or real. The mechanism behind the system is a constant Minmax game between the two networks, where the generator constantly tries to maximize the discriminator's loss (fool the discriminator into thinking a generated image is a part of the real dataset) and minimize its own (generator) loss. This is done through constant weight updates within the neural network. The discriminator utilizes the same method in its own network to counter the generator and output a more accurate probability of the input being real.

2 Neural Networks

The diagram below shows a simplified display of typical networks within the generator and discriminator. The input layer takes the provided information (usually whether or not certain factors are present) and associates the info with a value ranging from 0 to 1 indicating no or yes, respectively. Nodes (circles) from the input layer are connected to the second layer using channels (arrows in the diagram) that each have an assigned weight (W_1), which indicates how important a piece of information is. The following layers of the network are hidden layers, where most of the math required for there to be an output is computed. Notice how some of the nodes are connected by more than one channel. In this case, the value sent to the hidden layer is the sum of the weight times the assigned values of nodes from the input layer. The hidden layer nodes all contain a bias

value, which can be thought of as the constant in a linear equation. Finally, the bias is added to the previously obtained sum and passed through a threshold. If the threshold is met (value $\geq$ threshold), then the node lights up and continues passing input onto the next layer. This process is repeated until the output layer is reached.



In the training process, the weight of each neural network is updated through optimization and regularization methods. Optimization minimizes the cost/loss function of a system by testing various parameter values (for example, weights), while regularization can speed up this process by placing restrictions on the range of values chosen for parameters. In other cases, regularization prevents overfitting: a common problem in training, where the model overgeneralizes based on the given training information, and assumes an incorrect pattern for non-existent data.

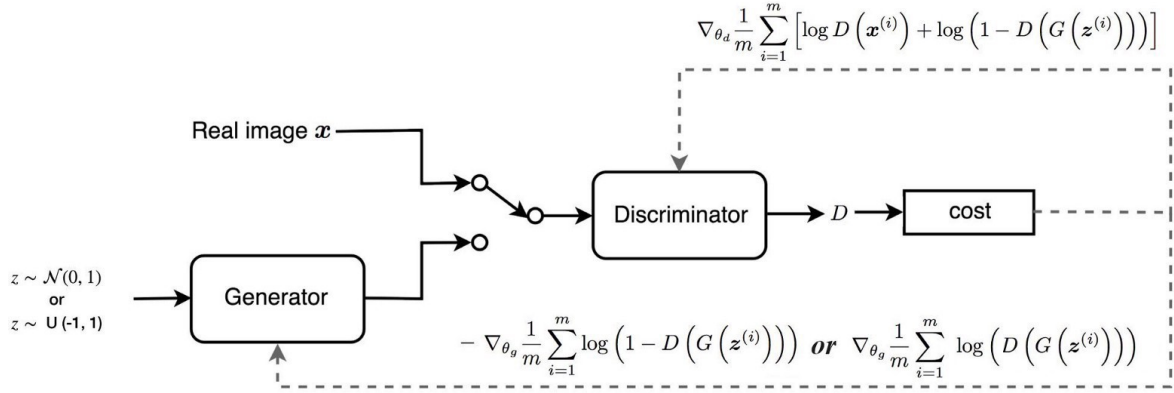
3 The Math Behind GANs

GANs use a minmax loss function that is very similar to the binary cross entropy (*BCE*) function, which penalizes networks for falsely labeling or classifying data. Based on the loss value, weights within the networks are adjusted accordingly. GAN labels data from the preexisting, true dataset with a value of 1 and generated data with a value of 0. It takes values from the generator ($G(z)$) and discriminator ($D(x)$), and multiplies it by the expected value over all instances of a dataset (E). The function $G(z)$ assigns the random noise, z , passed through the generator, a value. Likewise, the function $D(x)$ assigns the data (real or generated) passed through the discriminator, x , a value representing the discriminator's prediction. In *BCE*, Y_i is the ground truth or real value, and $\hat{Y}_i$ is the predicted value. Thus, replacing Y_i and $\hat{Y}_i$ in $Y_i \times \log \hat{Y}_i$ with 1 and $D(x)$, respectively, gives you the first half of the minmax function, or the discriminator's prediction on the real data. To get the second half of the function, or the discriminator's prediction on the fake data, simply replace Y_i and $\hat{Y}_i$ in $(1 - Y_i) \times \log(1 - \hat{Y}_i)$ with 0 and $D(G(z))$ (since the discriminator takes the generator's output as its input for the prediction), respectively. Because the minmax function is taking the whole dataset into account, it must multiply each variable by E , the expected average minmax value of each individual data point. Lower loss values correspond to higher accuracy. Gradient Descent, an optimization method, is then used to minimize the loss function and improve training.

$$L_{BCE} = -\frac{1}{n} \sum^n (Y_i \times \log \hat{Y}_i + (1 - Y_i) \times \log(1 - \hat{Y}_i))$$

$$\min_G \max_D V(D, G) = E_{x \sim p_{data}(x)} [\log D(x)] + E_{z \sim p_z(z)} [\log(1 - D(G(z)))]$$

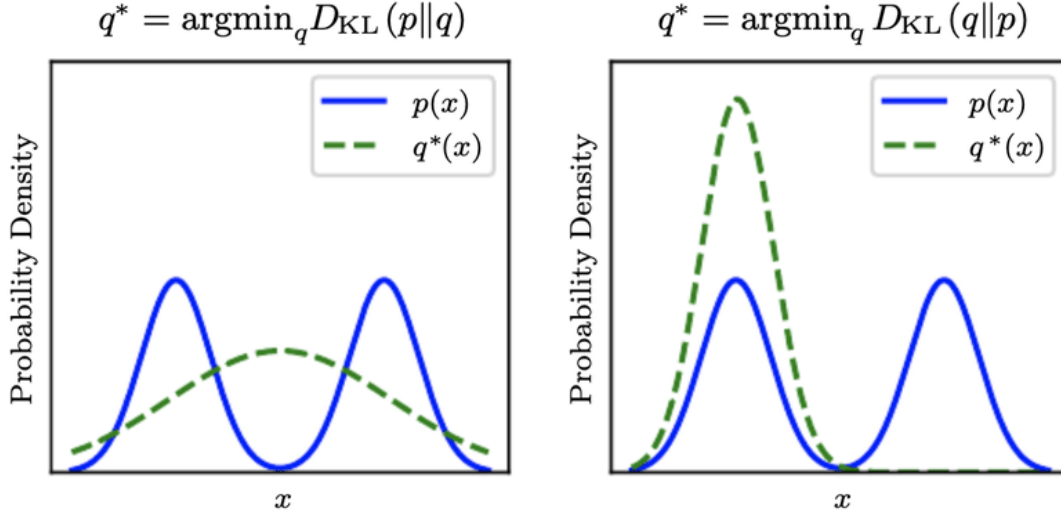
Binary Cross Entropy Function over trials.



Minimax GAN value function.

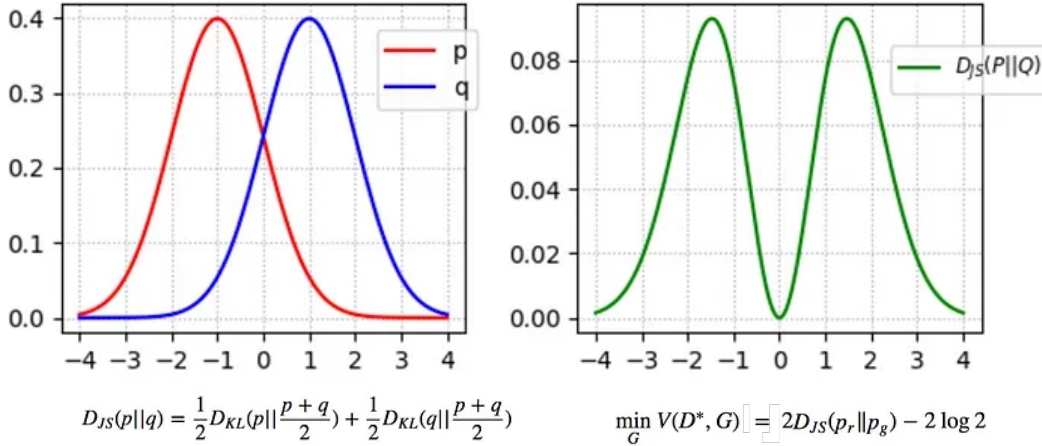
4 Kullback-Leibler Divergence

The Kullback-Leibler divergence measures the difference between two probability distributions of continuous random variables (a variable that has an infinite amount of possible values) using the function $D_{KL}(p||q) = \int_x p(x) \log[p(x)/q(x)] dx$. Simply put, the difference between two distributions can be found by subtracting one from another. The difference can be rewritten in terms of log, where $\log p(x) - \log q(x)$ is the same as $\log[p(x)/q(x)]$. This is then multiplied by a weight, $p(x)$. For continuous random variables, the values can be summed using an integral since each possible value of the variable has a probability of zero, but the regions of values have nonzero probabilities. However, a problem with the KL divergence is that the outcomes differ depending on the distribution used for reference. In other words, $D_{KL}(p||q) \neq D_{KL}(q||p)$. This is especially problematic for datasets that have little to no overlap, or use different parameters, as the cost is extremely low where values of p (actual distribution) are greater than zero and values of q (predicted distribution) approach zero, as shown in the graph below. This produces less diverse samples of higher quality. In the opposite case where p approaches zero and q is greater than zero, the quality is higher, but less diverse samples are produced. At other times, KL divergence approaches infinity. Furthermore, the asymmetrical property of this divergence causes unstable training.



5 Jensen-Shannon Divergence

The Jensen-Shannon Divergence is based on the KL divergence, with an improvement. JS divergence is smoother and has symmetry $D_{KL}(p||q) = D_{KL}(q||p)$, making the training process more stable. In addition, this divergence takes the average of the distributions, and has a range of $[0, 1]$, which solves the problem KL divergence has with approaching infinity when two datasets do not overlap. Interestingly, this is related to Claude Elwood Shannon's Information Theory presented in his article *A Mathematical Theory of Communication*. Information states that unlikely, unexpected, or chaotic events hold more information that is represented using an entropy value, and can be defined as mutual information $I(X; Z) = H(X) - H(X|Z)$, where, in divergence, X is a random variable set to a distribution between P and Q , and Z is a binary indicator variable. The range is a result of constraints due to $H(Z) = 1$ in $\log_2$.



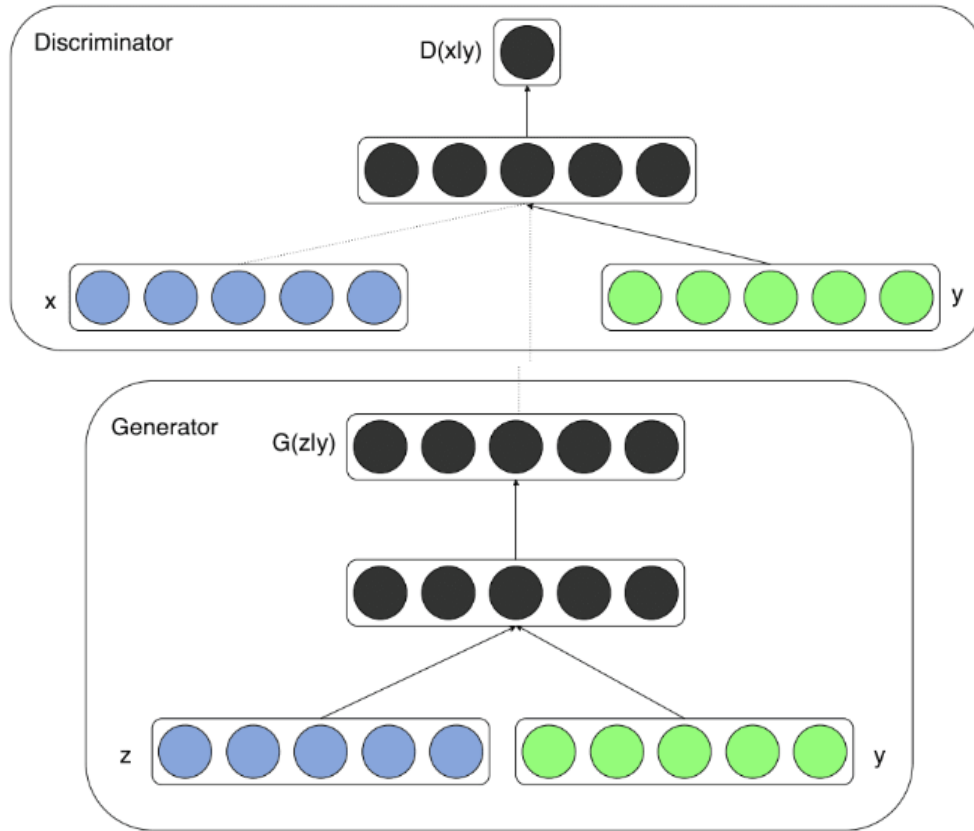
Nonetheless, as long as the data distributions do not overlap at all, the output value will still be the same for two sets of data that are slightly closer to each other, and two sets of data that

have significantly different distributions. To solve this problem, Russian mathematician Leonid Vaseršteĭn's concept of earth mover's distance, or Wasserstein distance, is used. This will be discussed in further detail later on in the article.

6 Types of GANs

6.1 Conditional Generative Adversarial Network (cGAN)

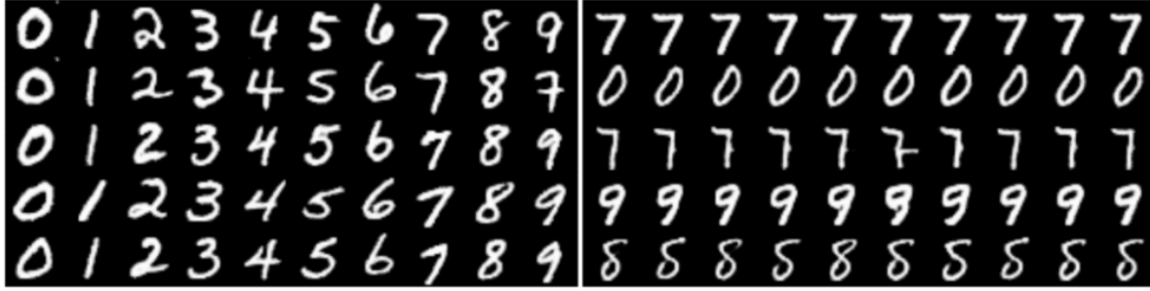
cGANs take additional input such as class labels during the training process, shown by the additional y parameter in the diagram. The additional parameter allows the generator to generate more specific information while assisting in better classification by the discriminator. Conditional Generative Adversarial Networks gave rise to the Pix2Pix model that performs image-to-image translation, which includes tasks such as photo reconstruction and image colorization. This type of network can be useful for synthesizing detailed, high quality photos using text and descriptions, as well as predict upcoming frames in a video based on previous instances.



6.2 Information Maximizing Generative Adversarial Network (InfoGAN)

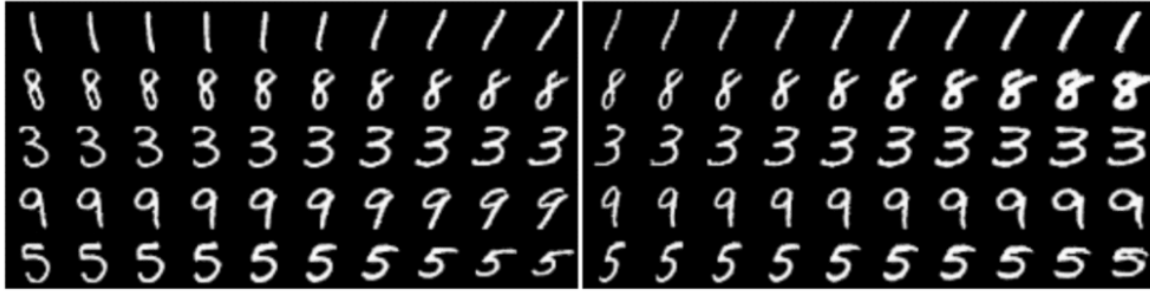
Different from GAN, InfoGAN permits control over the images the generator produces through additional control variables and the use of information theory. Information theory states that the amount of information learned from random variables X and Y is the mutual information between X and Y . To calculate the Mutual Information (MI) loss, the entropy (rate of transfer

of information) of control variable (c) given the generated image using noise (z) and the control variable (c) is subtracted from the entropy of the control variable alone. This mutual information is then used to train the generator in a new model. An example of the usage of InfoGAN can be seen in generated images from the MNIST dataset, where the thickness and rotation of digits within the image is purposely varied.



(a) Varying c_1 on InfoGAN (Digit type)

(b) Varying c_1 on regular GAN (No clear meaning)



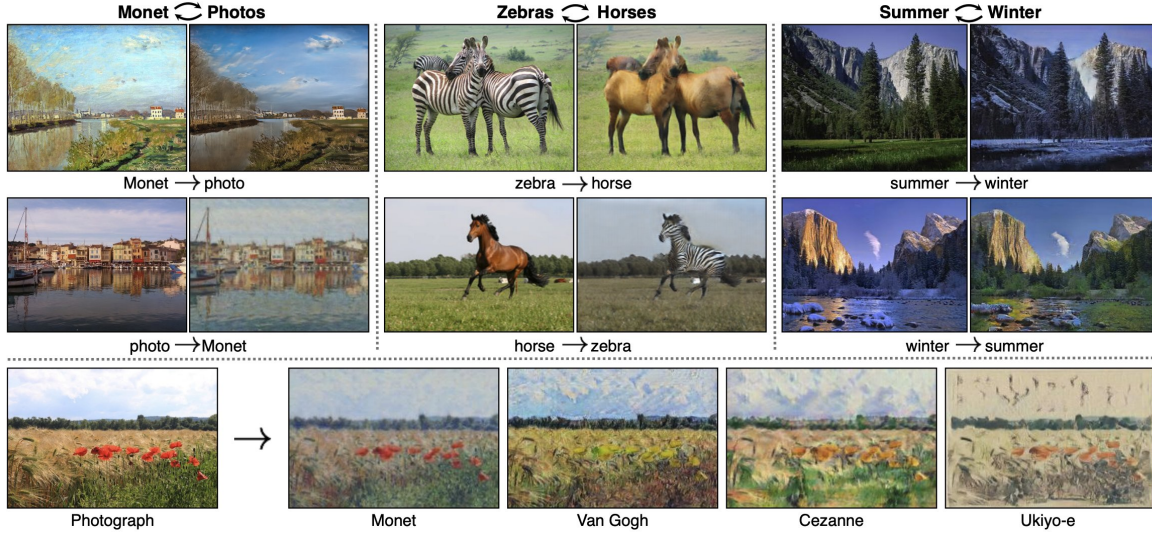
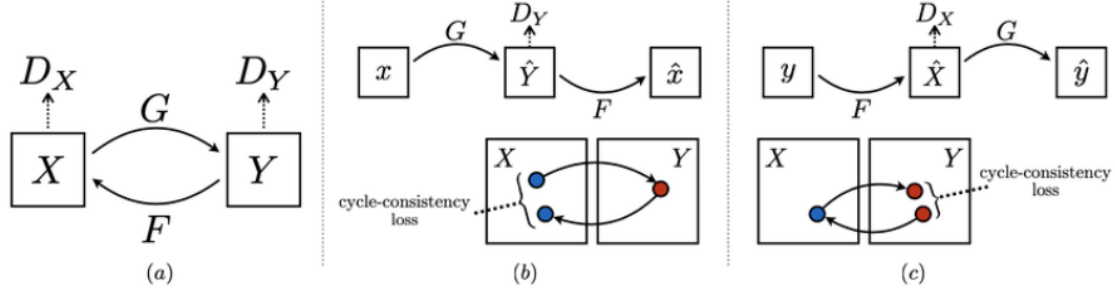
(c) Varying c_2 from -2 to 2 on InfoGAN (Rotation)

(d) Varying c_3 from -2 to 2 on InfoGAN (Width)

Generated images using infoGAN with control variables c_1 (thickness of the digit's stroke) and c_2 (angle of the digit's rotation)

6.3 CycleGAN

CycleGANs are used to perform image to image translation. This is typically done in order to change the style of the image, creating effects like changing seasons, generating paintings from realistic photos, or object transfiguration and applying filters. The math behind this GAN involves the sum of three losses, because two functions are simultaneously being performed. Apart from the usual min-max game between the generator and discriminator, an inverse mapping function is used to map the generated data back onto the original image. There are, in fact, two discriminators and two generators in this model, and they output values based on given domains (ex. winter and spring). One generator uses existing data for domain one to output images for the second domain, while the other generator generates images for the first domain using existing data for the second domain. Abstractly speaking, one generator would be taking pre-existing images of fields in the spring and outputting images of the same fields in the winter, and vice versa. The discriminators then evaluate images and their corresponding datasets.



6.4 Wasserstein Generative Adversarial Network (WGAN)

In WGAN, the discriminator is referred to as the critic, and Wasserstein distance is used instead of K-L or J-S divergence to stabilize training. A Wasserstein distance measures the distance between corresponding data points in separate data sets (the minimum of two data sets, for example), and is calculated using the equation:

$$W(p_{data}, p_G) = \inf_{\gamma} E_{(x,y) \sim \gamma(x,y)} [|x - y|]$$

The expected value of the distance between y and x where y and x are distributed according to the optimal choice of joint distribution γ with marginals $\gamma_X(x) = p_{data}(x)$ and $\gamma_Y(y) = p_G(y)$. However, finding a γ that minimizes the distances between p_{data} and p_G is hard to find directly due to complicated distributions and calculations. Instead, the Kantorovich-Rubenstein theorem regarding optimal transport problems is used to produce the following function:

$$W(p_{data}, p_G) = \sup_{\|f\|_{L \leq 1}} E_{p_{data}}[f(x)] - E_{p_G(x)}[f(x)]$$

where a restraint is placed on f by supremum $\|f\|_{L \leq L}$, and $f(x)$ is similar to that of a lagrange multiplier (a strategy for finding the local maximum and minimum of a function in constrained optimization problems). Essentially, restraint requires the expectation to be calculated over 1

lipschitz ($|f(x) - f(y)| \leq |x - y|$, which places a limit on the slope or speed of the function on how fast one data point can travel to another. This way, a small change in the input only results in a small change in the output. Other forms of lipschitz continuous functions can also be used for bounds that are not equal to 1. The discriminator could be updated using weight clipping, but that requires very specific and exact values. Therefore, it is easier to set a bounded slope similar to before using a lipschitz continuous gradient. The generator is updated to maximize the value of $E_{z \sim p(z)}[f_\theta(G(z))]$.

$$\max_{w \in W} E_{x \sim P_r}[f_w(x)] - E_{z \sim p(z)}[f_w(g_\theta(z))]$$

Below is the original proposed algorithm from the Wasserstein GAN paper:

Algorithm 1 WGAN, our proposed algorithm. All experiments in the paper used the default values $\alpha = 0.00005$, $c = 0.01$, $m = 64$, $n_{\text{critic}} = 5$.

Require: : α , the learning rate. c , the clipping parameter. m , the batch size. n_{critic} , the number of iterations of the critic per generator iteration.

Require: : w_0 , initial critic parameters. θ_0 , initial generator's parameters.

```

1: while  $\theta$  has not converged do
2:   for  $t = 0, \dots, n_{\text{critic}}$  do
3:     Sample  $\{x^{(i)}\}_{i=1}^m \sim \mathbb{P}_r$  a batch from the real data.
4:     Sample  $\{z^{(i)}\}_{i=1}^m \sim p(z)$  a batch of prior samples.
5:      $g_w \leftarrow \nabla_w [\frac{1}{m} \sum_{i=1}^m f_w(x^{(i)}) - \frac{1}{m} \sum_{i=1}^m f_w(g_\theta(z^{(i)}))]$ 
6:      $w \leftarrow w + \alpha \cdot \text{RMSPProp}(w, g_w)$ 
7:      $w \leftarrow \text{clip}(w, -c, c)$ 
8:   end for
9:   Sample  $\{z^{(i)}\}_{i=1}^m \sim p(z)$  a batch of prior samples.
10:   $g_\theta \leftarrow -\nabla_\theta \frac{1}{m} \sum_{i=1}^m f_w(g_\theta(z^{(i)}))$ 
11:   $\theta \leftarrow \theta - \alpha \cdot \text{RMSPProp}(\theta, g_\theta)$ 
12: end while
```

7 Disadvantages of The GAN Model

Although GANs can produce surprising results when trained properly, there are many factors and issues these models are prone to during the process. Certain presented problems include: Mode collapse, diminishing gradients, non-convergence, and sensitivity to hyperparameters. Mode collapse occurs when the generator finds a particular weakness in the discriminator. If the discriminator struggles with classifying a distinct type of data, then the generator will only go on to generate data of that type for the rest of the training duration. For instance, if the discriminator struggles with distinguishing real versus fake data for the number 6 when training for number identification, the generator will only produce images of the number 6, leading to a lack of data variation. Diminishing gradients occur when the initial layers of nodes of the network are not updated properly, resulting in less efficient training processes and failure to find proper parameters for deeper layers. Non-convergence is usually a result of the generator being unable to maximize discriminator loss. In other words, the discriminator is doing too well of a job at distinguishing between generated and real data, and thereby unable to tell the generator specific areas in need of improvement. Finally, GANs, like many other networks, are sensitive to hyperparameter tuning.

Examples of hyperparameters include learning rate, number of nodes in a network layer, and the number of network layers. A small change in any of these can have a massive impact on training and data convergence.

8 Conclusion

Although the development of such technology has been particularly useful in healthcare and cybersecurity, its ethics is also highly debated. On one hand, GANs can synthesize medical imaging scans to compensate for the lack of existing data and improve accuracy rates for early cancer detection. This adversarial approach to machine learning can protect systems from attacks and create a safer environment without having to learn from real, preexisting attacks. On the other hand, GANs can also be used to produce realistic drawings, sketches, previously nonexistent song covers, and other art forms at an industrial pace. This presents problems for those in the field, and is inevitably going to lower employment rates.

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Regressional NLP and Proximal Policy Optimization

Naowal Rahman

1 Introduction

The release of ChatGPT has produced a newfound public interest in AI and machine learning. ChatGPT uses natural language processing (NLP) to respond to prompts in ways that are optimized for conversation. In this article, we'll first look at traditional algorithms that can be used to train a chat bot and then dive into how ChatGPT was trained. We'll discuss logistic regression, how that can be extended to softmax regression, and the gist of proximal policy optimization, a new family of algorithms that OpenAI uses.

2 An Overview of Neural Networks

In case you aren't familiar with neural networks or need a review, here is an overview of how they work.

Neural networks produce an output given some input. They consist of groups of neurons called layers. Each layer serves a different purpose. Everything starts at the input layer, where input variables are passed. Each neuron in this layer will represent one input variable. If we want to pass in multiple inputs, we need multiple neurons. The next set of layers are called hidden layers. Here, weights are applied to the inputs and are passed through an activation function, which produces outputs that introduce non-linearity to the data. Each layer is fit to produce a specific type of output, and subsequent hidden layers can work in conjunction and use a previous layer's data to produce new output.

2.1 Exclusive-Or Classification Model

To further clarify, let's look at an example. Using Tensorflow's neural network visualizer we can see all of the elements of a neural network aimed at learning an exclusive-or (xor) function, which returns 1 if either x_1 or x_2 is true, and false otherwise.

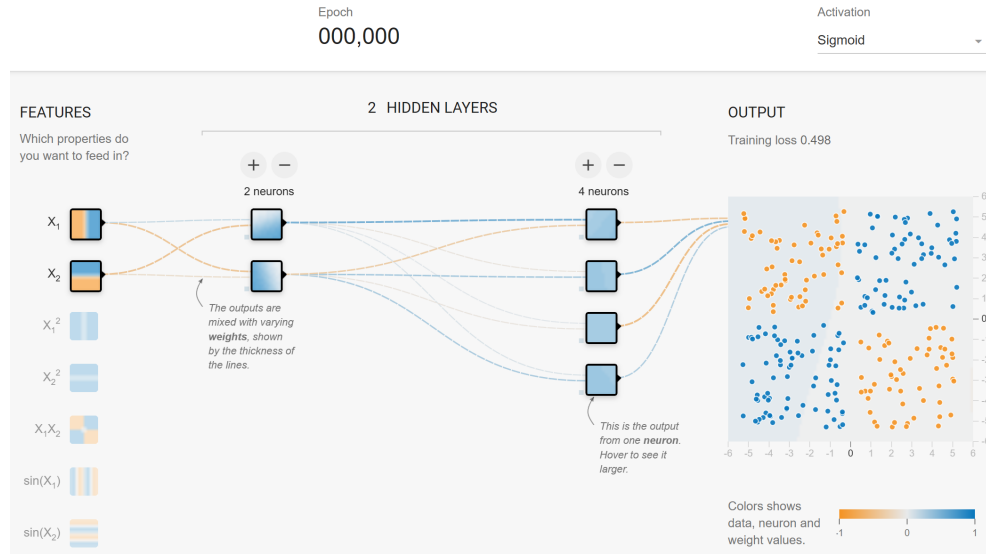


Figure 1: Setup for the xor neural network

Input variables x_1 and x_2 are passed, as those are the two variables used in the xor function. While any number of hidden layers and neurons within those layers can be used, two hidden layers work well, one with two neurons and another with four neurons. This is because we expect our graph to be split into around four sections, with the first hidden layer splitting the graph in half and the second layer distinguishing further from there.

We can see how the model starts off with a loss value close to 1 (meaning high error) and the graph covered entirely around white (0). The neural network starts off with this set of inaccurate values and improves them over time.

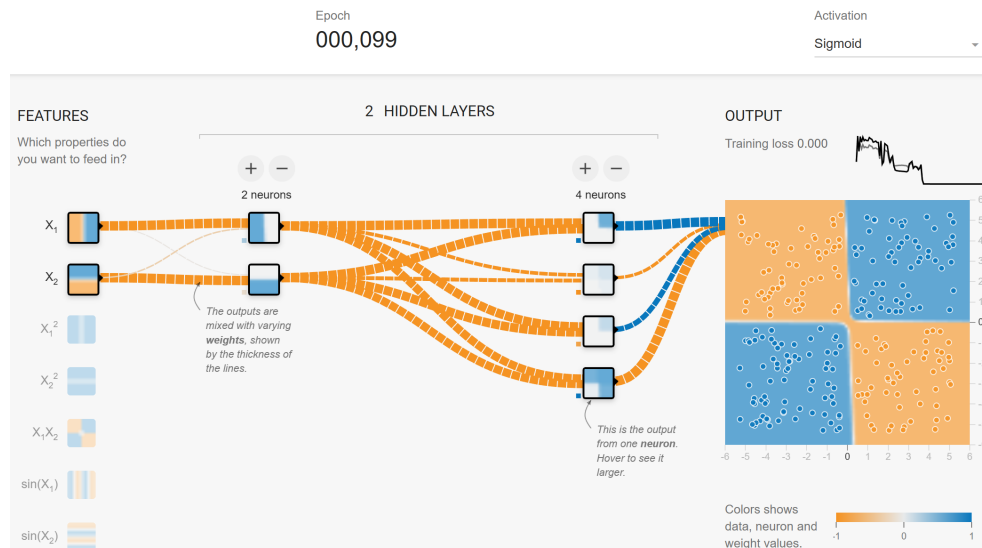


Figure 2: Trained xor model

As expected, this configuration of neurons worked well and the model was trained with a final loss of 0.

3 Logistic Regression

Generally, an NLP algorithm that produces the next most probable output will have to make choices between 2 or more possible outputs. The output is generally an array of probabilities summing to 1 – we pick the index with the highest probability as our model’s choice.

3.1 Sigmoid Function

Not directly used in NLP but essential to generating a probability based on a single input is the sigmoid function:

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

This is perfect for generating probabilities, as $\sigma(x) \in (0, 1)$, and $x \in (-\infty, \infty)$. No matter our real-number input, the output will always be suited to be a probability. Graphing the function, we see the familiar S-curve, where $\sigma(x) \rightarrow 0$ as $x \rightarrow -\infty$ and $\sigma(x) \rightarrow 1$ as $x \rightarrow \infty$.

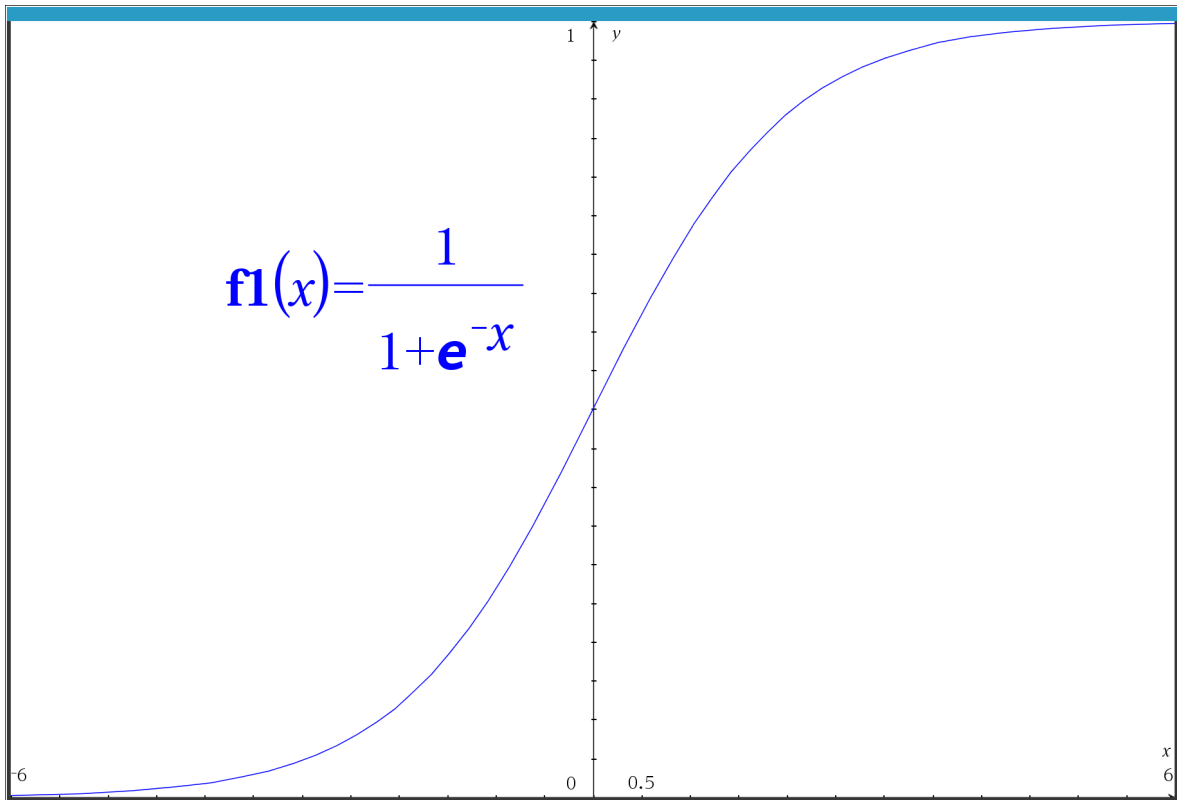


Figure 3: Graph of $\sigma(x)$

This is frequently used to add non-linearity to a model, where the function operates on each of the available inputs and corresponding weights.

3.2 Extending σ to a Multi-Input Scenario

Using σ , we arrive at logistic regression, a basic model that may be used in an NLP program. It generates probabilities for a binary outcome given multiple inputs, and is defined as

$$\begin{aligned} p(y = 1|x) &= \frac{1}{1 + e^{-(b+w_1x_1+w_2x_2+\dots+w_nx_n)}} \\ &= \frac{1}{1 + e^{-(\mathbf{w}^T \cdot \mathbf{x} + b)}} \end{aligned}$$

where

1. $x_1, x_2, \dots, x_n$ are the independent variables.
2. $w_0, w_1, w_2, \dots, w_n$ are the model's weights.
3. We simplify with linear algebra and thus take the dot product of the transpose of the weight vector $\mathbf{w}^T$ and the input vector $\mathbf{x}$.
4. b is the bias term.

Given x , logistic regression produces the probability that $y = 1$ (y is true).

While this isn't sufficient for a chat bot, we can use this for determining properties of text, like whether a sentence is positive or negative.

3.2.1 Exercise

Let $z = \mathbf{w}^T \cdot \mathbf{x} + b = \ln(a)$. Find a .

3.2.2 Solution

Substituting z into $p(x)$, we get

$$p(x) = y = \frac{1}{1 + e^{-(\mathbf{w}^T \cdot \mathbf{x} + b)}} = \frac{1}{1 + e^{-z}}$$

meaning we can solve for z . So, we have

$$\begin{aligned} y &= \frac{1}{1 + e^{-z}} \\ \Rightarrow 1 + e^{-z} &= \frac{1}{y} \\ \Rightarrow -z &= \ln\left(\frac{1}{y} - 1\right) = \ln\left(\frac{1-y}{y}\right) \\ \Rightarrow z &= -\ln\left(\frac{1-y}{y}\right) = \ln\left(\left(\frac{1-y}{y}\right)^{-1}\right) = \ln\left(\frac{y}{1-y}\right) \end{aligned}$$

$$\therefore \boxed{a = \frac{y}{1-y}}$$

3.2.3 Remark

This means that z , the input to $p(x)$, can be represented as the natural logarithm of the probability that y is true divided by the probability that y is false. Thus, z is referred to as the *log-odds ratio*, or $\text{logit}(y)$.

4 Softmax Regression

As logistic regression produces a probability for a binary outcome, softmax regression extends this idea into multi-class scenarios. Thus, rather than returning a single probability $p \in [0, 1]$, softmax returns a vector $\mathbf{p}$ of n probabilities such that $\sum_{i=1}^n p_i = 1$.

4.1 Implementation

We can implement this by adding an extra hidden layer just before the output layer. This hidden layer must have the same number of neurons as the output layer, or the number of classes. Each of these neurons produces a logit value for their respective class to be passed into the sigmoid function that produces probabilities in the output layer.

Mathematically, we represent this as an extension of the logistic regression equation:

$$\begin{aligned} p(y = j|\mathbf{x}) &= \frac{e^{(w_j^T \mathbf{x} + b_j)}}{\sum_{k=1}^K e^{(w_k^T \mathbf{x} + b_k)}}, \quad \text{for } j = 1, 2, \dots, K \\ &= \frac{e^{\text{logit}(y_j)}}{\sum_{k=1}^K e^{\text{logit}(y_k)}}, \quad \text{for } j = 1, 2, \dots, K \end{aligned}$$

Thus, each probability y_j is calculated as the ratio of class j 's exponentiated logit score over the sum of all K classes' exponentiated logit scores.

4.2 Caveat

Softmax regression can only be used under the assumption that all input belongs to exactly one class. This is because the highest probability in the generated output vector is considered the choice; if

$$\mathbf{x} = [0.25, 0.12, 0.15, 0.31, 0.17]$$

then it makes sense to take only the index of $\max(\mathbf{x})$ as the model's choice.

5 Proximal Policy Optimization

Proximal Policy Optimization (PPO) is OpenAI's new class of reinforcement learning algorithms. They are useful because they perform better than current approaches whilst being simpler to implement and modulate.

5.1 Types of Machine Learning

PPO differs distinctly from softmax regression because they are classified as different types of learning algorithms. It is important to know the differences between them before proceeding, so here are the 3 types of learning algorithms: supervised, unsupervised, and reinforcement.

5.1.1 Supervised Learning

Supervised learning algorithms use labeled datasets to make a mathematical model for future datasets. Such algorithms are used for classification problems because they can produce a discrete value, deciding the specific class that the input belongs in. They may also be used for regression problems, where discrete relationships between data need to be created.

Softmax regression, being a classification problem, is a supervised learning algorithm.

5.1.2 Unsupervised Learning

In contrast to supervised learning, unsupervised learning involves the use of *unlabeled* data. Essentially, such algorithms find patterns in data without being trained in a strict input-output manner. They are used for problems where datasets need to be clustered and analyzed.

5.1.3 Reinforcement Learning

Reinforcement learning algorithms are not deliberately trained. Rather, they are continuously training themselves with data from their environment. You may see why this is good for NLP problems or ChatGPT - as more users use the service, the model becomes better and better. Other use cases can include gaming or self-driving cars, scenarios where there are continuous streams of data.

5.2 Training ChatGPT

ChatGPT was trained using both supervised and reinforcement learning. The model was improved over GPT-3.5, an improved version of GPT-3. First, in supervised learning, human trainers provided the model with responses resembling those from both the user and the model. Then, in the reinforcement learning step, the model generated responses that were rated by the human trainers. The ratings were used to further improve the model through PPO 2.

5.3 Explanation

PPO is used to train policies for decision-making problems. This means that the objective function must maximize some expected "reward" obtained from following a given policy over a trajectory of states and actions. A good example of this is that of self-driving cars – the states involve the properties of car at a given timestamp, like velocity and acceleration, and the actions involve what is done by the car at those states, like steering at a certain angle or braking. The associated reward may be related to trip time or energy used.

Expressing this computationally means using a clipped surrogate objective function that limits how the policy is updated to a trust region of allowable updates. We define this trust region through a hyperparameter (a parameter external to the model's control) called a "clipping parameter" to the model. Then, we must factor in the loss of our model and an entropy bonus term that encourages the model to explore different actions.

5.4 Implementation

We can express our clipped surrogate objective as follows:

$$L^{\text{CLIP}}(\theta) = \hat{\mathbb{E}}_t \left[\min \left(r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right) \right]$$

where

1. θ is the policy parameter.
2. $\hat{\mathbb{E}}_t$ is the average empirical expectation at timestamp t , taken from a batch of finite samples.
3. r_t is the ratio of the old probability to the new probability.
4. $\hat{A}_t$ is the estimate of the current advantage.
5. ϵ is the clipping parameter.
6. $\text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t$ clips $r_t(\theta)$, keeping it between $[1 - \epsilon, 1 + \epsilon]$.

Then, we can define our squared-error loss function:

$$L^{\text{VF}}(\theta) = \left(V_\theta(s_t) - V_t^{\text{targ}} \right)^2$$

where

1. V_θ is the value estimate predicted by the policy with parameter θ .
2. s_t is the state in question at timestamp t .
3. V_t^{targ} is the current target value.

Finally, we have our entropy bonus S . For some policy and state configuration, the entropy bonus is denoted as $S[\pi_\theta](s_t)$, where π_θ is the policy.

Combining these 3 terms, we get our full objective function:

$$L_t(\theta) = \hat{\mathbb{E}} \left[L_t^{\text{CLIP}}(\theta) - c_1 L_t^{\text{VF}}(\theta) + c_2 S[\pi_\theta](s_t) \right]$$

where c_1 and c_2 are coefficients. Using stochastic gradient descent, this objective can be tuned with different hyperparameters and optimized, finding the best policy.

6 Conclusion

We have covered a wide variety of machine learning algorithms, from ones used in classification problems to those of OpenAI. We started with the foundations of producing probabilities in machine learning using the sigmoid function. From there, we discussed logistic regression, its extension into multi-class scenarios as softmax regression, and onto different types of learning. We looked at the types of learning ChatGPT uses, and dove into how PPO is used, forming a final objective function for a generic PPO algorithm. Each algorithm may be used to create some form of an NLP program, but the applications correspond to the capabilities of each algorithm. While logistic regression can decide a binary outcome, softmax regression decide between multiple classes. On the other hand, PPO, being a reinforcement learning algorithm, can make improvements over

time, making use of the reward-model nature of reinforcement learning.

I hope this article has taught you more about extending common functions to NLP. I also hope you've learned the power of existing learning algorithms and the promise of PPO. As a challenge, try using TensorFlow to program your own chatbot, keeping the equations in mind, or mess around with TensorFlow's neural network playground and get a feel for what's happening. Good luck!

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How Does AI Audio Generation Work?

Caden Khuu

1 Introduction

In the rapidly developing field of AI technologies, one of the most recent and intriguing applications is the generation of music and speech. Services such as Uberduck have taken the internet by storm with the seemingly miraculous ability to generate any phrase the user desires in the voice of cartoon characters or celebrities. Jukebox AI has made it possible to create "extensions" of songs given snippets less than 20 seconds in length. At the core of AI music, audio, and speech generation is the utilization of neural networks, which are computational models inspired by the human brain's interconnected neurons. These networks are trained on vast datasets of existing musical compositions, spoken words, or diverse audio samples, enabling them to learn complex patterns and structures inherent in the data. As the models learn, they become capable of generating new, original content that often exhibits a remarkable level of creativity and coherence. In the realm of music generation, AI systems can compose melodies, harmonies, and rhythms across various genres. They can analyze the styles of different composers and create music that emulates their unique characteristics or blend influences to produce entirely novel compositions. Similarly, in the field of audio synthesis, AI algorithms can recreate lifelike sounds or generate entirely new ones, providing unprecedented possibilities for sound design in industries such as gaming, film, and virtual reality. The application of AI in speech generation has yielded systems that can produce natural-sounding human speech, often indistinguishable from recordings of real voices. These technologies find applications in voice assistants, audiobooks, and voiceovers, contributing to enhanced user experiences across various platforms.

2 History of Development

The evolution of AI music generation traces its roots back to the mid-20th century when researchers began exploring algorithmic composition as a means of creating music through mathematical rules. Pioneering works such as the 1957 composition ILLIAC Suite, the first string quartet composed entirely on a computer program, laid the foundation for early experiments in computer-generated music. The advent of MIDI, a standard file format to store music, in the 1980s marked a significant milestone, allowing musicians to transmit musical ideas digitally with a number of virtual instruments and paving the way for digital synthesis. In the 1990s, a new wave of innovation emerged with the application of evolutionary algorithms and genetic programming to music composition. This period saw the development of interactive installations like "Emily Howell," an early attempt at AI-generated classical-style music by David Cope. These endeavors hinted at the potential of AI in the creative realm. The 2010s witnessed a paradigm shift with the rise of neural networks and deep learning. Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks became popular for sequence generation tasks, including music. Google's Magenta project, launched in 2016, played a pivotal role by providing tools and models for AI music generation, making the technology more accessible to both researchers and artists. The integration

of AI-generated music into popular culture gained momentum during this period. Platforms like AIVA and Jukedeeck offered AI-generated music for commercial use, showcasing the technology's potential in the mainstream music scene. As we entered the 2020s, transformer models, exemplified by OpenAI's GPT, revolutionized natural language processing and, consequently, music generation. These models could understand and generate more complex musical structures, expanding the possibilities of AI-generated compositions. Style transfer techniques and hybrid models combining rule-based approaches with deep learning further enriched the expressiveness of AI-generated music, allowing the mimicry of specific genres or artists. Real-time collaboration between human musicians and AI systems became a focal point of innovation, exemplified by platforms like Amper Music and IBM's Watson Beat. These tools demonstrated the potential for collaborative creativity, where AI serves as a dynamic and responsive partner in the creative process. AI-generated music also found applications in education, assisting learners in composition, arrangement, and exploration of diverse musical styles. The continuous interplay between technological advancements, artistic exploration, and the quest to push the boundaries of musical creativity defines the rich history of AI music generation. As AI continues to evolve, its role in shaping the future of music composition and performance promises to be even more transformative.

3 How it Works

One of the most popular services for AI generated audio is Uberduck, a popular online service that allows users to effortlessly convert text into speech by selecting from a diverse range of voices, including those of cartoon characters and celebrities. The platform is just one of many that utilize AI vocal imitation and voice cloning techniques, a process wherein an extensive database of a specific character's voice serves as the foundation. The training data of audio clips of the voice is inputted into the system and a neural network learns to mimic the patterns, tone, and nuances of the voice by analyzing the acoustic features, intonation, and speech patterns present in the given audio. In this case, Uberduck uses a transformer model architecture, commonly used for natural language processing, to analyze the text prompt and convert it into audio format. Using similar techniques to this, machine learning developers have been able to create a wide array of AI-assisted audio manipulations, from editing the pitch so that an AI voice could sing a song, to being able to record your own speech and playing it back with a different tone and accent. In addition to human voices, AI is also capable of generating music. Jukebox AI, exemplified by models such as OpenAI's implementation, revolutionize music generation through its advanced generation of diverse musical genres and styles. At its core, a user inputs a text prompt of what they wish the outputted song to sound like, and the system begins by meticulously collecting and analyzing extensive datasets encompassing a wide array of musical compositions. The training data is encoded by a generative mode called VQ-VAE, which compresses the audio into discrete codes. The underlying architecture, often rooted in the Generative Pre-trained Transformer (GPT) framework, such as GPT-3.5, forms the foundation for the model's ability to comprehend intricate musical elements. Through an intensive training process, the AI learns patterns associated with melody, harmony, and rhythm, culminating in a comprehensive understanding of musical composition. When tasked with music generation, users provide a seed input, such as a genre or mood, guiding the AI's creative output. The resultant compositions showcase the versatility of the model, offering original and coherent pieces across a spectrum of genres. Notably, some jukebox AI implementations allow for fine-tuning and customization, permitting users to influence specific elements such as tempo or instrumentation. In more recent developments, rudimentary singing was also implemented as well as AI "extensions" of a short user-inputted audio clip.

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Securing Your Cyber Security: What is Hacking?

Aditya Anand

1 Introduction

People often hear stories of AI taking over the world, or of malicious software posing a threat to your personal information. However, the individuals behind such cyber attacks are sometimes overlooked. With the internet being such a vast space, hackers seek the opportunity to take advantage of any opening they can find. Here are some of the most common techniques hackers use to steal your information.

2 Five Common Strategies Used by Hackers

2.1 Malware

It is important to understand how hackers attack their virtual prey. One of five strategies is usually used, with the most common being the utilization of malware. Malware is short for malicious software, and it consists of harmful code or programs. These programs are attached to websites or emails, and when downloaded, can be used to access personal files and steal data, such as login credentials. Malware can enter the computer in multiple ways, including downloads from websites and removable storage hard drives.

2.2 Social Engineering

Another method commonly used is called social engineering. This involves manipulating “individuals to unknowingly disclose confidential information or personal data, which hackers use for fraudulent purposes.”[1] After convincing the victims that the hackers are a legitimate and trustable source, the hackers can also send links and make use of malware. Common examples of social engineering include scam emails promising free rewards or fake organizations asking for contact information.

2.3 Hacking

Hackers may also use a strategy simply known as “hacking”. As established, hacking is much broader than one method, but this method is often the first step in a cyberattack and involves the use of vulnerabilities in the operating system, such as the lack of encryption, as well as the physical device. This allows hackers to gain unauthorized access to programs, data, networks, or computers, which they use to steal money, make unauthorized purchases, and sell personal information to third parties.

2.4 Credential Compromise

While many methods focus on downloading software to gain access to private information, “credential compromise” targets password managers, such as the Google Password Manager. This form of hacking is completed through an application that automates entry by trying thousands of possible passwords per minute. This allows hackers to view valuable information such as social security numbers, bank accounts, and passwords. Once hackers have access to your password manager, they are able to commit identity theft, transfer money, and access private information.

2.5 Distributed Denial of Service

A fifth method is used more commonly for industrial and political gain. “DDos” (Distributed Denial of Service), is the process of overwhelming a website with more traffic than the server or network can accommodate. Traffic is the number of web users who visit a website, and higher traffic is typically beneficial for the business. However, an unmediated rise in traffic leaves the targeted website inoperable, which can have a variety of impacts. DDos may be used to cause an organization to lose money or popularity, benefitting rival organizations.

3 Conclusion

Overall, there are a variety of methods that hackers can use to steal personal information, money, and passwords. However, there are also multiple ways to avoid such cyber attacks. Downloading software updates, using virtual private networks (VPNs), and avoiding suspicious links and ads are all effective forms of protection. While AI could pose a threat to humanity, it is the people behind these attacks that we must be aware of and defend against.

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